TOP 200

AZURE

Interview Questions

HAPPY RAWAT

PREFACE

ABOUT THE BOOK

This book contains Top 200 Azure interview questions and answers.

ABOUT THE AUTHOR

Happy Rawat has more than 15 years of experience in software development.
He helps candidates in clearing technical interview in tech companies.

Azure Chapters

Azure Fundamentals

- Introduction to Cloud & Azure
- Azure Resource Manager
- Azure Compute Services
- Azure Storage Services

Azure for Developers

- Azure Functions
- Azure Database Services
- DevOps, CI/CD & IaC
- Identity, Security & Access
- Monitoring & Logs
- Event-Driven Messaging
- Performance Optimization

Azure Advance

- Azure Networking - Part 1
- Azure Networking - Part 2
- Real-Time Analytics
- Azure Cognitive Services & ML

1. Introduction to Cloud & Azure

Q. What is Cloud computing? What is Azure? **V. IMP.**

Q. What are the 5 advantages of using Azure over on-premises server?

Q. What are the main cloud service models? Explain the difference between IaaS, PaaS, and SaaS with Azure example? **V. IMP.**

Q. What is Azure Portal? What are the top 20 Azure services?

Q. What is Hybrid Cloud?

Q. Scenario based Real Application Questions

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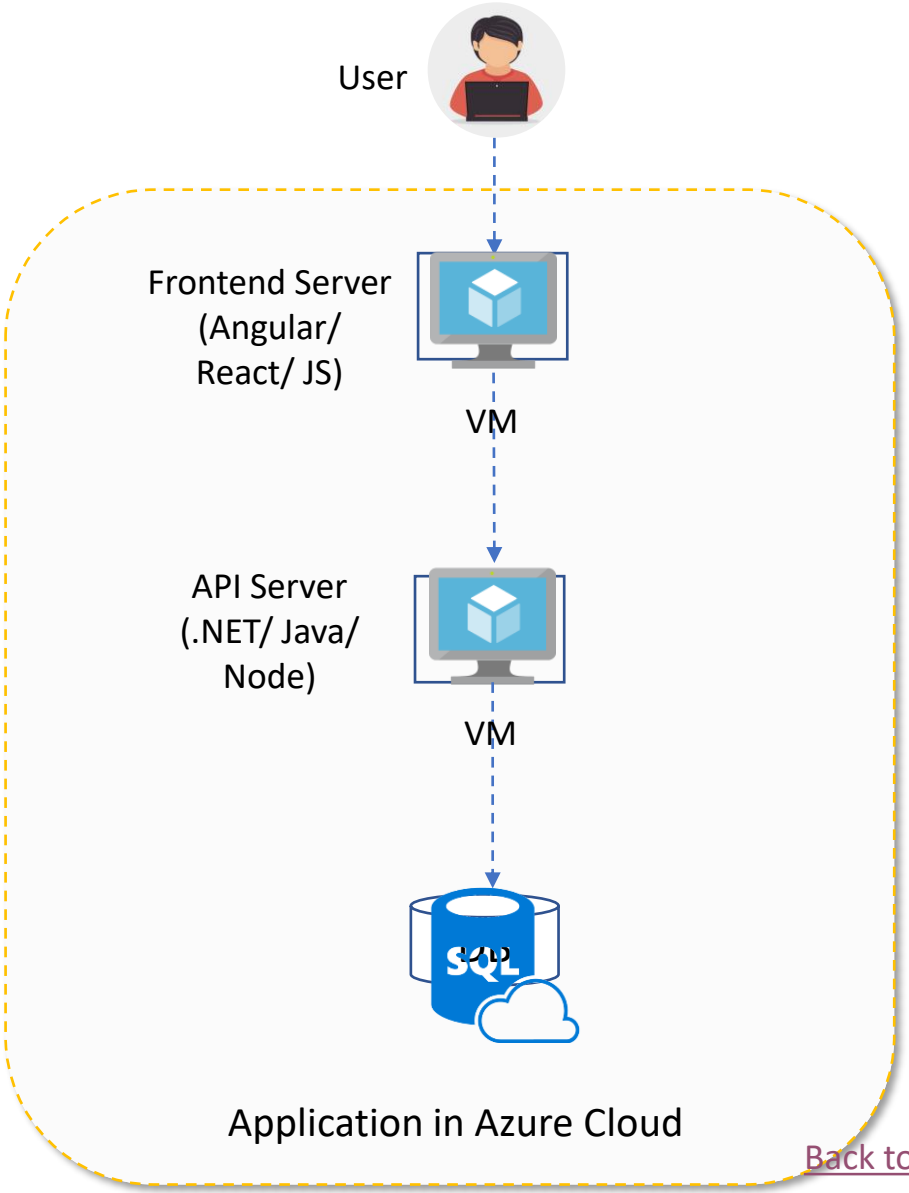
Q. What is Cloud computing? What is Azure? V. IMP.

- Cloud computing is the **delivery of computing services like servers, storage, databases, networking, software, and AI over the internet (the cloud)** instead of using physical hardware or local servers.

- Azure is a **cloud computing platform** provided by Microsoft, offering a wide range of services for building, deploying, and managing applications and infrastructure.



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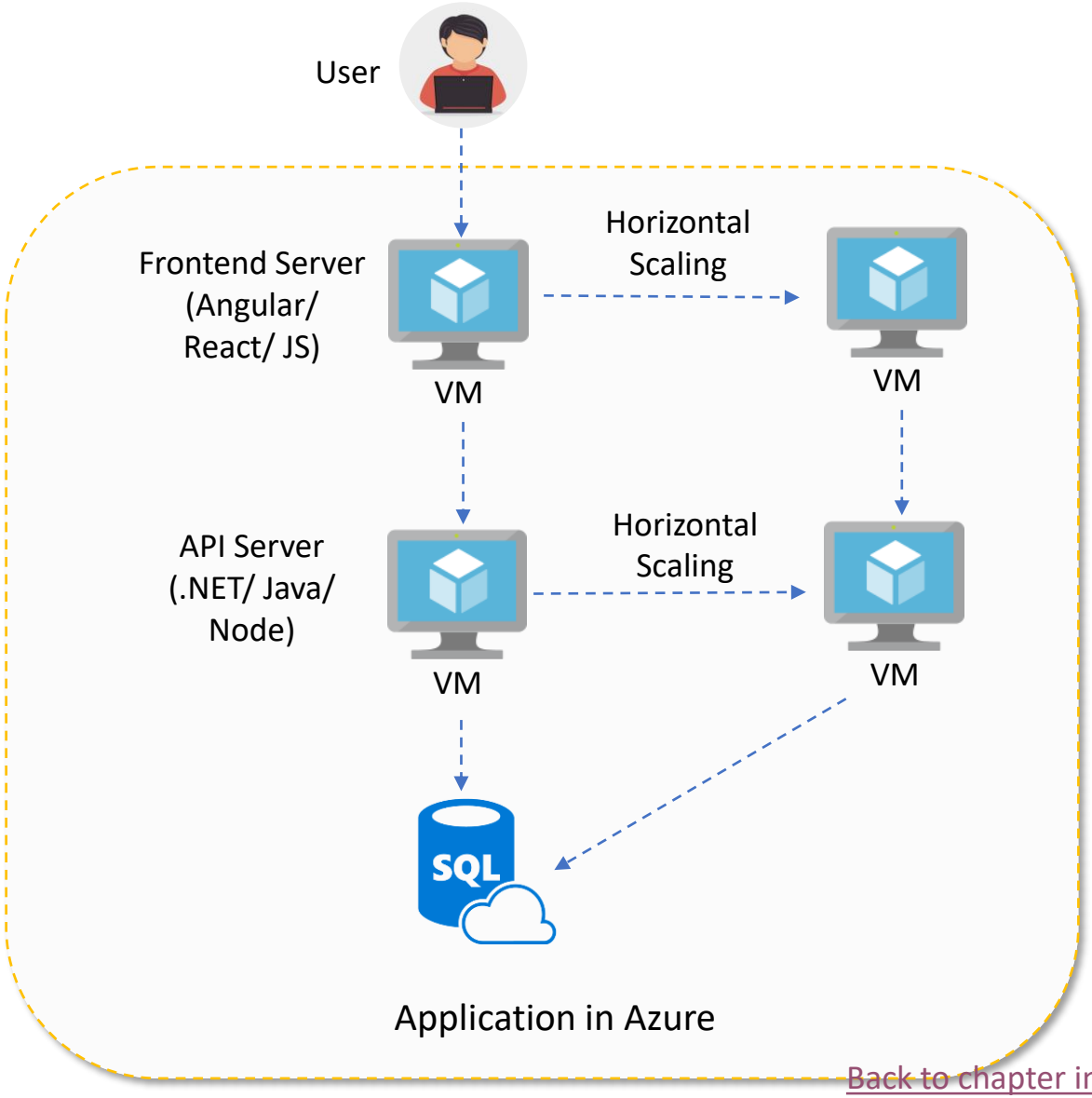
Q. What are the 5 advantages of using Azure over on-premises server? V. IMP.

❖ Advantages of using Azure cloud:

1. **Cost Savings:** Pay-as-you-go model as no need to buy expensive hardware; pay only for what you use. No maintenance costs.
2. **Auto-scaling & Flexibility:** Automatically increases or decreases resources based on demand. Instant loads can be handled.
3. **High Availability:** 99.99% uptime SLA, means more reliable than traditional servers.
4. **Disaster Recovery:** Automated disaster recovery and data backup because data is stored across multiple data centers.
5. **Advanced security:** Azure uses AI-driven threat detection, encryption, and multi-layer security.



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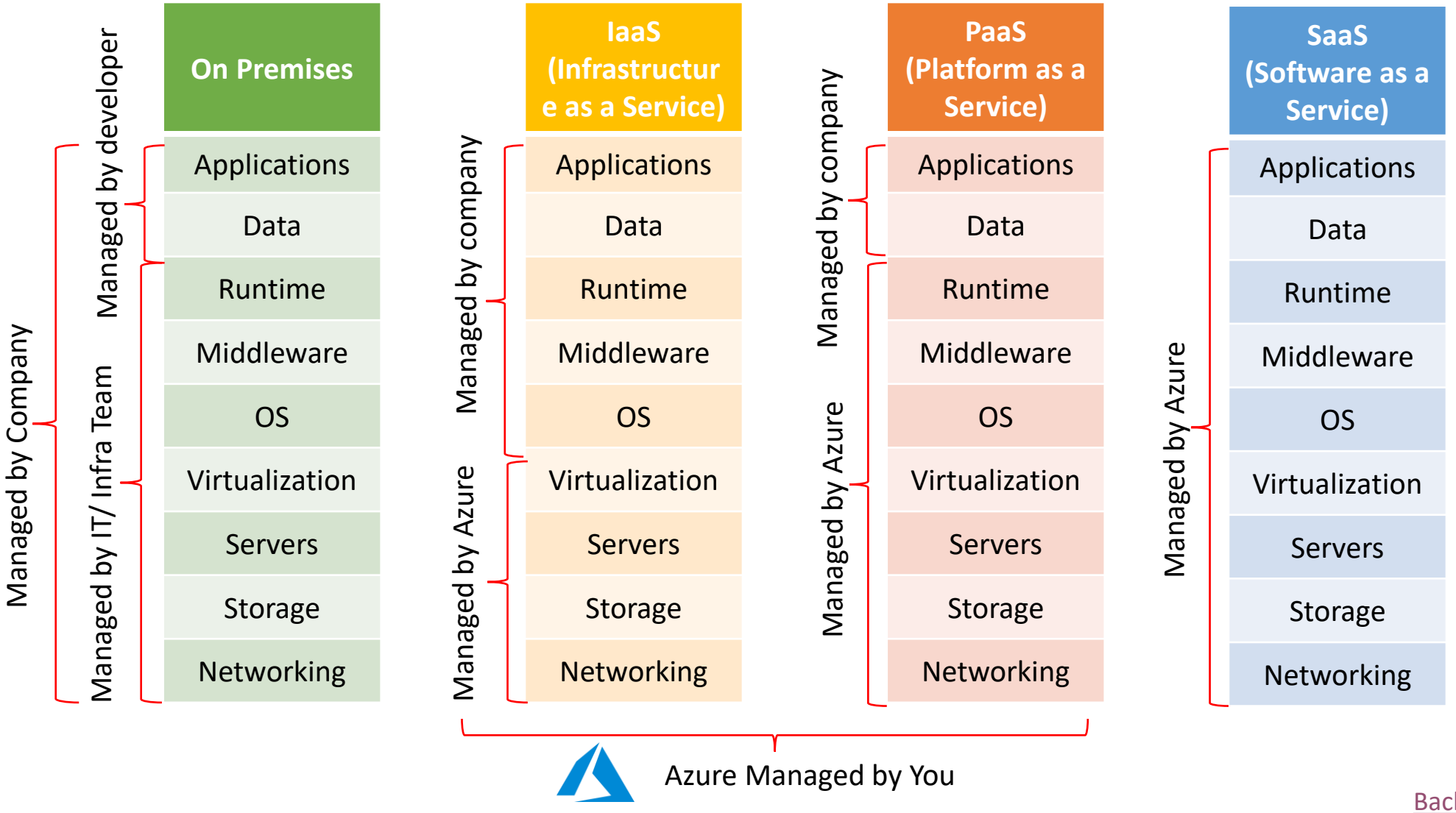
Q. What are the main cloud service models? Explain the difference between IaaS, PaaS, and SaaS with Azure example? V. IMP.



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Q. What are the main cloud service models? Explain the difference between IaaS, PaaS, and SaaS with Azure example? V. IMP.

Category	Definition	Example
Application	Software programs that users interact with.	Web apps, mobile apps.
Data	Information stored, managed, and processed.	Customer records.
Runtime	Environment where applications run.	CLR for .NET, JVM for Java apps.
Middleware	Software that connects applications and systems.	APIs, databases, messaging systems.
OS	Software that manages hardware resources.	Windows, Linux, macOS.
Virtualization	Virtual versions of hardware.	VMware, Hyper-V, Virtual Machines.
Servers	Physical or virtual machines that run applications.	Data center servers, cloud-hosted virtual machines.
Storage	Storing and retrieving digital information.	Hard drives, cloud storage, SSDs.
Networking	Connecting computers, devices, and cloud resources.	Wi-Fi, Ethernet, VPN, cloud networks.

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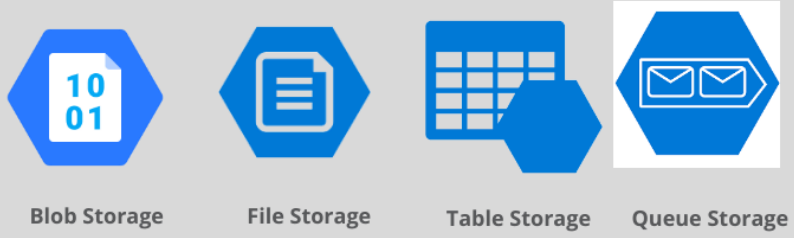
Q. What is Azure Portal? What are the top 20 Azure services?

Application

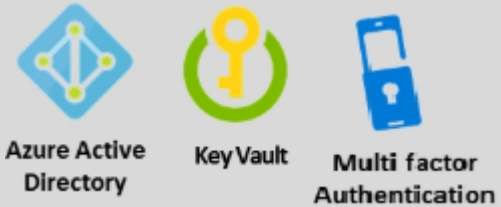
Database

Storage



Infra/ DevOps

Security



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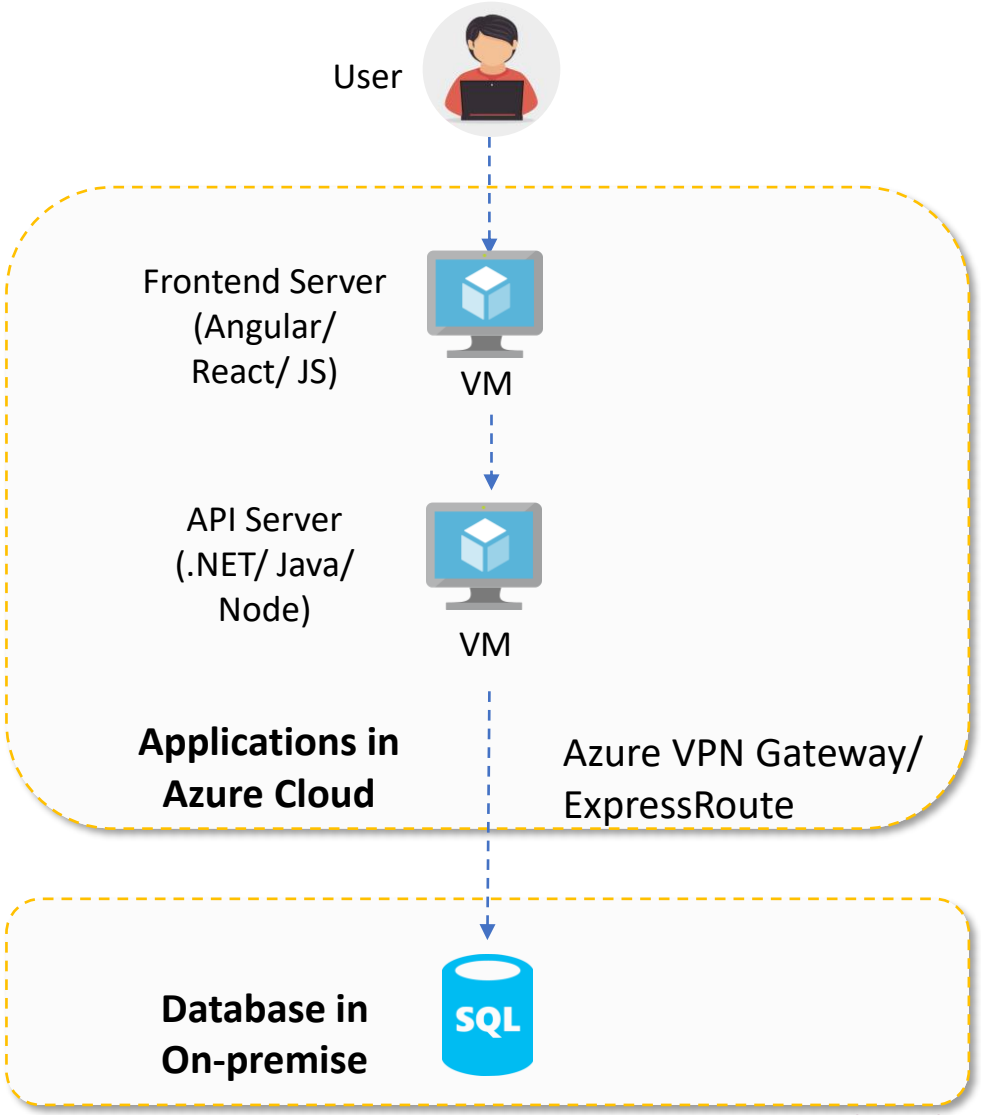
Q. What is Hybrid Cloud?

- Hybrid Cloud = **On-premises + Azure cloud**
- Hybrid Cloud means combining on-premises infrastructure (like your own data center or servers) with Azure cloud services, so they work together.



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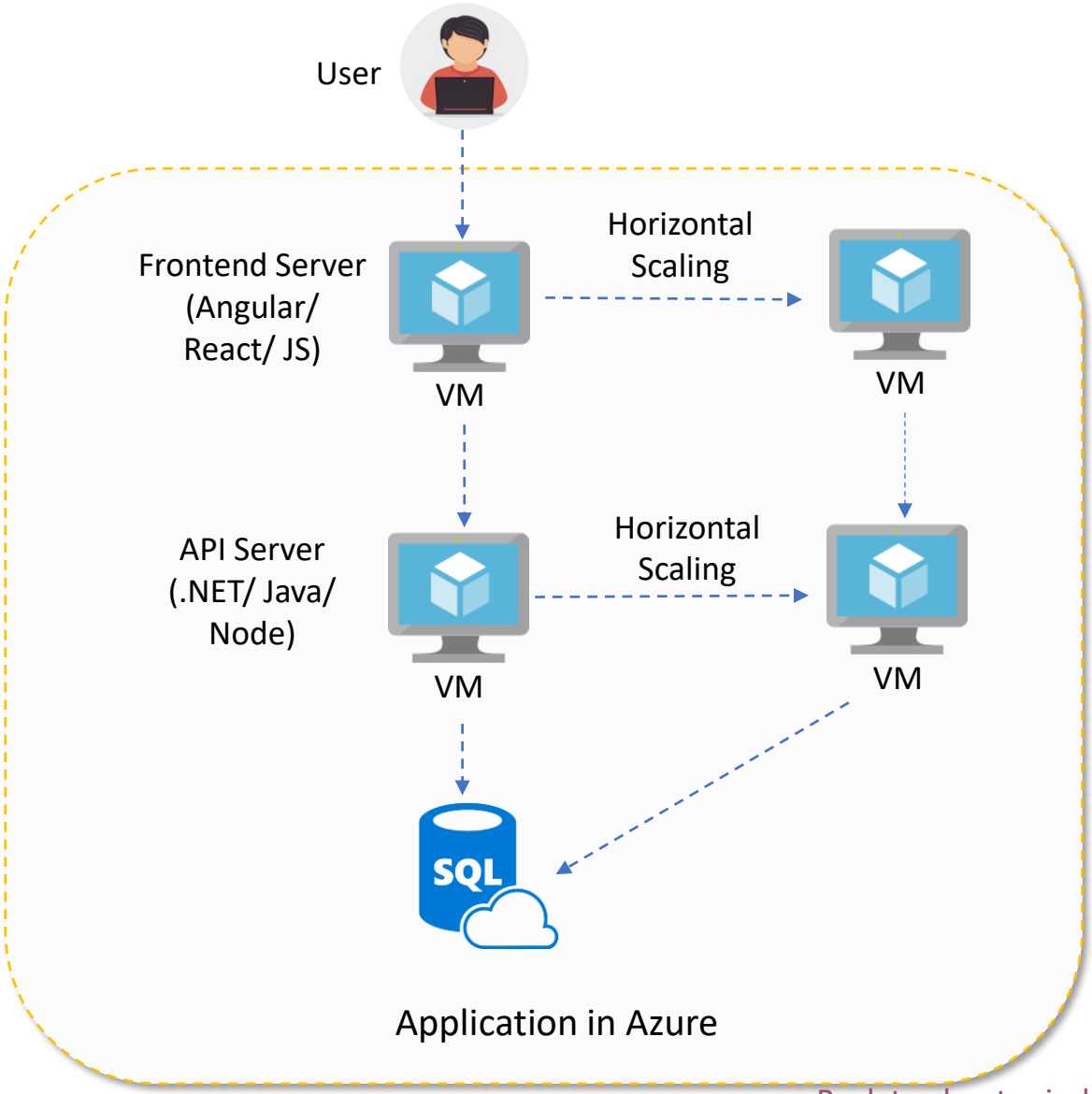
Introduction to Cloud & Azure
(Scenario Based Questions)



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Q. Your team wants to avoid purchasing servers for a new app. What Azure model would you suggest?

- Recommend **PaaS** (Platform as a Service) like Azure App Service.
- It removes the need to manage servers and allows the team to focus on code while Azure handles hosting, scaling, and updates.

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Q. Your client wants to use their existing on-prem server with Azure. What cloud model fits best?

- Suggest **Hybrid Cloud**.
- It allows seamless connection between on-prem servers and Azure resources like VMs or databases using VPN or ExpressRoute.

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Q. Who is responsible for patching the OS in IaaS, PaaS, and SaaS?

- In IaaS, customer/ you/ your company handles patching.
- In PaaS, Azure handles OS patching.
- In SaaS, Azure manages everything including the app itself.

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Q. Your team wants to deploy a web app without managing OS. What Azure offering is ideal?

- Azure App Service (**PaaS**) is perfect.
- Developers focus on code; Azure manages the OS, runtime, and scaling automatically.

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Q. A client needs full OS control and custom software installation. Which compute service fits?

- Use Azure Virtual Machines (**IaaS**).
- They offer full control over OS, configurations, and custom installations.

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Q. What's the difference between Public Cloud and Hybrid Cloud?

- Public Cloud is fully on Azure.
- Hybrid Cloud mixes Azure with on-prem systems for flexibility.

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Q. Your company wants to try Azure without billing. What do you suggest?

- Use **Azure Free Tier**.
- It includes popular services like Azure Functions, VMs, and Storage with limited usage for free.

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Q. Can you access Azure resources using mobile?

- **Yes**.
- Use the Azure Mobile App to monitor, restart services, check status, and receive alerts on the go.

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2. Azure Resource Manager

Q. What is a Resource in Azure? How is it different from a Service in Azure? **V. IMP.**

Q. What are Resource Groups in Azure? Why do we use them in projects?

Q. How do you decide which resources to keep in the same resource group?

Q. What are Azure Subscriptions?

Q. How does access control differ between Subscription level and Resource Group level?

Q. What are Management Groups in Azure? Can you give an example?

Q. What is Azure Resource Manager, and what is its role? **V. IMP.**

Q. Can you explain Azure Regions and Availability Zones? How are they different?

Q. Scenario based Real Application Questions

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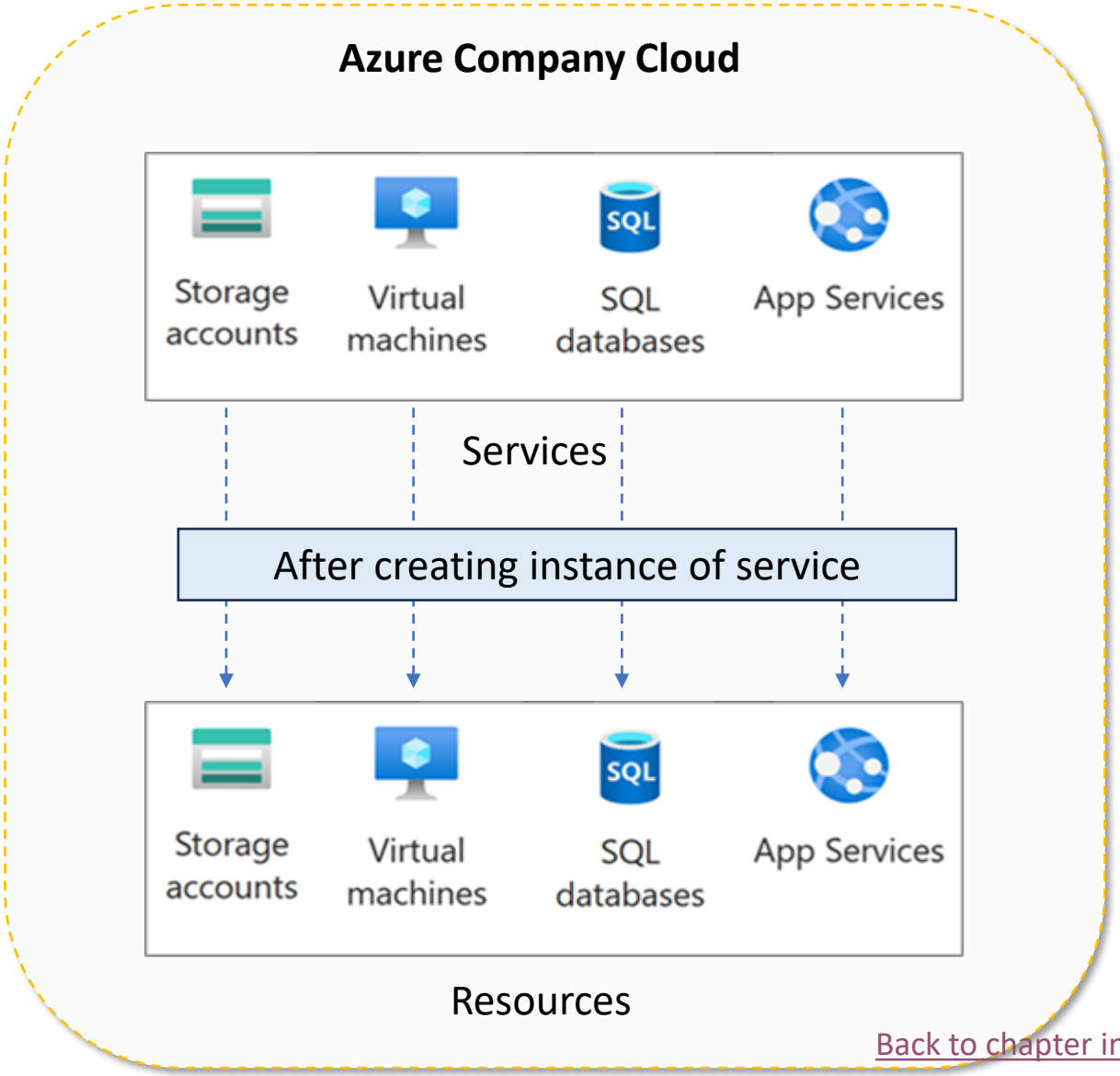
Q. What is a Resource in Azure? How is it different from a Service in Azure? V. IMP.

- **Azure Resource** - A resource in Azure is any individually manageable component (an instance of service) that you can create, configure, and use in the cloud.



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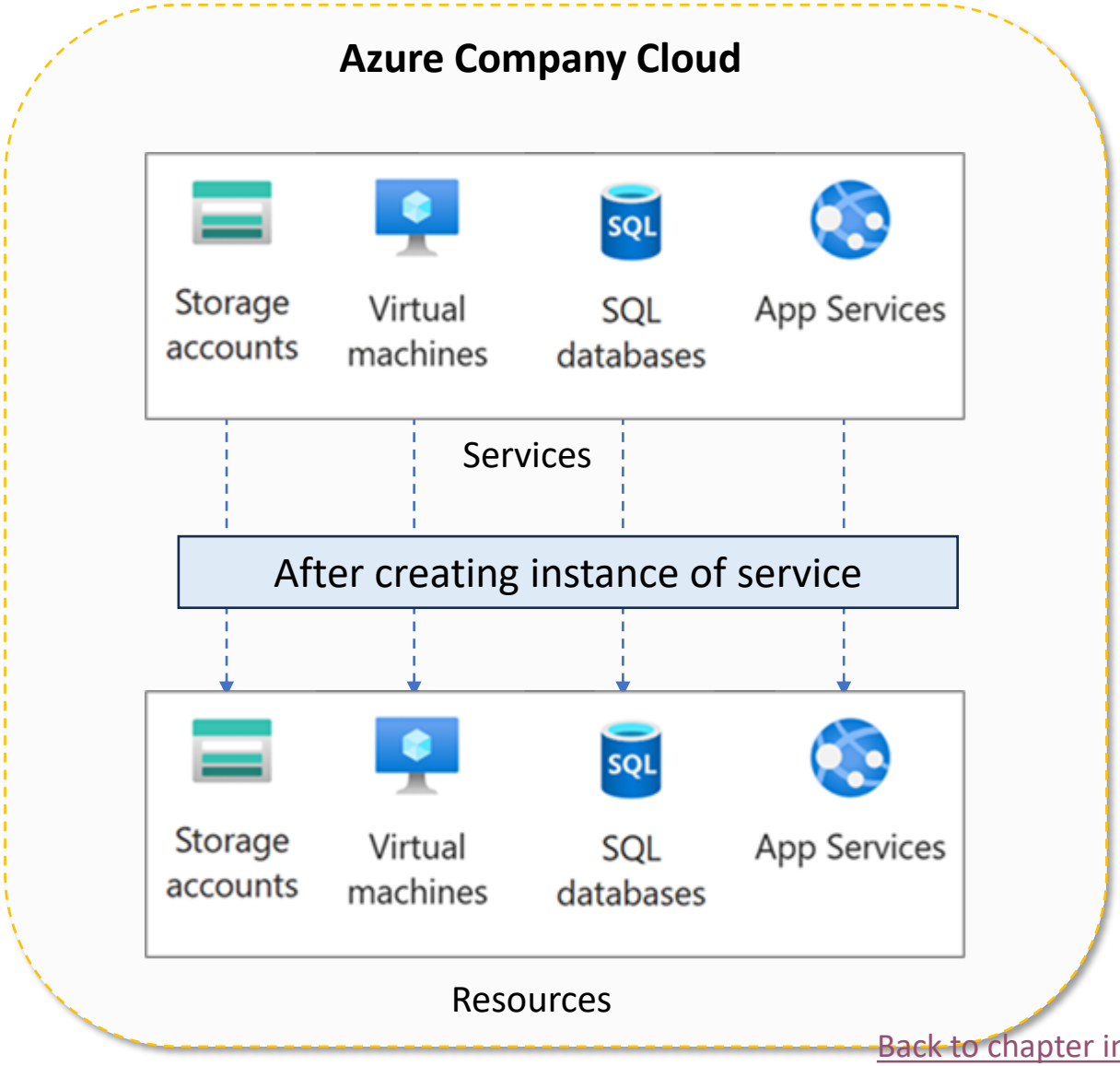
Q. What is a Resource in Azure? How is it different from a Service in Azure? V. IMP.

❖ **Difference between a Resource and a Service?**

1. Azure Service refers to the overall offering (e.g., Virtual Machines, App Services), whereas an Azure Resource is an **individual instance** you create within a service (e.g., a specific VM, a specific storage account)

2. For example, Azure Service is like a class and Azure Resource is like an object or instance of that class.



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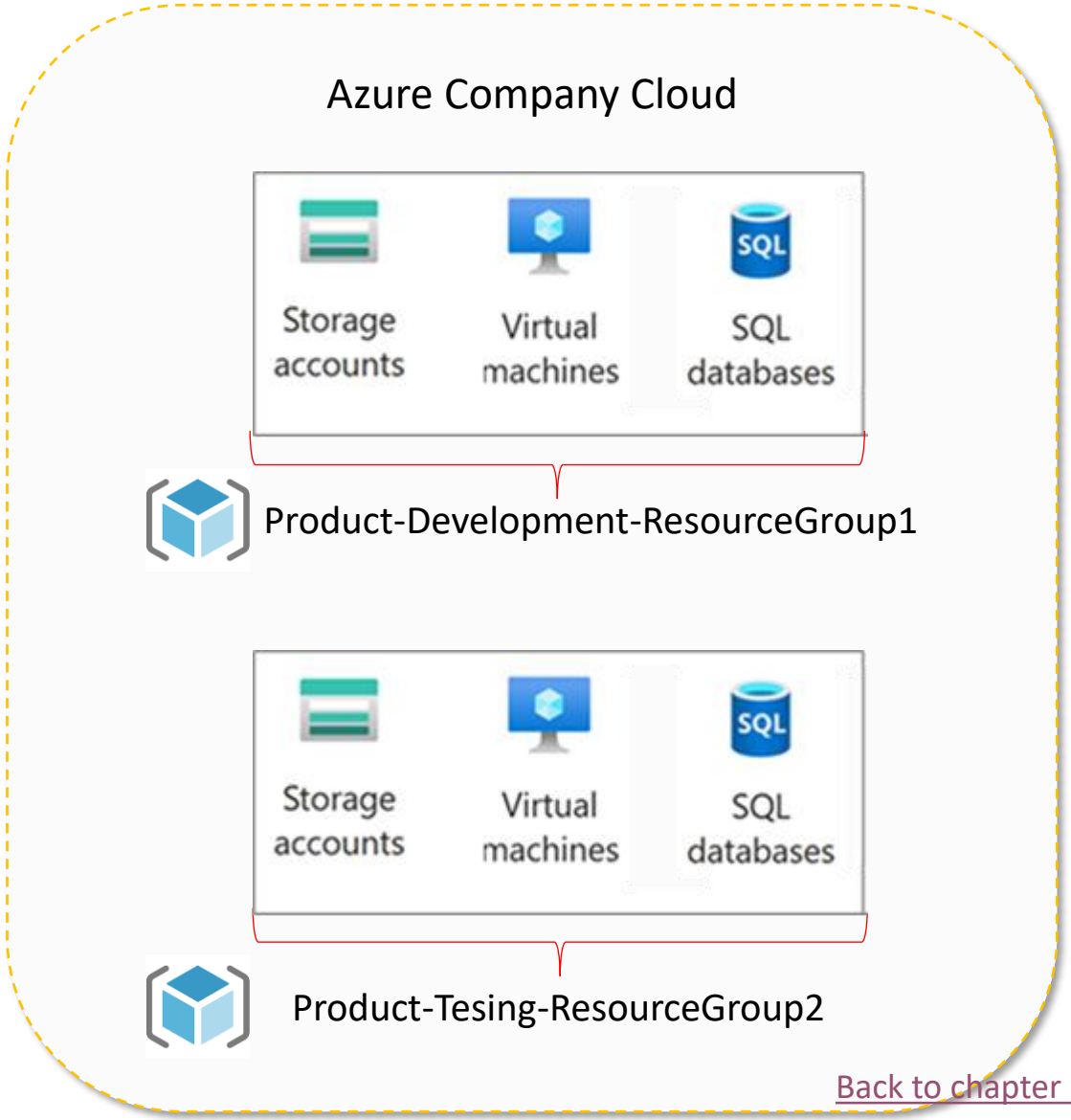
Q. What are Resource Groups in Azure? Why do we use them in projects?

- **Resource Group** - A Resource Group (RG) in Azure is a logical container that holds related Azure resources.
- **Why to use?** - It allows you to manage, organize, and control resources efficiently.



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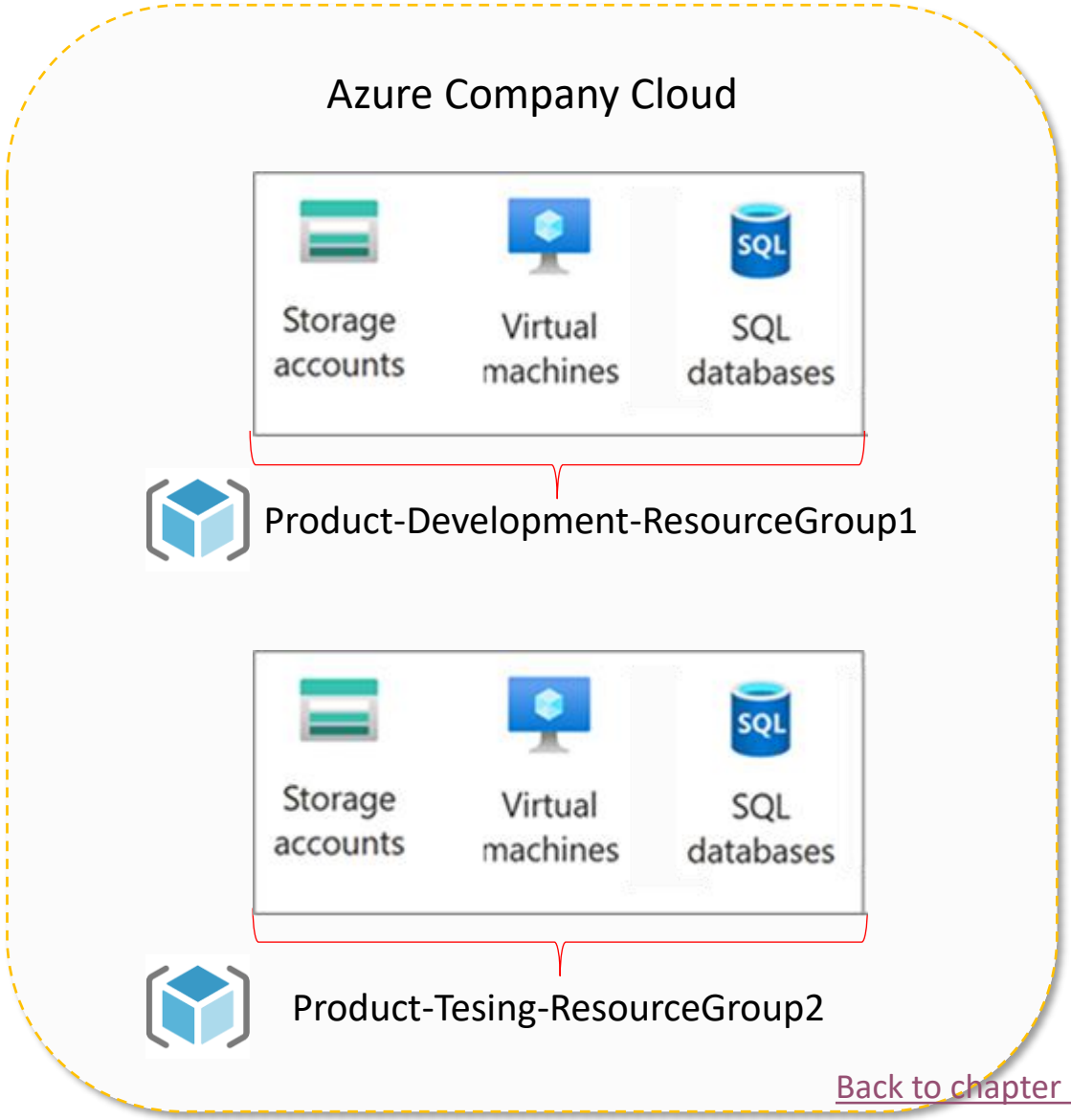
Q. How do you decide which resources to keep in the same resource group?

❖ **On what basis you will group resources?**

1. **Environment-Based Resource Groups (Most Common)** - Companies create separate Resource Groups for different environments to keep production, testing, and development isolated. For example, Product-Dev-RG, Product-Test-RG, Product-Staging-RG, Product-Prod-RG
2. **Business Unit-Based Resource Groups** - A company with multiple departments or business units may structure RGs based on teams.



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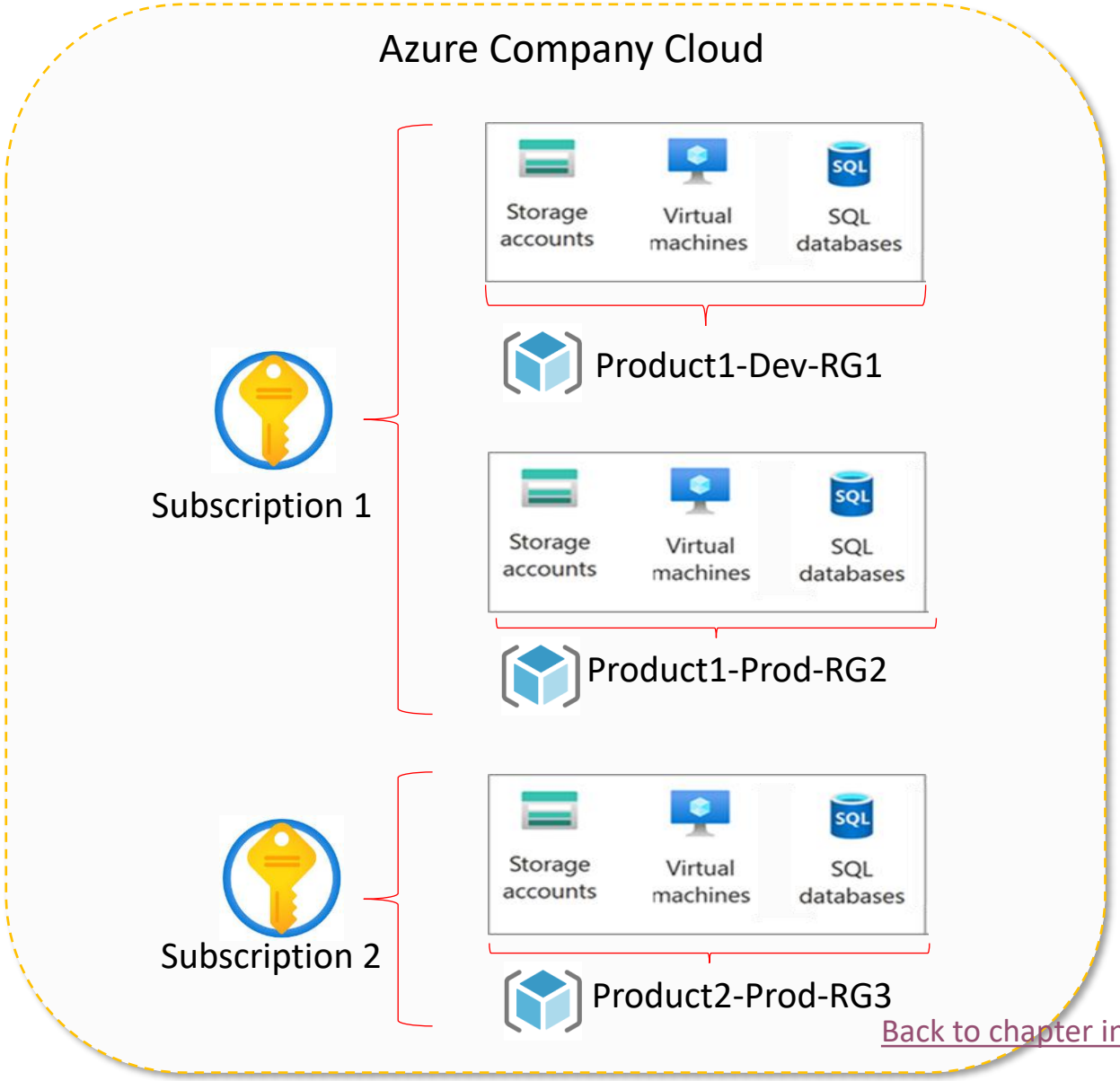
Q. What are Azure Subscriptions?

- **Subscriptions** - An Azure Subscription is a container for Azure resources that **defines billing**, access control, and resource limits.



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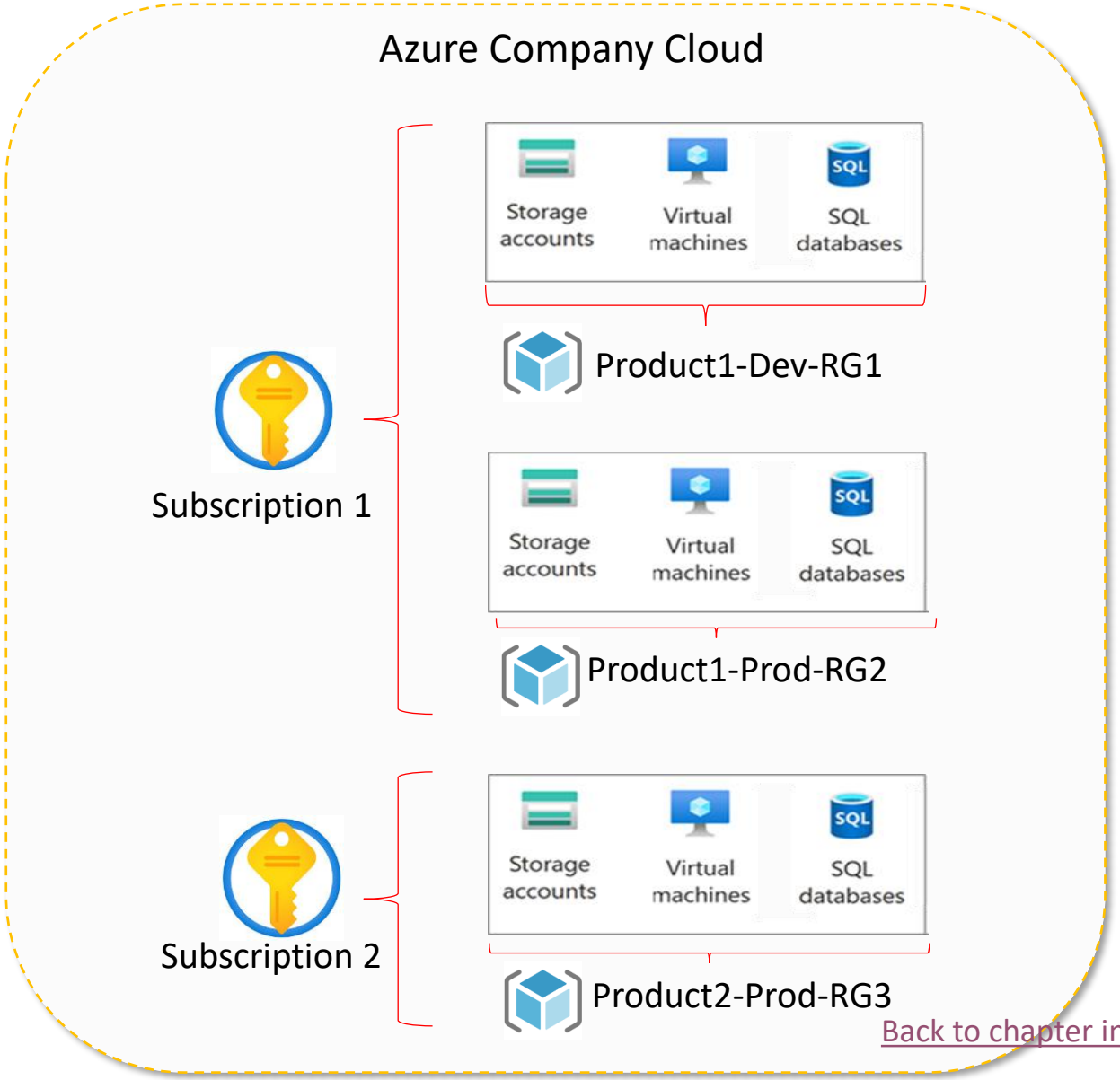
Q. How does access control differ between Subscription level and Resource Group level?

❖ **How Does Access Differ at the Subscription Level vs. Resource Group Level?**

If a person is given access at the Subscription level, they automatically get access to all Resource Groups inside it. Typically, managers or leads have Subscription-level access to manage billing and overall control. Developers usually get access to specific Resource Groups (one or two) to prevent unintended changes across all resources.



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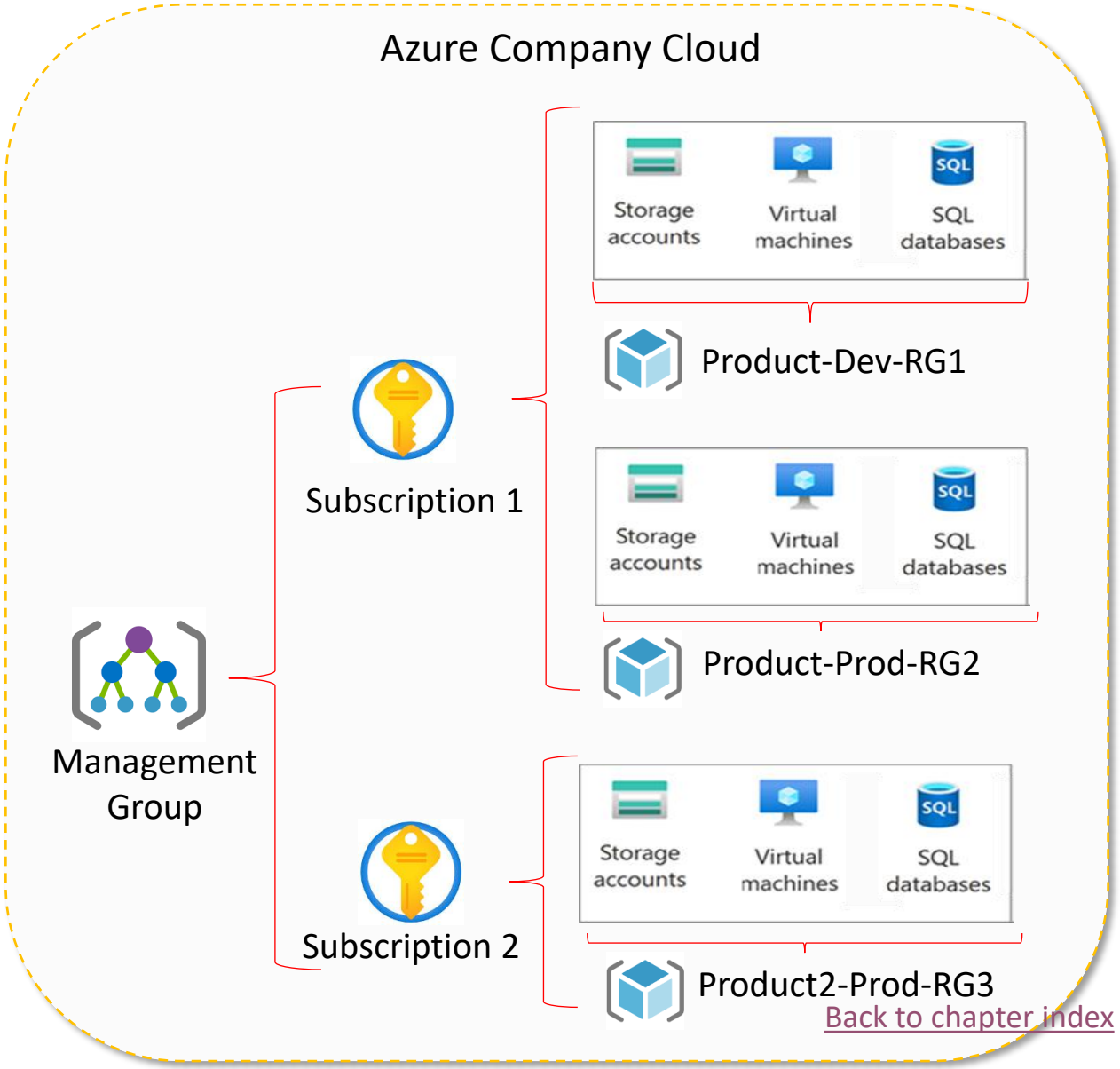
Q. What are Management Groups in Azure? Can you give an example?

- **Management Groups** - Azure Management Groups are top-level containers used to organize multiple subscriptions under a common hierarchy for governance, security, and policy enforcement.

- Management Group = CTO/ DU Lead
- Subscription = Managers/ Leads
- Resource Group = Developers/ Testers



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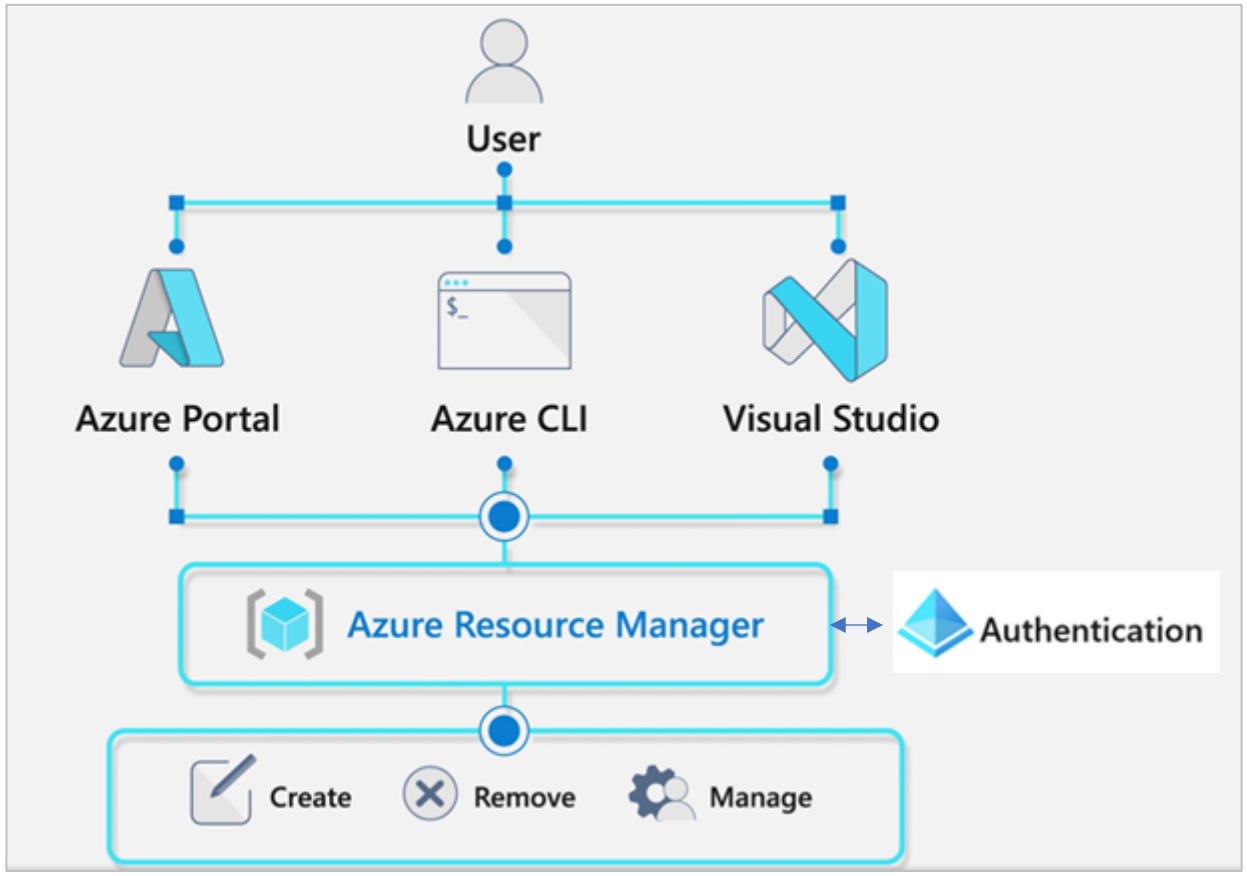
Q. What is Azure Resource Manager, and what is its role? V. IMP.

- **Azure Resource Manager:** Azure Resource Manager (ARM) is the central management layer in Azure that enables users to **deploy, manage, and organize cloud resources in a structured way**.
- It ensures that all resources are created, updated, and deleted in a consistent and controlled manner.



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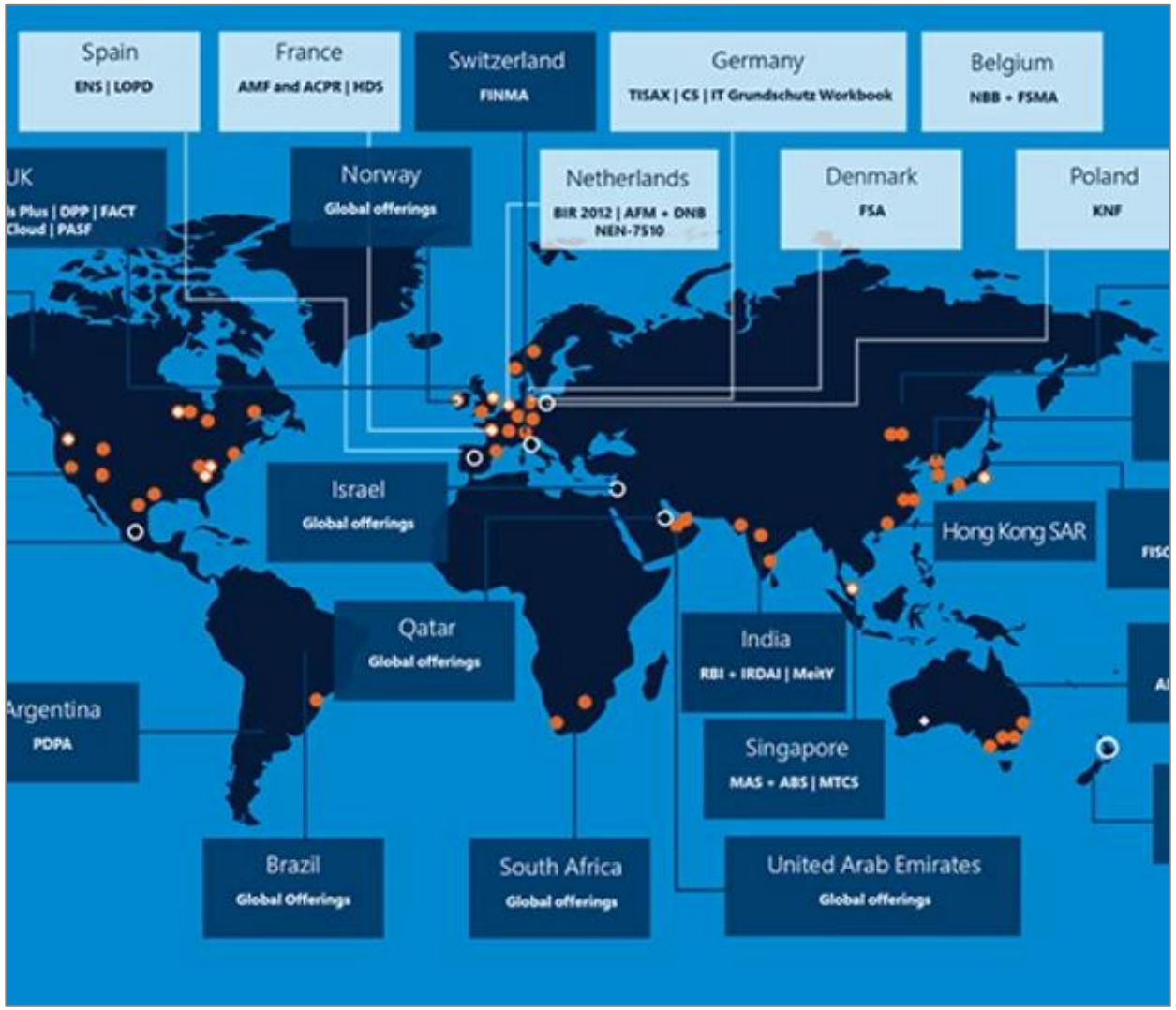
Q. Can you explain Azure Regions and Availability Zones? How are they different?

- **Azure Regions:** A **geographical area** where Azure has multiple data centers. It provides cloud services close to users to improve performance and compliance.

- **Azure Availability Zones:** A physically separate **data center** within an Azure region. Ensures high availability by protecting against failures in a single data center.



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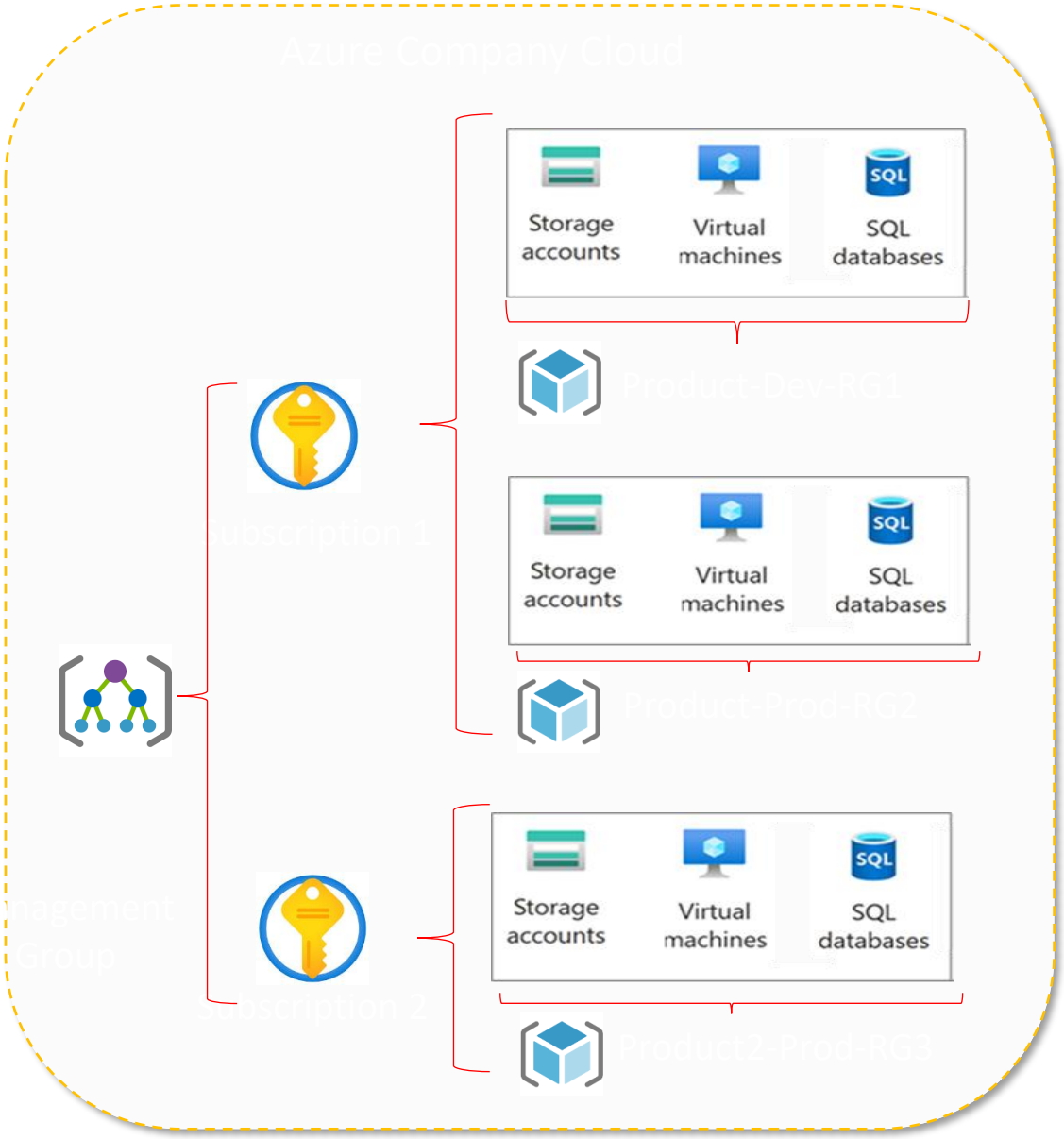
Azure Resource Manager
(Scenario Based Questions)



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Q. You want to apply a single policy across multiple subscriptions. What Azure feature should you use?

- Use **Management Groups**.
- They allow you to organize multiple subscriptions and apply RBAC or policies at one place for consistent governance.

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Q. Your team accidentally deletes a resource group. What happens to its resources?

- All resources inside that Resource Group get **deleted permanently**.
- Always use role-based access to prevent accidental deletions.

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Q. You want to allow a user to manage resources in one project but not another. What's the best way?

- Create separate Resource Groups for each project.
- Assign role-based access (RBAC) only to the required group to limit access.

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Q. What is the main benefit of grouping all resources like VMs, databases, and storage into one resource group?

- It enables easy management, cost tracking, and common lifecycle actions like deletion or access control for all related resources.

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Q. Can one resource be part of multiple resource groups?

- **No**.
- A resource can belong to only one Resource Group at a time, but it can still communicate with resources in other groups.

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Q. How is access control different at subscription vs. resource group level?

- Subscription-level roles apply across all resource groups inside it.
- Resource group-level roles only apply to the resources within that group.

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Q. You are asked to deploy 10 VMs in different regions. Can you use the same resource group?

- **Yes**.
- A Resource Group is not tied to a specific region. You can deploy resources in any region within it.

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Q. What is the difference between a Resource and a Service in Azure?

- A Service is a capability Azure provides (e.g., Storage).
- A Resource is a **specific instance** of that service (e.g., myStorage123).

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Q. How does Azure Resource Manager (ARM) help in deployments?

- ARM is the **deployment engine**.
- It ensures declarative templates, resource grouping, role-based access, and policy enforcement during deployment.

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Q. Your app needs high availability within a region. What should you use: Region or Availability Zone?

- **Use Availability Zones**.
- These are isolated physical locations within a region to protect against datacenter-level failures.

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3. Azure Compute Services

Q. What are Azure Compute Services? Name a few of them?

Q. What are Azure Virtual Machines? When would you use them in a project? **V. IMP.**

Q. What are the differences between horizontal and vertical scaling? When to use what?

Q. What are VM Scale Sets? When will you use them in your project?

Q. What is Azure App Service? For what purpose can you use them? **V. IMP.**

Q. What are the 5 main features or advantages of using Azure App Services?

Q. What is the difference btw Azure VMs and Azure App Services? When to use what?

Q. Imagine your application gets more traffic on weekends and less on weekdays. How would you handle that in Azure?

Q. What are different ways to deploy apps in Azure?

Q. What are Deployment Slots in App Service? Why are they useful?

Q. Scenario based Real Application Questions

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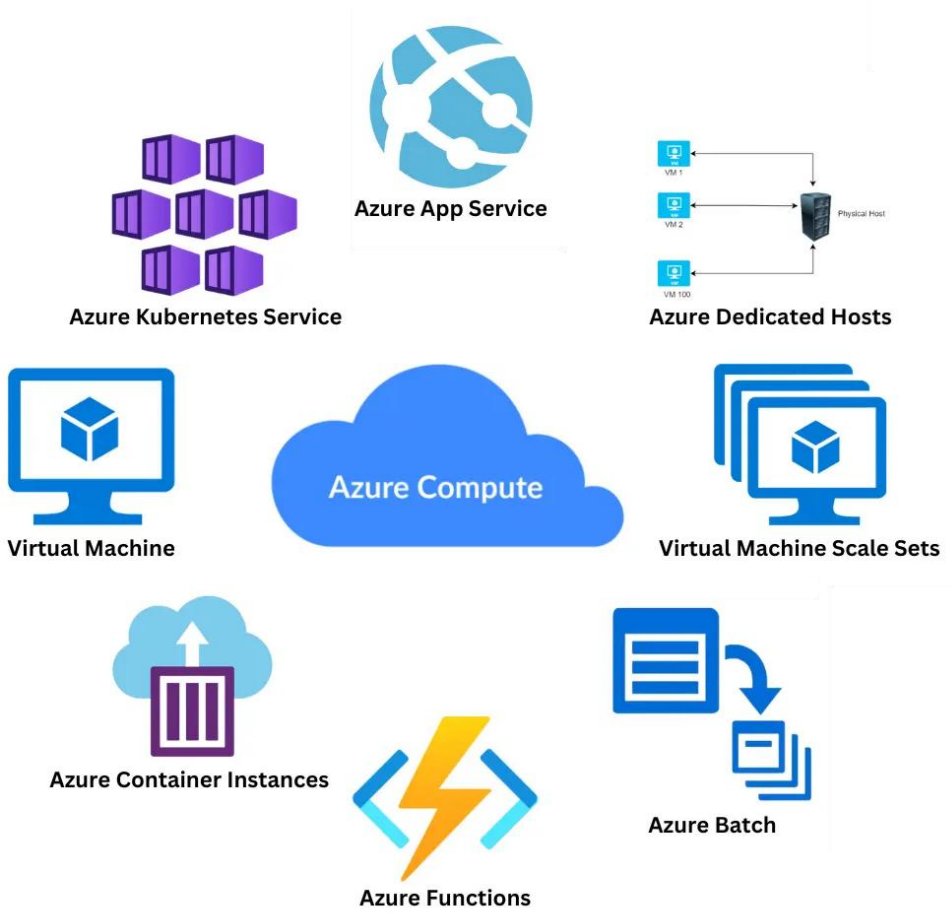
Q. What are Azure Compute Services? Name a few of them?

- Azure Compute Services provide cloud-based processing power, scaling, and execution environments for applications.
- Compute Services because they provide the computing power needed to run applications, process workloads, and manage virtual environments in the cloud.



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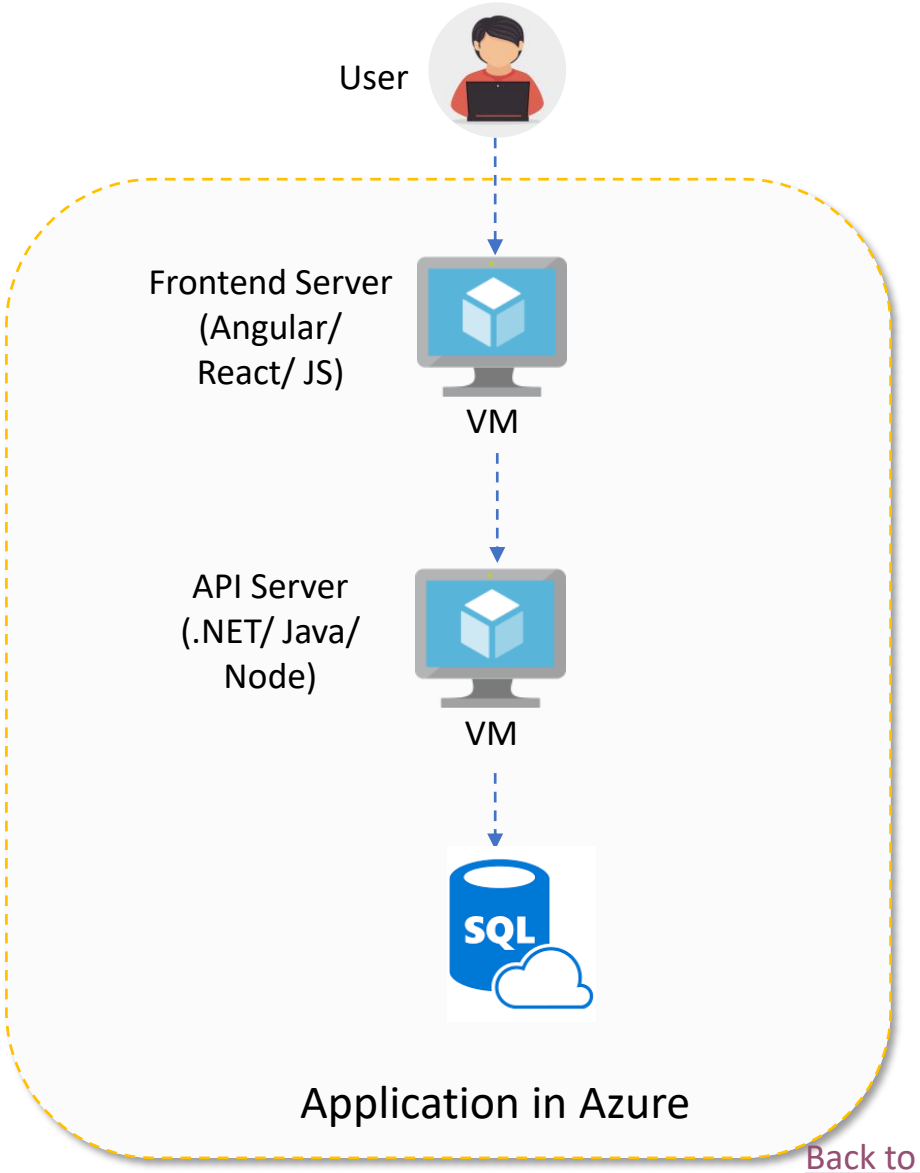
Q. What are Azure Virtual Machines? When would you use them in a project? V. IMP.

- Azure Virtual Machines (VMs) are cloud-based **virtual servers** that allow users to run applications without needing physical hardware.

💡 Azure VM = A Computer in the Cloud
Just like a physical computer needs an OS, CPU, RAM, and storage, an Azure VM provides the same, but in the cloud.



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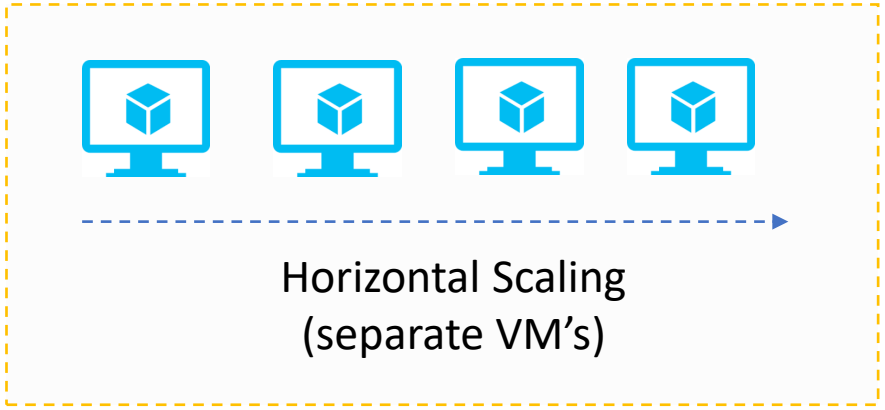
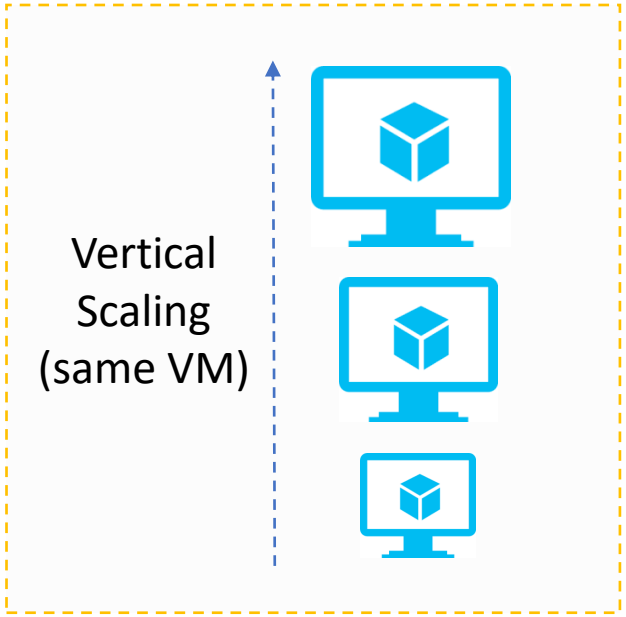
Q. What are the differences between horizontal and vertical scaling? When to use what?

Horizontal Scaling (Scale-Out/In)	Vertical Scaling (Scale-Up/Down)
1. Adds more instances (VMs/Containers)	Increases the size of a single VM (Increase size of CPU/ RAM)
2. Best for **Stateless** applications (Microservices, Web Apps)	Best for **Stateful** applications (Monolithic Apps)
3. More cost-effective as auto scaling is possible because of VM scale sets	Expensive as auto scaling is not possible





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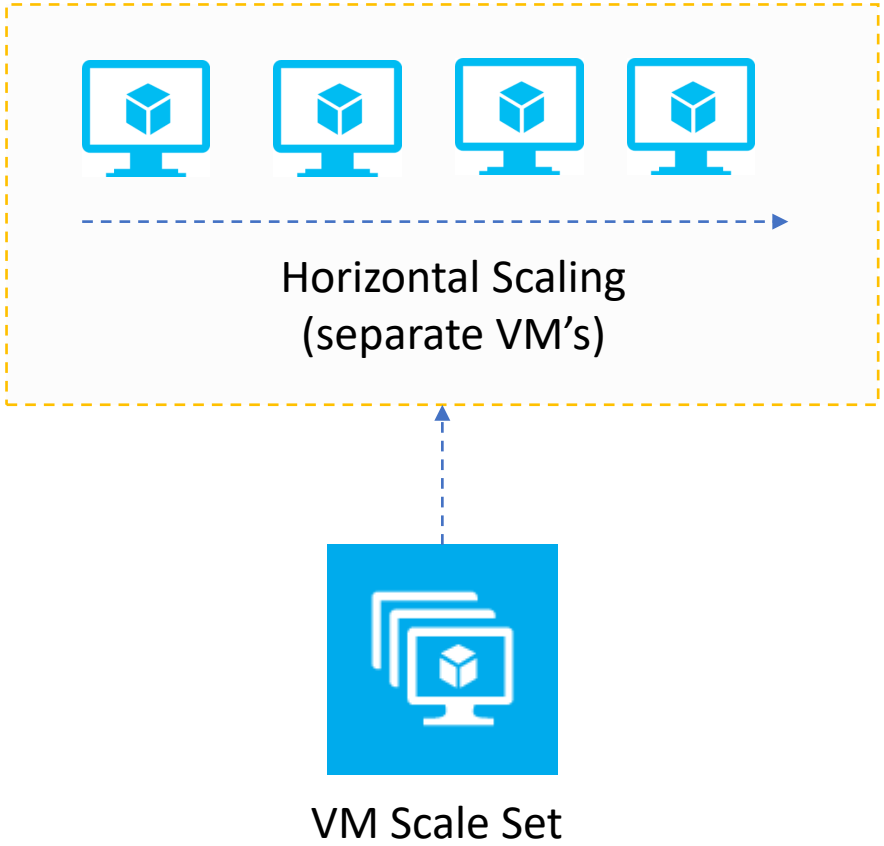
Q. What are VM Scale Sets? When will you use them in your project?

- Azure virtual machine scale sets let you create and manage a group of identical, load balanced VMs.
- The number of VM instances can automatically increase or decrease in response to demand.



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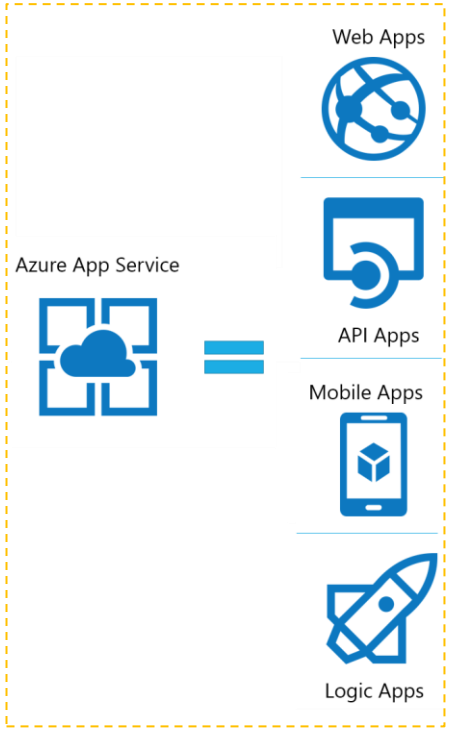
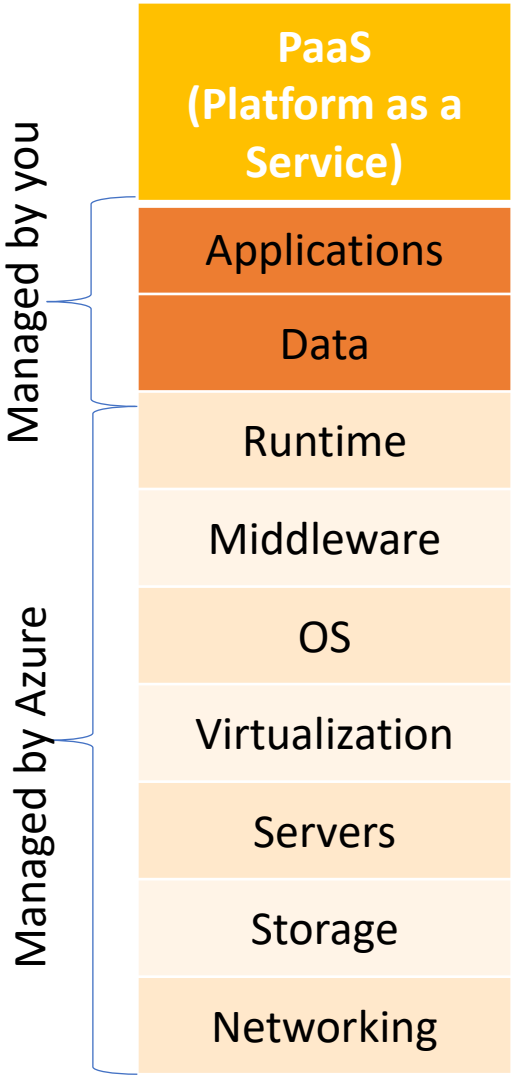
Q. What is Azure App Service? For what purpose can you use them? V. IMP.

- Azure App Services is a fully managed Platform-as-a-Service (PaaS) that enables developers to host web applications, APIs, and mobile backends without managing the underlying infrastructure.





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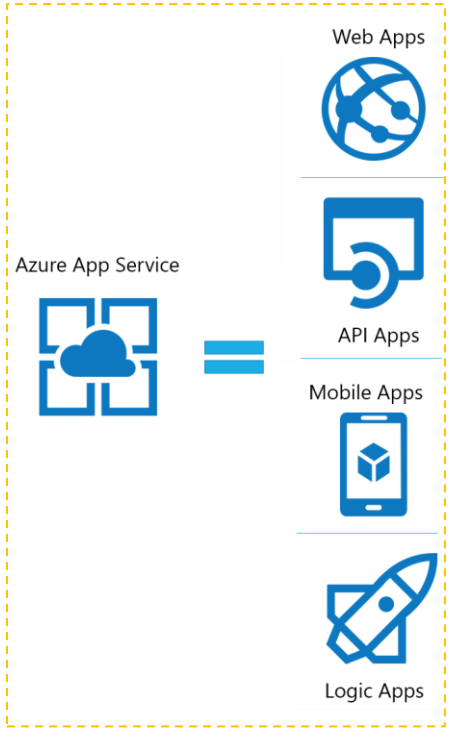
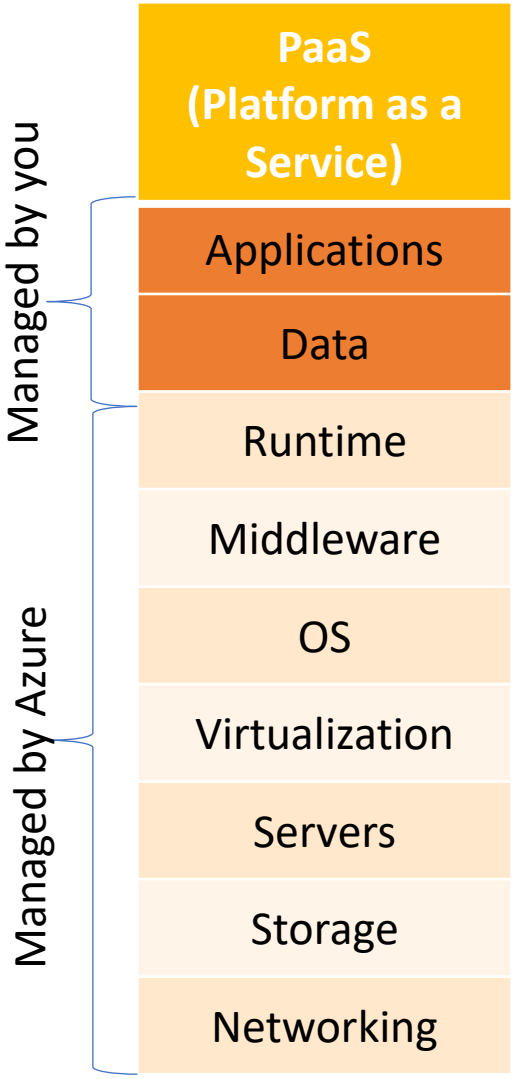
Q. What are the 5 main features or advantages of using Azure App Services?

❖ Features of Azure App Services:

1. Supports Multiple Programming Languages: .NET, Java, Python, Node.js, PHP, Ruby.
2. Built-in Scalability: Automatically scales up/down based on traffic.
3. Fully Managed: Microsoft handles OS updates, security, and maintenance.
4. High Availability: Supports auto-failover & backups.
5. Secure & Compliant: Built-in authentication.





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Q. What is the difference btw Azure VMs and Azure App Services? When to use what?

Azure Virtual Machines (VMs)	Azure App Services
1. Cloud-based **virtual servers** where you can install any software and manage configurations.	A Platform-as-a-Service (PaaS) that allows hosting web applications, APIs, and mobile apps **without managing the servers** or infrastructure.
2. User-managed: You handle OS updates, security, scaling, load balancing & application installation.	**Fully managed**: Microsoft handles OS, security, scaling and load balancing.
3. Use Case: Hosting **custom applications, databases, and workloads** that need full control.	Hosting **web apps, APIs, and mobile backends** with automatic scaling.

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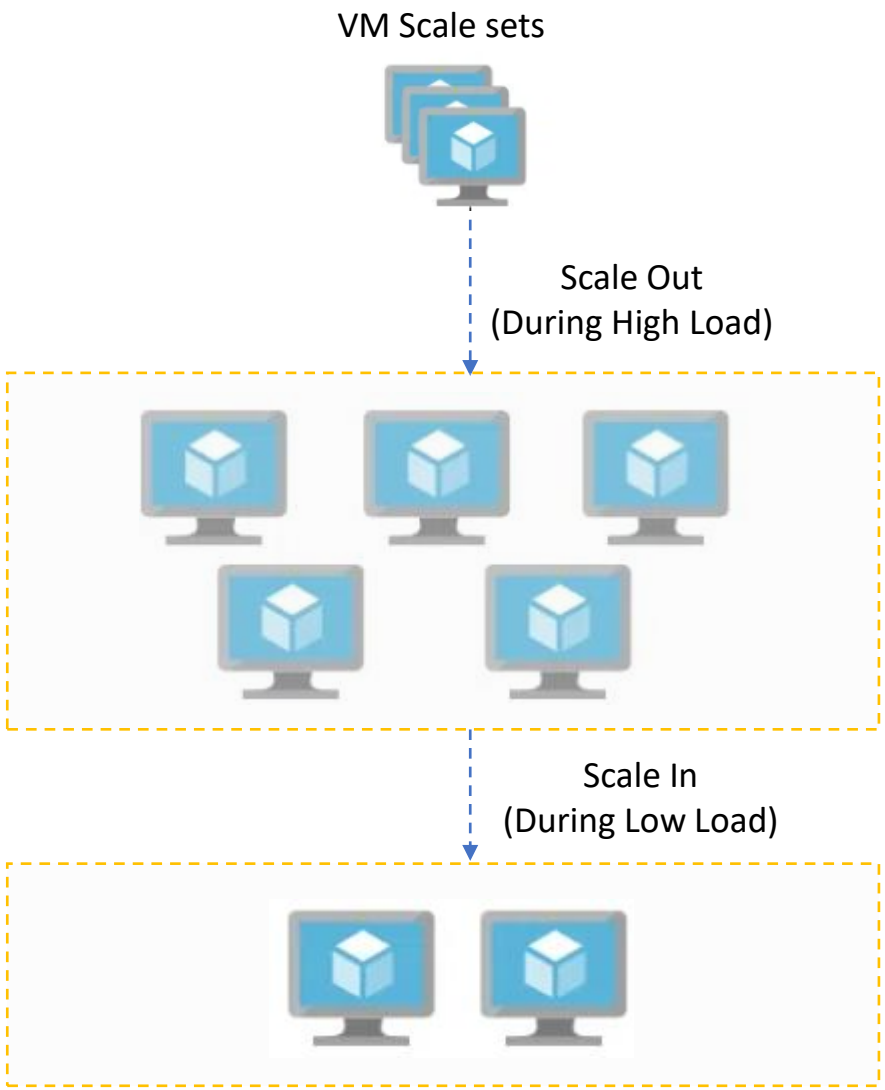
Q. Imagine your application gets more traffic on weekends and less on weekdays. How would you handle that in Azure?

- I will use VM Scale Sets or App Services to **enable autoscaling**.
- In VM Scale Sets, the traffic will automatically distribute evenly using load balancing.
- This is cost effective also as when not required the VM instance numbers will go down automatically.



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Q. What are different ways to deploy apps in Azure?

- 7 Different Ways to Deploy Applications in Azure:

1. Azure Portal: Manual Deployment
2. Azure CLI & PowerShell
3. Visual Studio / VS Code
4. Azure DevOps (CI/CD Pipelines) (For Developers)
5. Terraform / Bicep: Infrastructure as Code - (For Infra Team for Prod Deployment
6. ARM Templates
7. Docker & Kubernetes: Container-Based Deployment

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Q. What are Deployment Slots in App Service? Why are they useful?

- Azure Deployment Slots are live environments within an Azure App Service that allow you to deploy and test applications before making changes to production.

Deployment Slots

Deployment slots are live apps with their own hostnames. App content and configurations elem
deployment slots, including the production slot.

NAME	STATUS	APP SERVICE PLAN
b59-eus-webapp PRODUCTION	Running	ASP-b59euswebapp-a039
b59-eus-webapp-staging	Running	ASP-b59euswebapp-a039
b59-eus-webapp-dev	Running	ASP-b59euswebapp-a039
b59-eus-webapp-test	Running	ASP-b59euswebapp-a039

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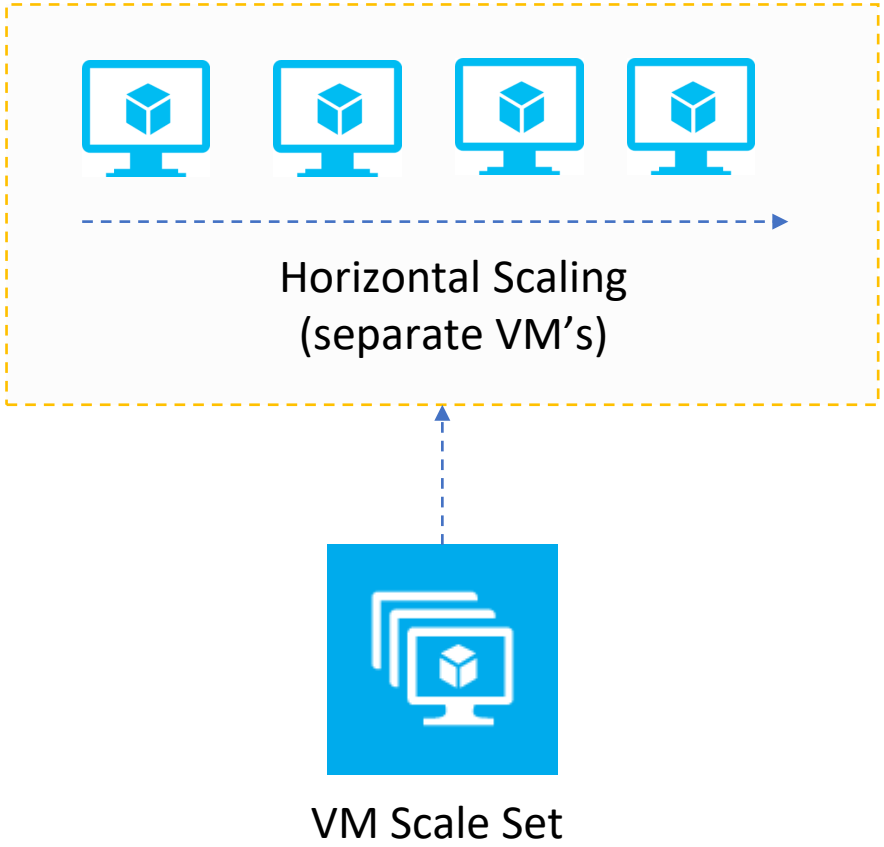
Azure Compute Services
(Scenario Based Questions)



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Q. Your app experiences unpredictable traffic spikes. What Azure compute option would you use?

- **Use Azure App Service** with Auto-Scaling or VM Scale Sets. Both scale out automatically based on metrics like CPU or requests.

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Q. Can you schedule VMs to automatically shut down at night to save costs?

- **Yes**.
- Use Azure Automation or Start/Stop VM during off-hours to reduce costs when VMs aren't in use.

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Q. What is the main benefit of VM Scale Sets over manually adding VMs?

- VMSS automates deployment and **scaling of VMs** across availability zones. You don't have to manually manage or configure each VM.

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Q. Your company wants to host a web app quickly with minimum infra management. What do you choose?

- **Use Azure App Service (PaaS)**.
- It lets you deploy with zero infrastructure overhead and built-in scaling, security, and CI/CD.

Back to chapter index

Q. What is the difference between Azure App Service and Azure Container Instances?

- App Service runs code in **managed environments** (great for web apps).
- Container Instances run **Docker containers** directly (better for microservices and jobs).

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Q. Your client wants automatic scaling across multiple regions. Which compute option supports this?

- **Use VM Scale Sets or App Service** Environments (ASE) with Traffic Manager or Front Door for multi-region scaling.

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Q. How can you ensure zero downtime while deploying a new version of your app?

- Use **Deployment Slots** in Azure App Service. Deploy to staging, test it, then swap with production with no downtime.

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4. Azure Functions

Q. What are Azure Functions? Have you ever used them in your project? Why? **V. IMP.**

Q. Why not include the Azure Function code directly within the main application?

Q. What exactly is Serverless in Azure? Is it truly server-less?

Q. What are Triggers in Azure Functions? Can you name a few types? **V. IMP.**

Q. What are the important steps to write code of an Azure function? Share example?

Q. How do you deploy an Azure Function to the cloud?

Q. Which services of Azure are Serverless?

Q. What is the difference between Azure Function and Function App?

Q. What are Azure Logic Apps? **V. IMP.**

Q. How Azure Logic Apps are different from Azure Functions? When to use what?

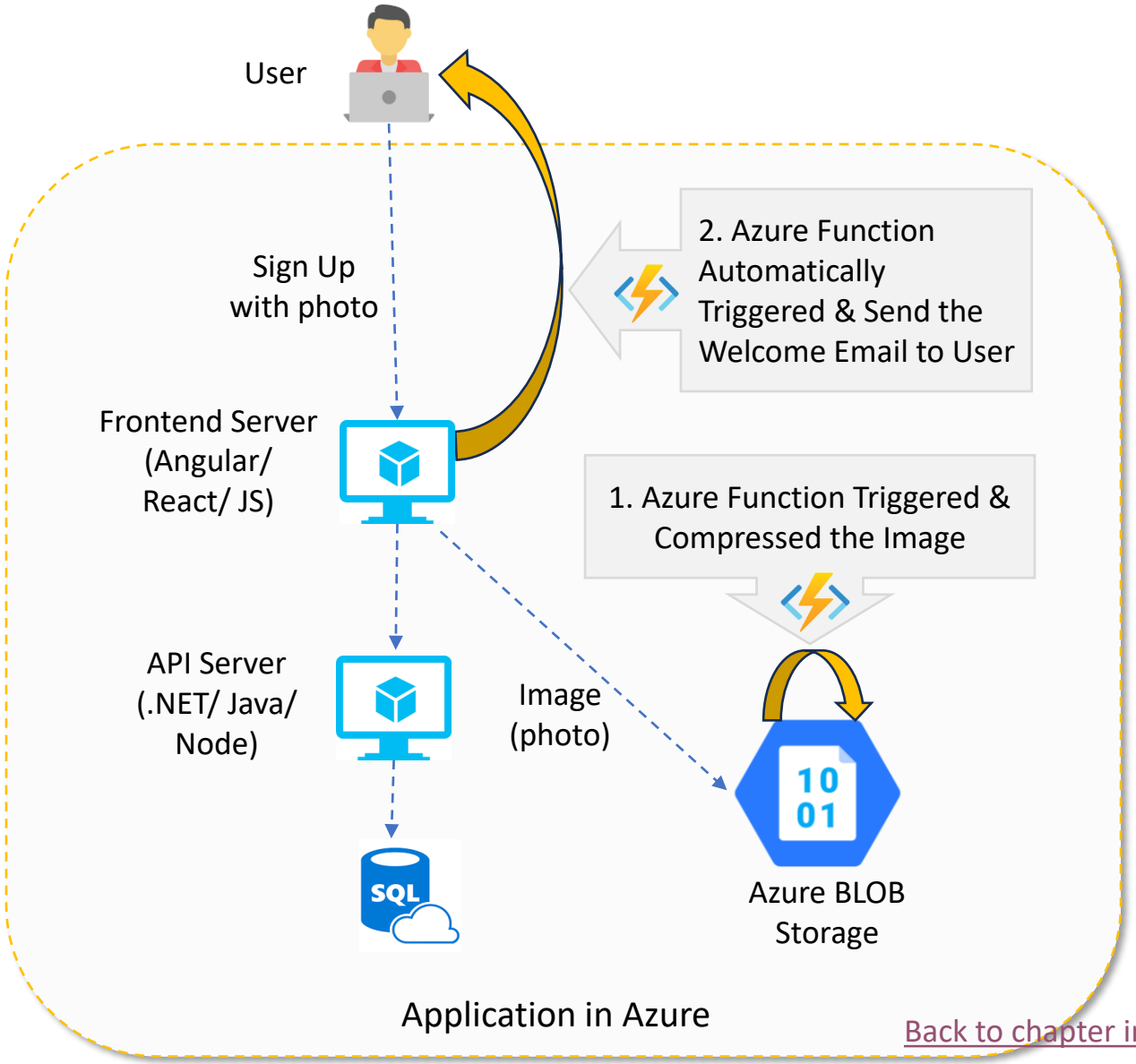
Q. Scenario based Real Application Questions

Back to main index

Q. What are Azure Functions? Have you ever used them in your project? Why? V. IMP.

- Azure Functions is a **serverless compute service** in Azure that lets you run code in response to events without managing infrastructure(servers).

- Advantage: It automatically scales and executes only when triggered, making it cost-efficient and lightweight.



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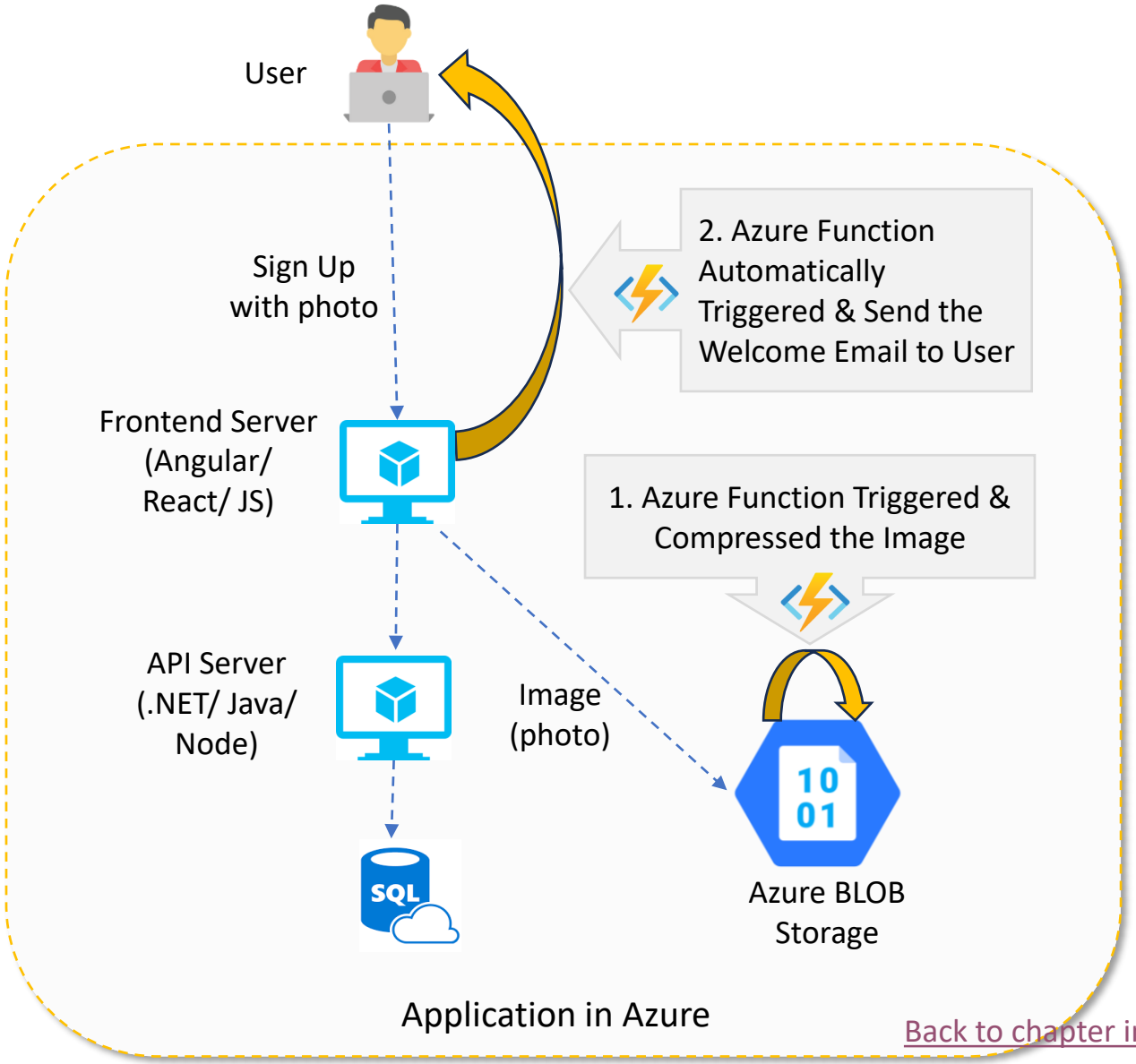
Q. Why not include the Azure Function code directly within the main application?

- Because Azure Function code runs independently, which helps improve the performance of the main application by keeping main application lighter and more focused with less code to manage.



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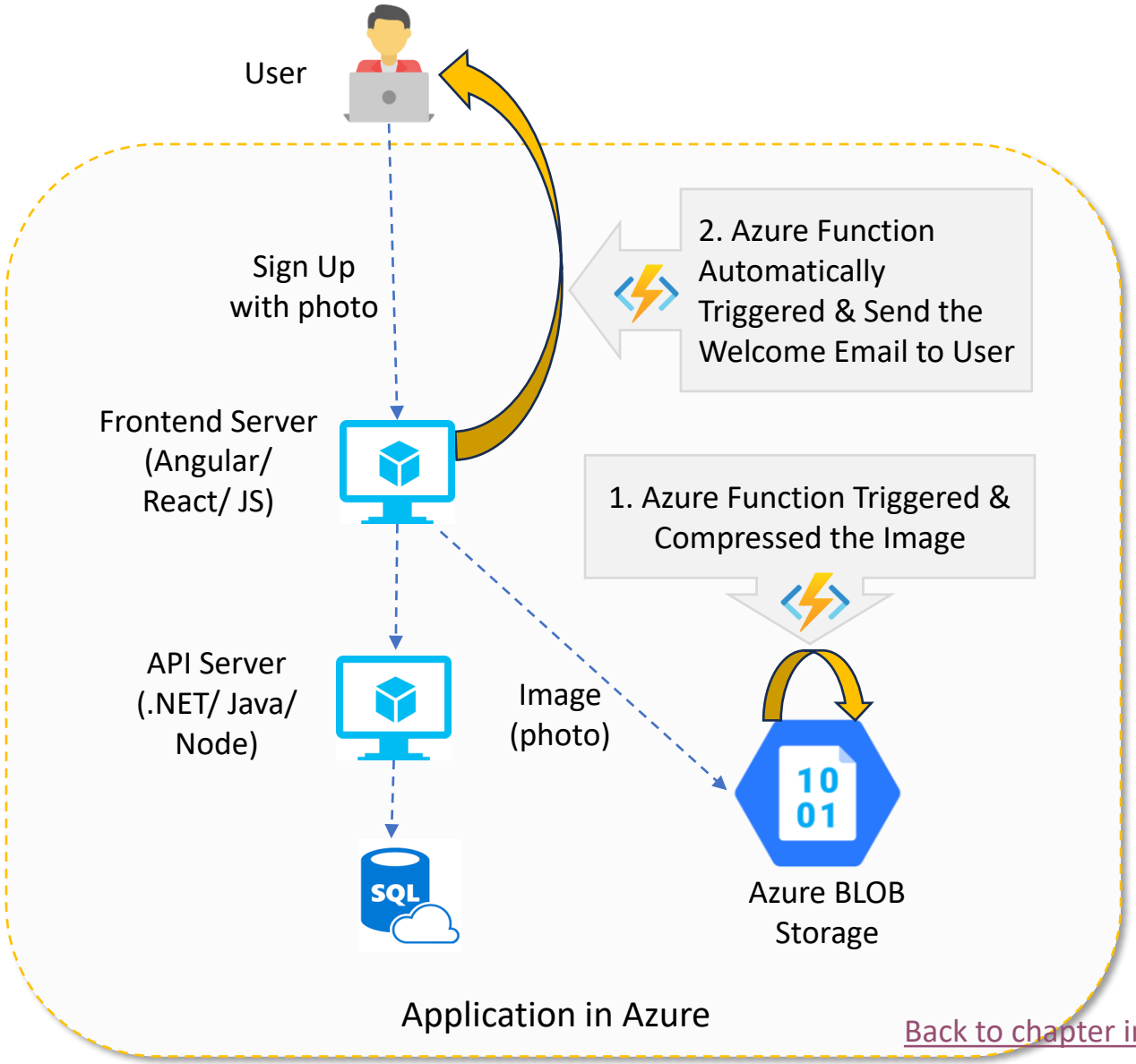
Q. What exactly is Serverless in Azure? Is it truly server-less?

- Serverless in Azure means running code without managing servers.

- No, it's not truly "server-less" — servers still exist, but you don't manage them.



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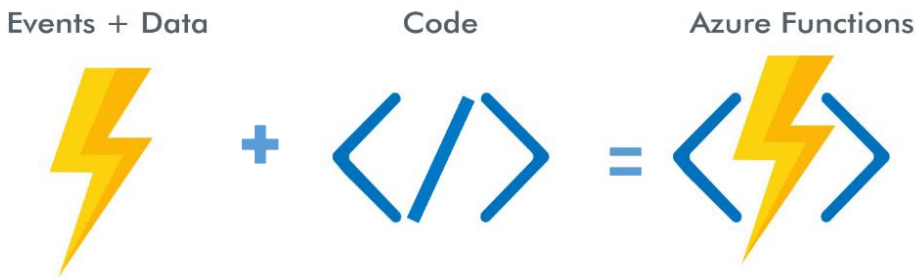
Q. What are Triggers in Azure Functions? Can you name a few types? V. IMP.

- Triggers are what start or invoke an Azure Function.
- When an event occurs, it triggers the Azure functions which then automatically executes, performs the task, and stops when done.



Types of Triggers

- HTTP Trigger (when an HTTP request comes)
- Timer Trigger (run at specific times)
- Blob Trigger (when a file is uploaded)

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Q. What are the important steps to write code of an Azure function? Share example?

❖ Important things in the code of Azure Functions:

1. Each function starts with a method marked by the [**FunctionName**("FunctionNameHere")] attribute.
2. **Run** is the method that executes when the function is triggered.
3. This Azure function is triggered (**HttpTrigger**) when the request is received from the browser.

```
using System.Net.Mail;
using Microsoft.AspNetCore.Mvc;
using Microsoft.Azure.WebJobs;
using Microsoft.Azure.WebJobs.Extensions.Http;
using Microsoft.AspNetCore.Http;

public static class SimpleEmailFunction
{
    [FunctionName("SimpleEmail")]
    public static IActionResult Run(
    [HttpTrigger(AuthorizationLevel.Function, "get", "post")] HttpRequest req)
    {
        string to = req.Query["to"];
        string name = req.Query["name"];

        MailMessage mail = new MailMessage("youremail@example.com", to);
        mail.Subject = "Hello from Azure Function";
        mail.Body = $"Hi {name}, this is a simple email.";

        SmtpClient smtp = new SmtpClient("smtp.example.com", 587);
        smtp.Credentials = new System.Net.NetworkCredential("a@abc.com", "pwd");
        smtp.EnableSsl = true;
        smtp.Send(mail);

         return new OkObjectResult("Email sent.");
    }
}
```

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Q. How do you deploy an Azure Function to the cloud?

- Azure Functions can be deployed using:

1. Azure Portal – Manually upload code.
2. Azure CLI – Run func azure functionapp publish <FunctionAppName>.
3. VS Code / Visual Studio – Directly publish from IDE.
4. CI/CD (GitHub Actions, Azure DevOps) – Automate deployments.

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Q. Which services of Azure are Serverless?

Azure Serverless Services

Compute

- Azure Functions
- Azure Logic Apps
- Azure Container Apps

Messaging & Events

- Azure Event Grid
- Azure Service Bus
- Azure Queue Storage

Data & Analytics

- Azure Cosmos DB (Serverless)
- Azure SQL Database (Serverless Tier)
- Azure Stream Analytics

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Q. What is the difference between Azure Function and Function App?

- Azure Functions: A single unit of execution (a function that runs based on a trigger)

- Function App: A container that groups multiple Azure Functions together

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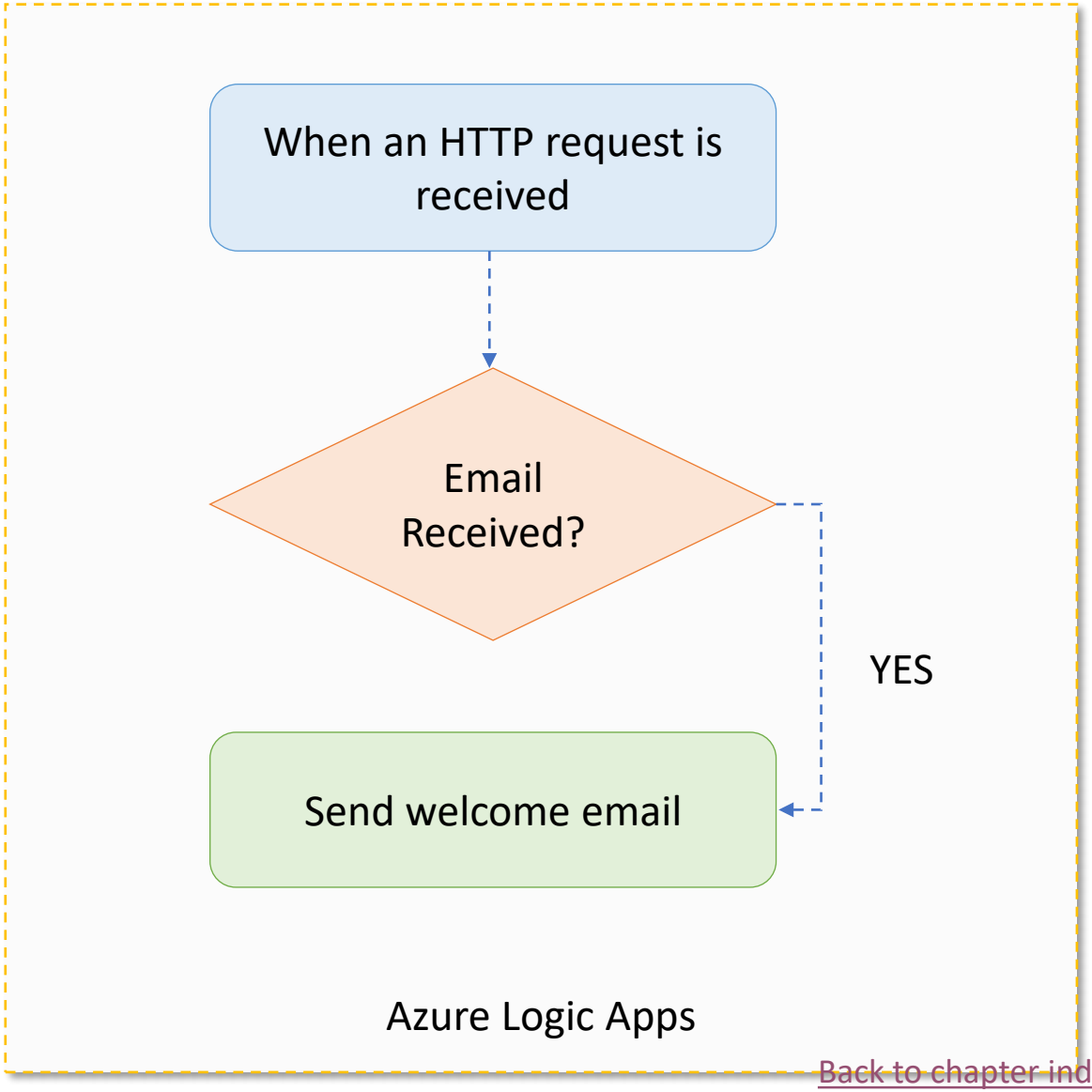
Q. What are Azure Logic Apps?

- Azure Logic Apps is a cloud-based **workflow automation service**.
- It allows you to connect and automate tasks across different systems, services, and applications - without writing much code.



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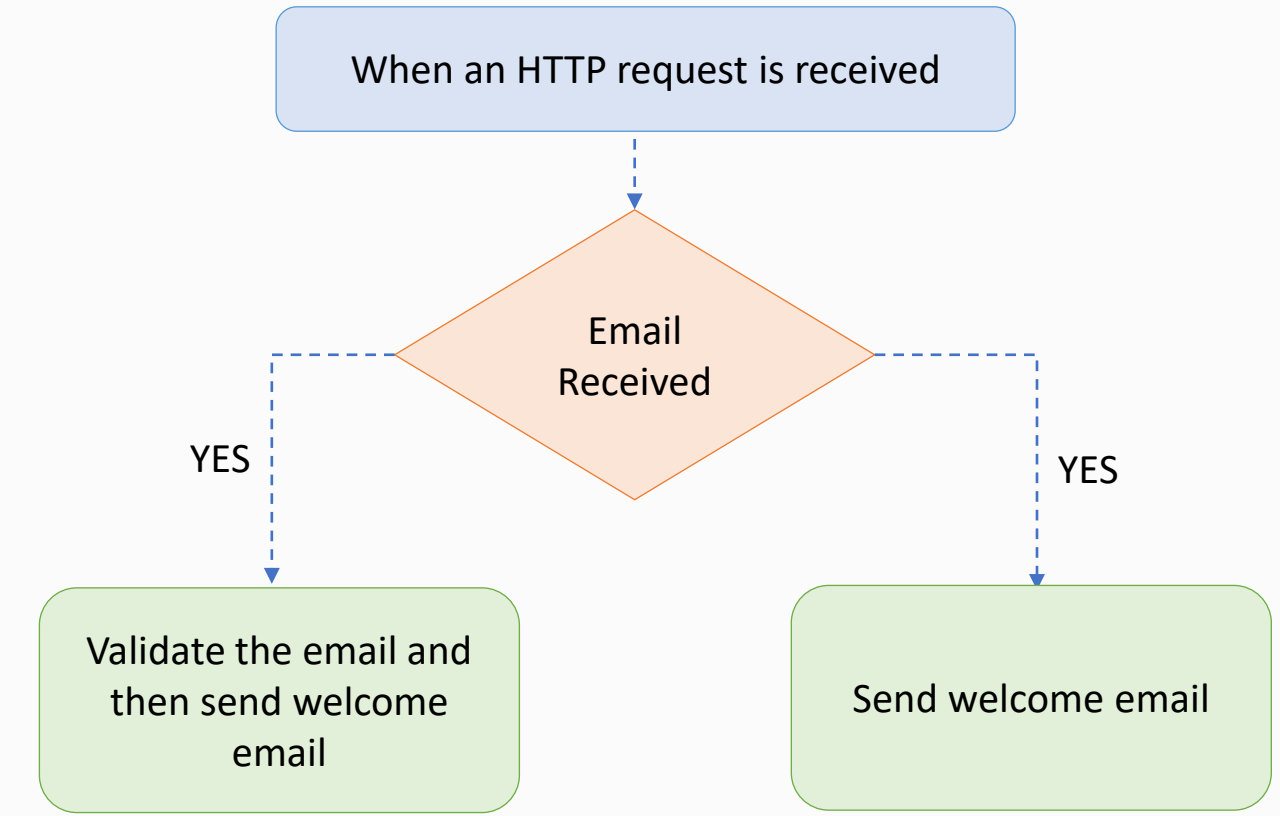
Q. How Azure Logic Apps are different from Azure Functions? When to use what? V. IMP.

- Azure Logic Apps is a low-code/no-code service used to automate workflows and integrate different systems and services.
- Azure Functions is a serverless compute service used to run custom code in response to events, such as HTTP requests, timers, or data changes.



use, Azure functions for custom logic and custom email template

use, Logic Apps for direct and simple email template

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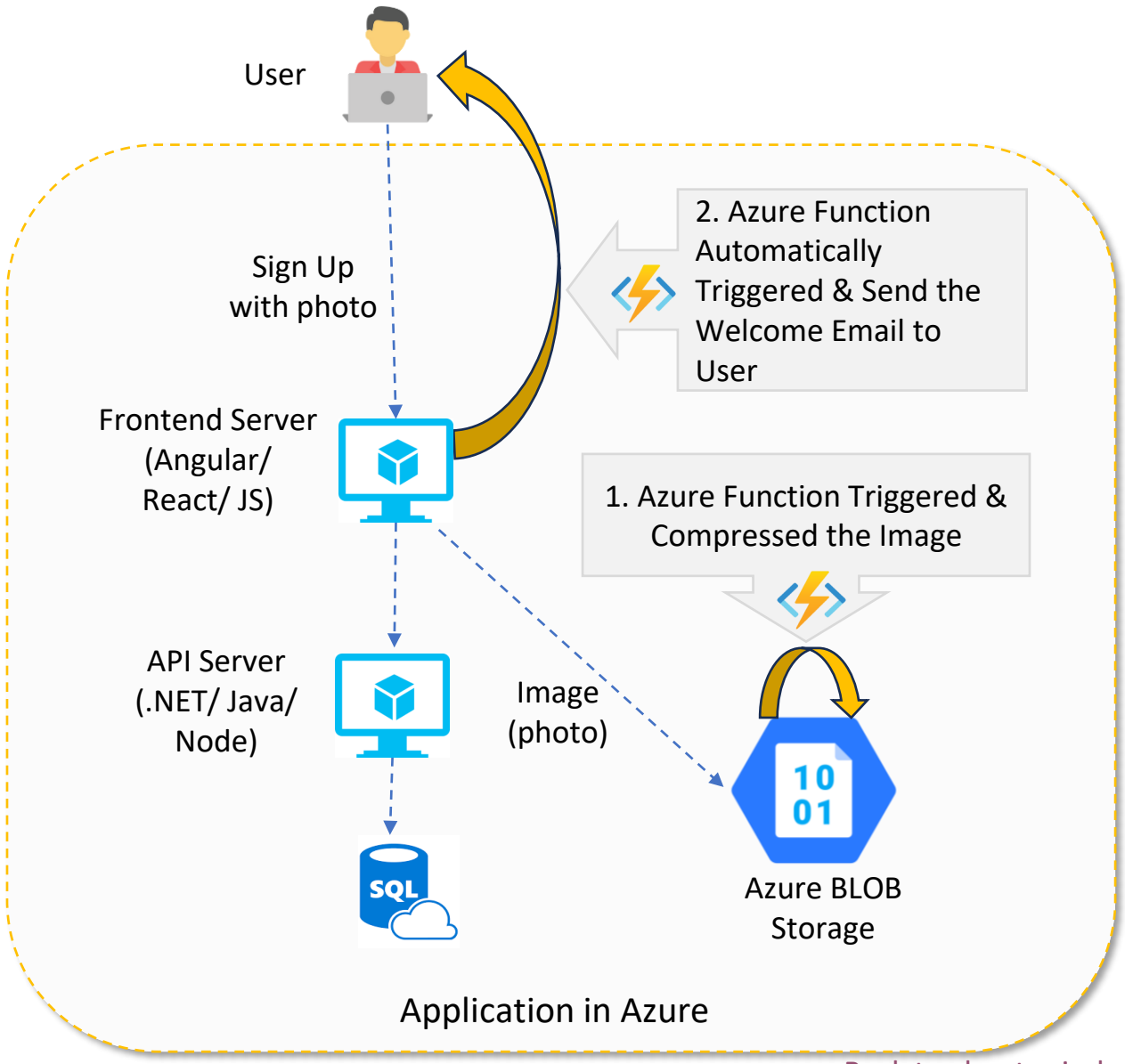
Azure Functions
(Scenario Based Questions)



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Q. You want to send an email when a file is uploaded in Azure Blob Storage. What would you use?

- Use an **Azure Function** with a **Blob Trigger**.
- It automatically executes code when a new file is uploaded to a container.

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Q. A developer adds Azure Function code inside their main app. What's the risk?

- Mixing function code inside the app increases coupling, hurts scalability, and **affects performance**. Functions should be isolated for better maintainability.

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Q. Your function needs to read data from SQL DB. How do you store credentials securely?

- Store credentials in **Azure Key Vault** and reference them using managed identity within the function for secure access.

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Q. What if your Azure Function needs to run every day at 9 AM?

- Use a **Timer Trigger** with to schedule the function daily at 9 AM.

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Q. Can you return a JSON response from an Azure HTTP Trigger Function?

- **Yes**.
- You can return a JSON using return new OkObjectResult(myObject); in C#. It's perfect for APIs.

Back to chapter index

Q. What is the main benefit of serverless Azure Functions in a billing context?

- You pay **only when the function runs**, not for idle time. Ideal for event-based, cost-sensitive workloads.

Back to chapter index

Q. How do Logic Apps differ from Functions in error handling?

- Logic Apps have **built-in retry policies** and visual workflows.
- Functions require **custom try-catch blocks** and manual error handling.

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Q. Can a Logic App call an Azure Function?

- **Yes**.
- Use the HTTP action in Logic App to call an Azure Function endpoint, enabling integration between the two.

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Q. You need a quick automation when a message arrives in a Queue. What should you use?

- Use an Azure Function with a **Queue Trigger** to process messages asynchronously without setting up a full backend.

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5. Azure Storage Services

Q. What is Azure Storage? What are the types of Azure Storage? **V. IMP.**

Q. Explain Azure Blob Storage? When would you use it in your project? **V. IMP.**

Q. Explain Azure File Storage? When would you use it in your project?

Q. Explain Azure Table Storage? When would you use it in your project?

Q. Explain Azure Queue Storage? When would you use it in a project?

Q. What format is used to store messages in Azure Queue Storage?

Q. What is Azure Disk Storage, and how does it support VMs?

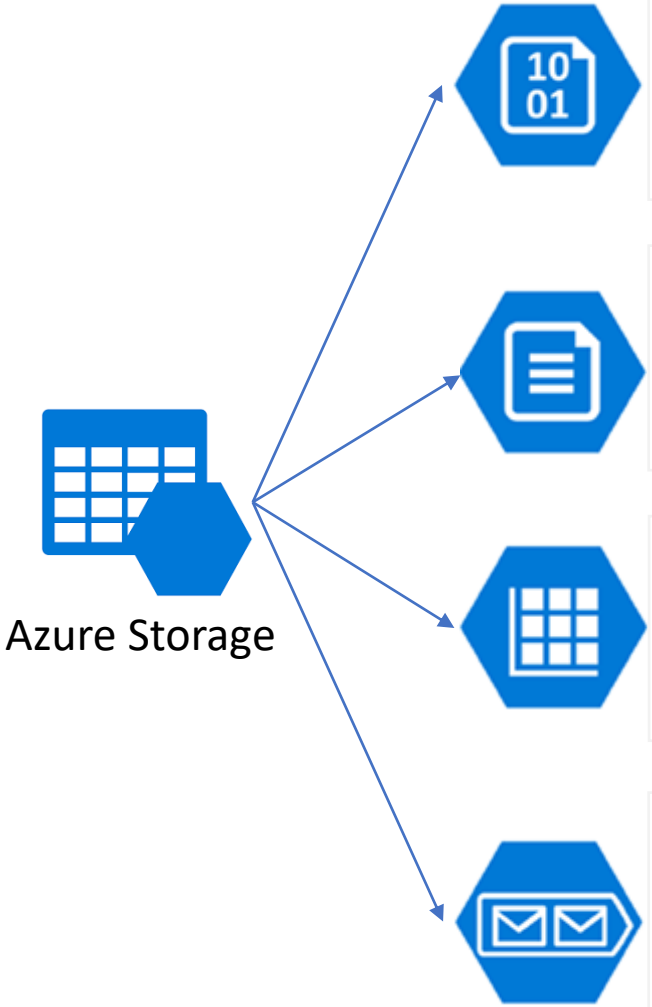
Q. What are the different disk types available in Azure?

Q. Scenario based Real Application Questions

Back to main index

Q. What is Azure Storage? What are the types of Azure Storage? V. IMP.

- Azure Storage is a cloud-based storage solution from Microsoft Azure that lets you store and access data over the internet.



- **Blob Storage:** Stores **unstructured data** like images, videos, and backups.
- **File Storage:** Provides cloud-based shared **file** storage.
- **Table Storage:** Stores large-scale NoSQL **key-value data** for fast lookups.
- **Queue Storage:** Manages **message-based** communication between applications.

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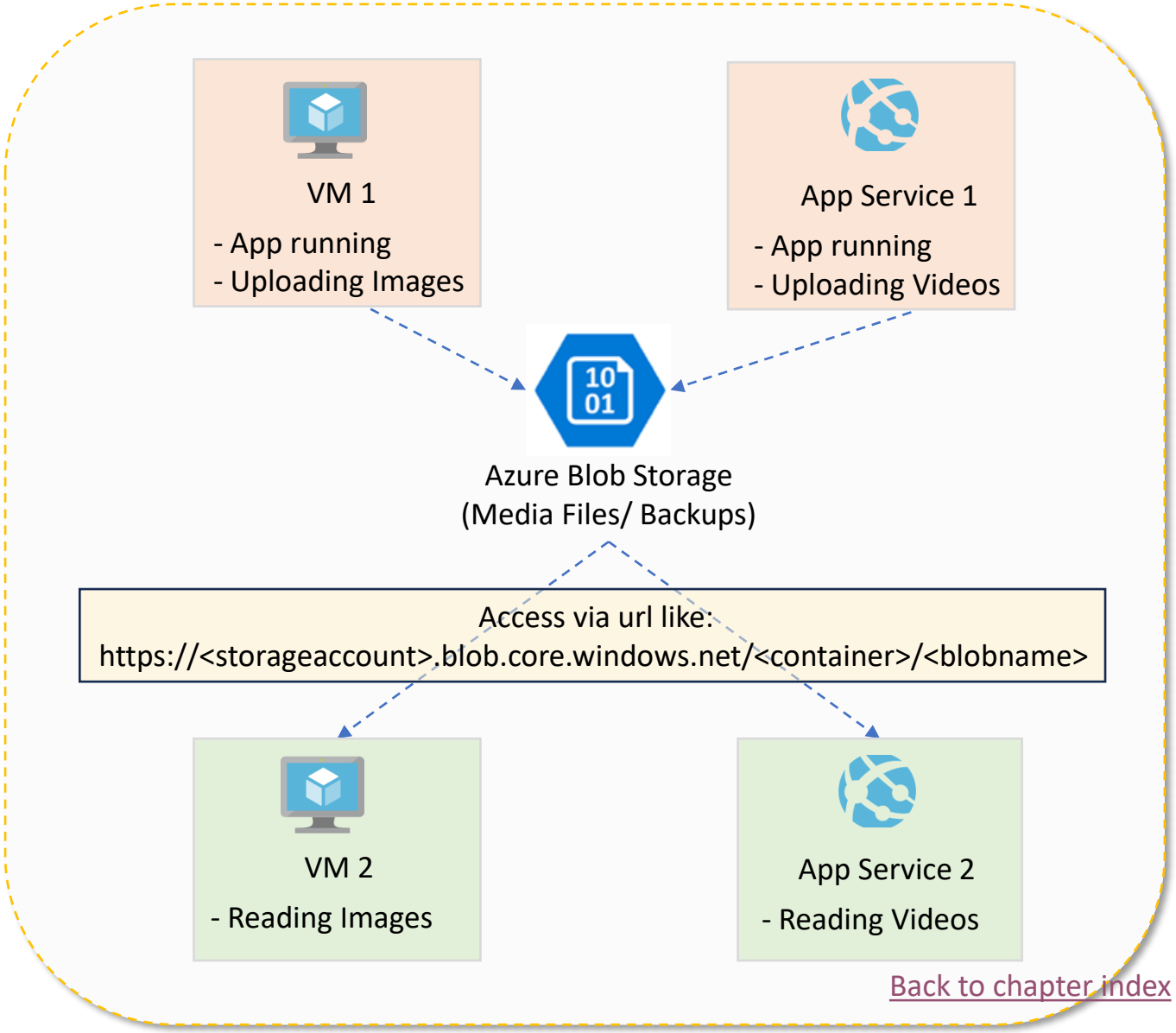
Q. Explain Azure Blob Storage? When would you use it in your project? V. IMP.

- You will use Blob Storage when you need to store large files like images, videos, documents, or backups, and want to access them through URLs or APIs - especially in web, mobile, or cloud applications.
- Editing files in Blob storage manually is difficult, because the files are stored as objects, not as regular files in a file system.



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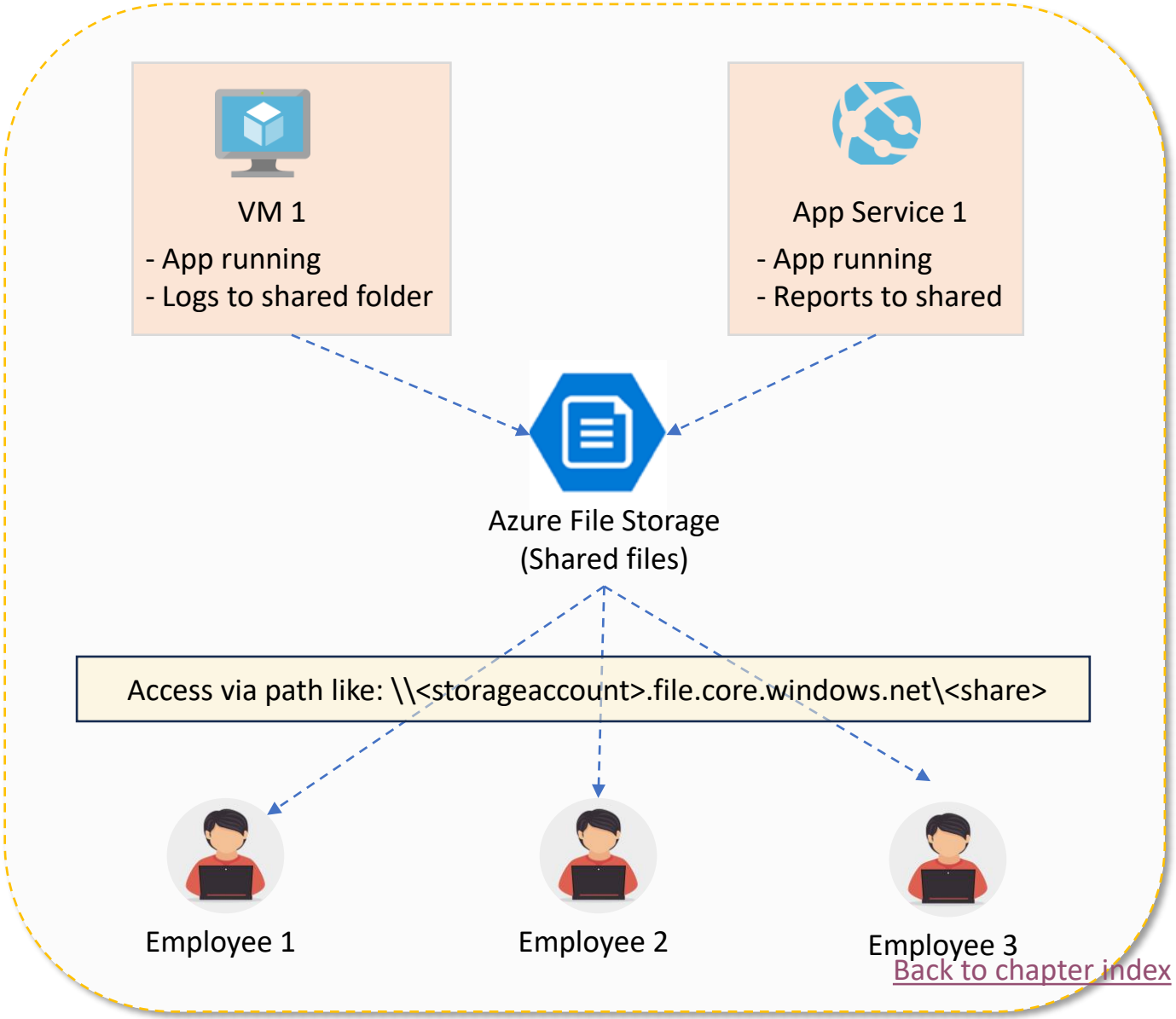
Q. Explain Azure File Storage? When would you use it in your project?

- You will use File Storage when you need to mount the storage as a **shared drive (like a network folder)** that your company employees can easily access using a simple path, just like opening folders in Windows Explorer.
- It is easy to open, edit and save files in file storage manually.



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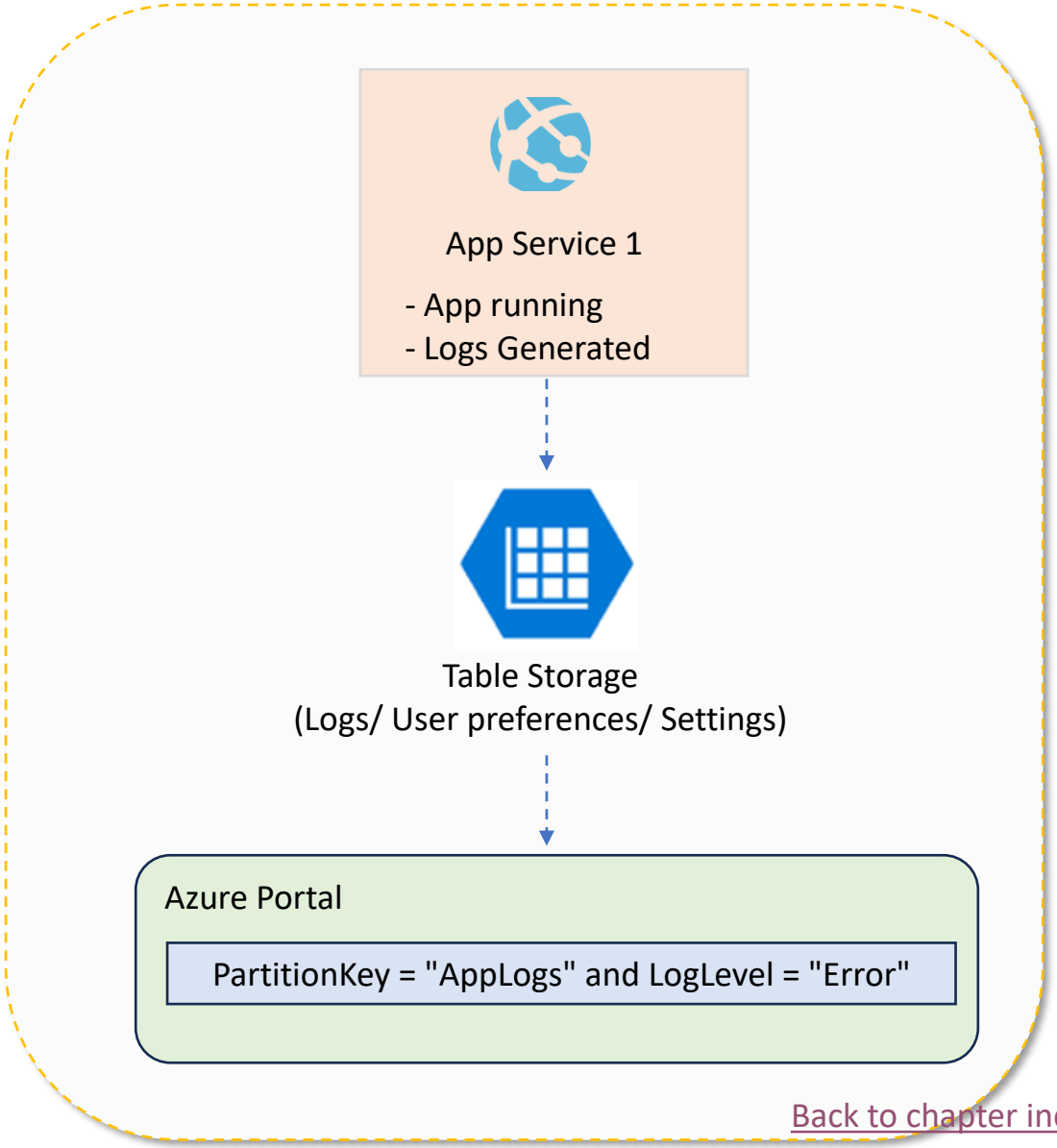
Q. Explain Azure Table Storage? When would you use it in your project?

- Use Azure Table Storage when you need lightweight, fast, and cheap NoSQL storage for massive amounts of simple, structured data, without needing complex querying or relationships.

- Log Example Stored in Azure Table Storage

PartitionKey	RowKey	LogLevel	Message
AppLogs	20240321T120045Z	Info	User login successful
AppLogs	20240321T120115Z	Error	DB connection failed



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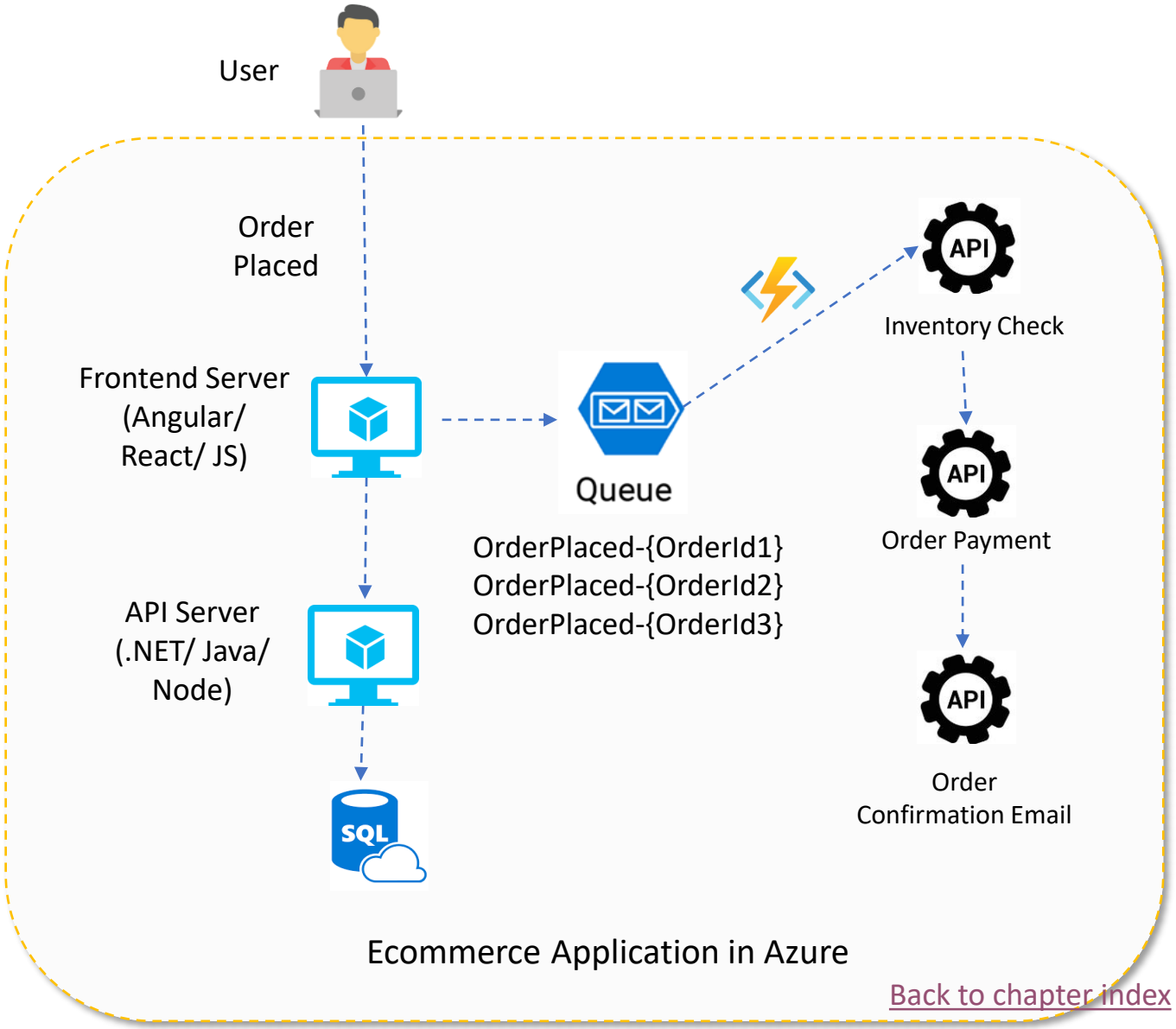
Q. Explain Azure Queue Storage? When would you use it in a project?

- Azure Queue Storage is used when you need to pass messages between different parts of your system, especially when they run asynchronously and independently.

- In E-commerce applications, Azure Queue Storage can be used to decouple the order processing workflow, ensuring a responsive UI and scalable backend processing.



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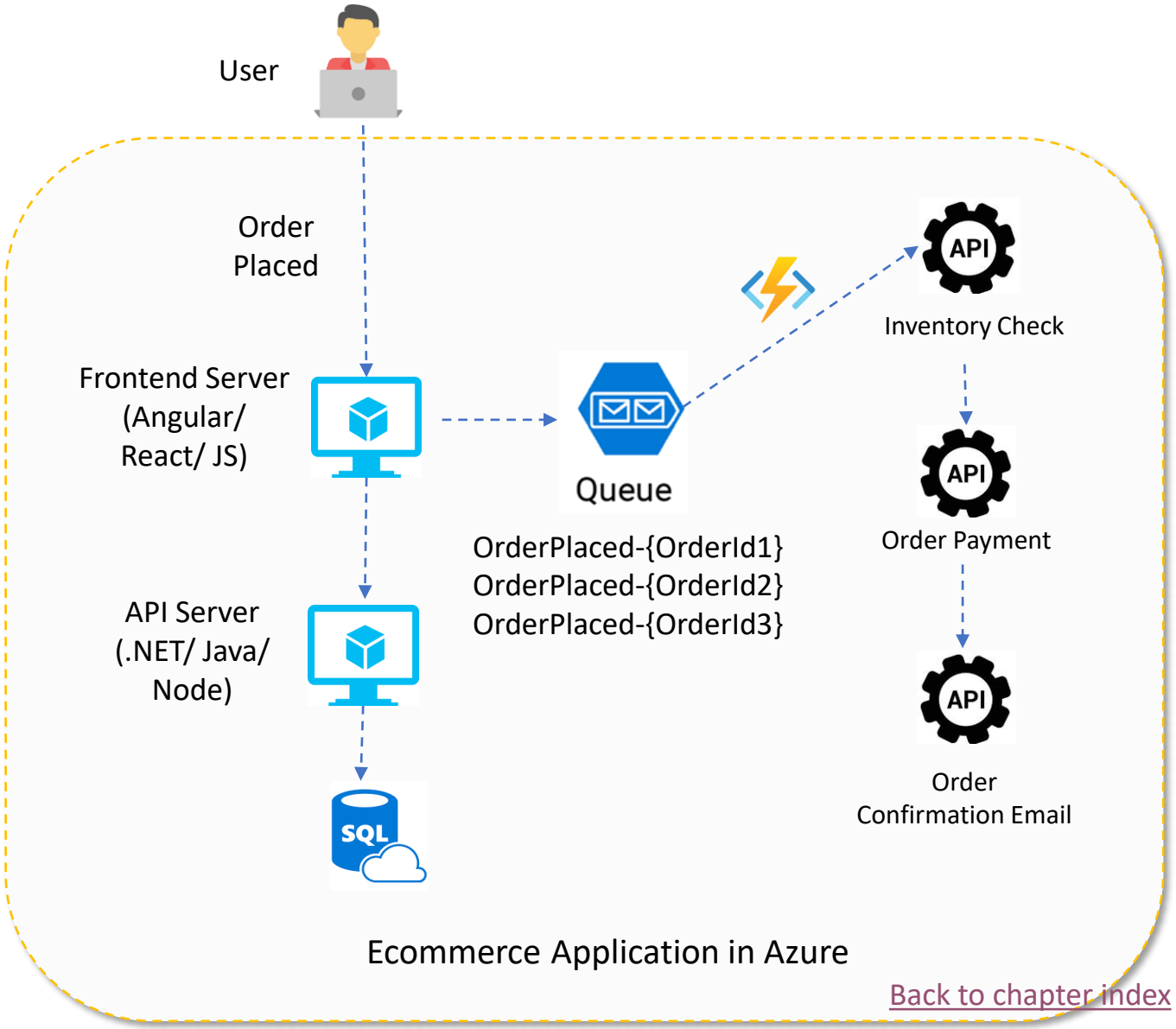
Q. What format is used to store messages in Azure Queue Storage?

- Commonly used format: JSON (as a string)

```
{
  "orderId": "12345",
  "customerId": "98765",
  "items": [
    { "productId": "A1", "quantity": 2 },
    { "productId": "B3", "quantity": 1 }
  ],
  "totalAmount": 1500,
  "paymentStatus": "Pending"
}
```



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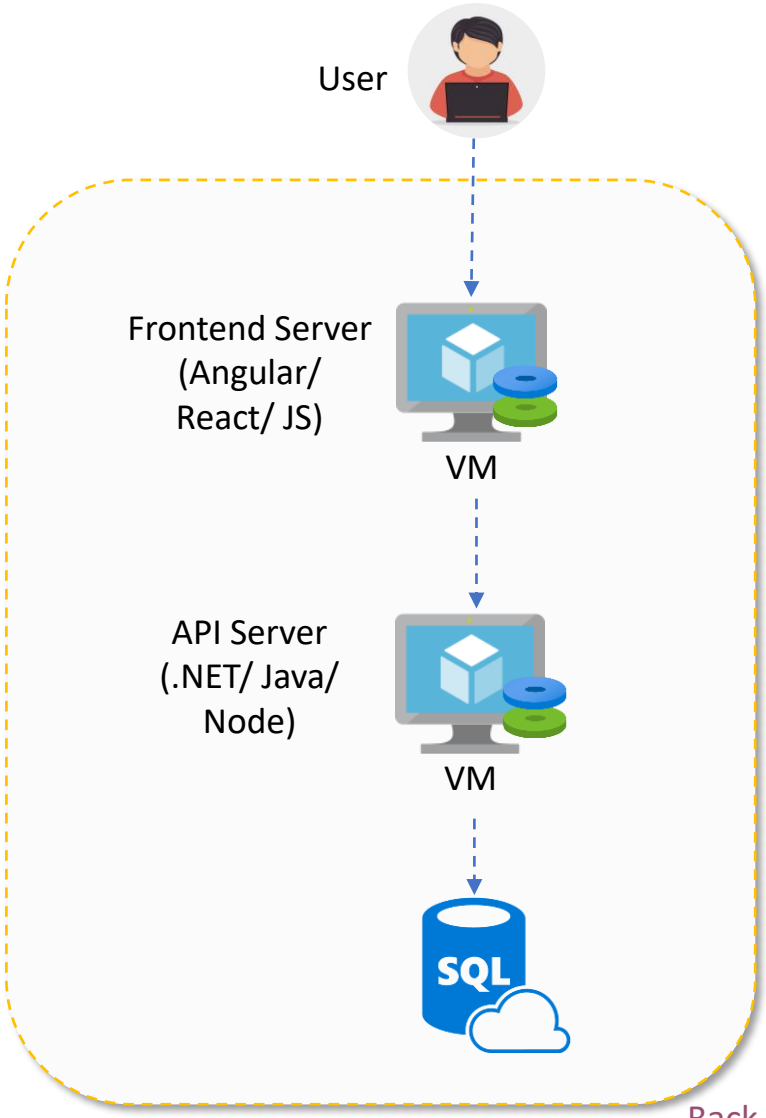
Q. What is Azure Disk Storage, and how does it support VMs?

- Azure Disk Storage is used to provide persistent storage to VMs, ensuring data is retained even when the VM is stopped or restarted.



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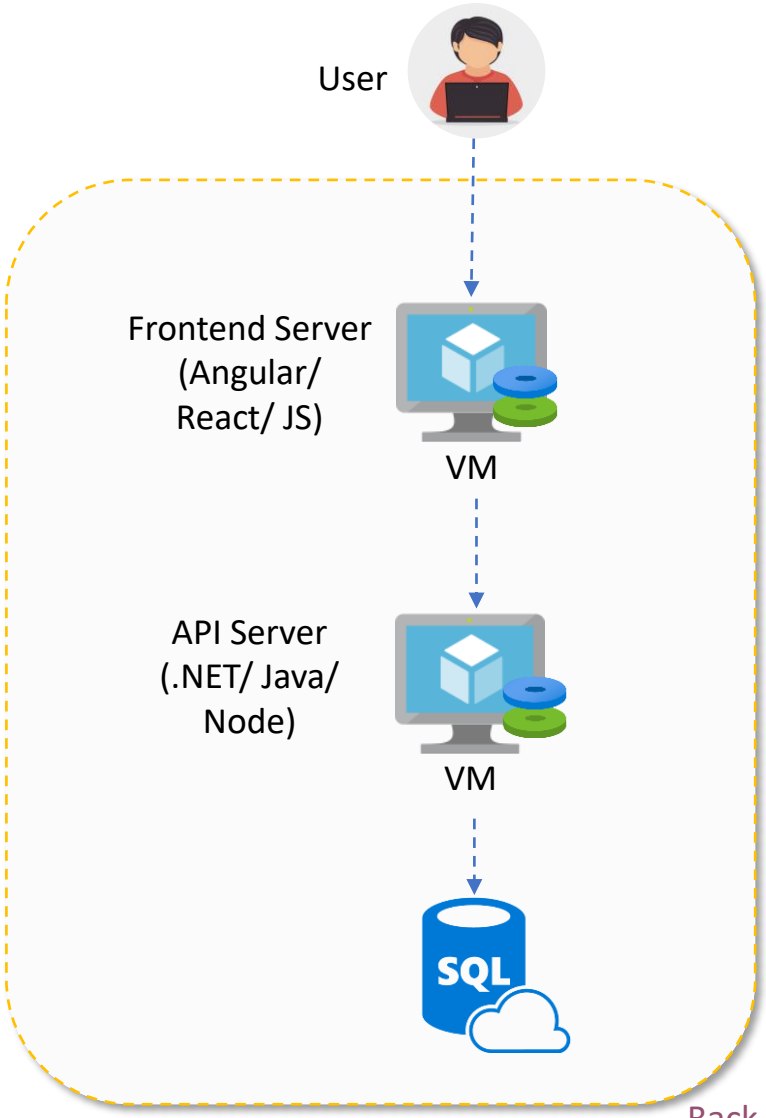
Q. What are the different disk types available in Azure?

❖ Types of Disk:

1. **OS Disk** (Attached by default when VM is created): Contains the VM's operating system.
2. **Data Disk**: Used for storing additional application data, databases, or logs.
3. **Temporary Disk**: Ephemeral storage for temporary data; data is lost when the VM is stopped/deallocated.



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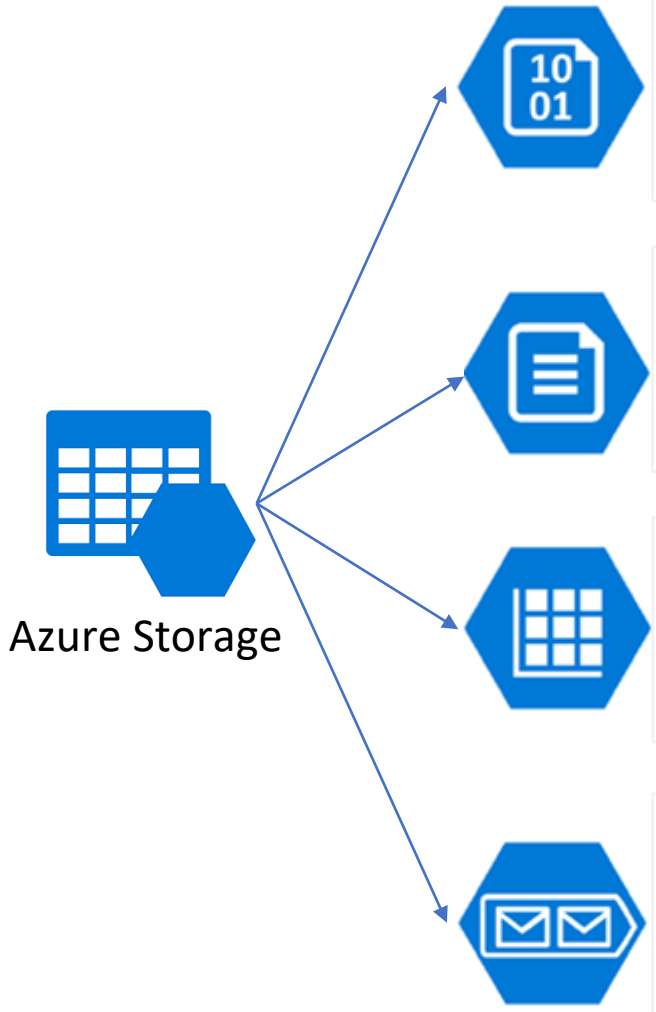
Azure Storage Service
(Scenario Based Questions)



- **Blob Storage:** Stores **unstructured data** like images, videos, and backups.
- **File Storage:** Provides cloud-based shared **file** storage.
- **Table Storage:** Stores large-scale NoSQL **key-value data** for fast lookups.
- **Queue Storage:** Manages **message-based** communication between applications.

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Q. Your app needs to store and stream user-uploaded videos. Which Azure Storage type fits best?

- **Use Azure Blob Storage**.
- It's optimized for unstructured data like images, videos, and documents, and supports large files and streaming.

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Q. Your team needs to share files between multiple VMs like a shared drive. What would you use?

- **Use Azure File Storage**.
- It supports SMB protocol, allowing multiple VMs to mount and access files just like a network drive.

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Q. Your app needs to store millions of log records with quick key-based access. What storage should you pick?

- **Use Azure Table Storage**.
- It's a NoSQL store, optimized for key-value data and large-scale structured datasets.

Back to chapter index

Q. What is the ideal Azure service to manage background task queues for microservices?

- **Use Azure Queue Storage**.
- It stores and manages messages between components for asynchronous processing.

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Q. You want to encrypt your storage data automatically. What should you enable?

- **Use Storage Service Encryption**.
- Azure encrypts data at rest automatically using Microsoft-managed keys or your own key in Key Vault.

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Q. How does Azure Disk Storage support Virtual Machines?

- Azure Disks are attached to VMs as OS or Data disks. They persist data across reboots and scale based on performance needs.

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Q. Can you access Azure Blob Storage over HTTP/HTTPS directly from the browser?

- **Yes**.
- Use SAS (Shared Access Signature) or configure the blob container to **public access** for direct browser access.

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6. Azure Database Services

Q. What is Azure SQL Database? How is it different from regular SQL Server?

Q. What is the difference between Azure SQL Managed Instance and Azure SQL Database? **V. IMP.**

Q. What is Azure Cosmos DB? When should you use it in your project? **V. IMP.**

Q. What is NoSQL?

Q. What is the difference between NoSQL & RDBMS?

Q. When to use Azure SQL (RDBMS) and when Azure Cosmos (NoSQL) in your project?

Q. Scenario based Real Application Questions

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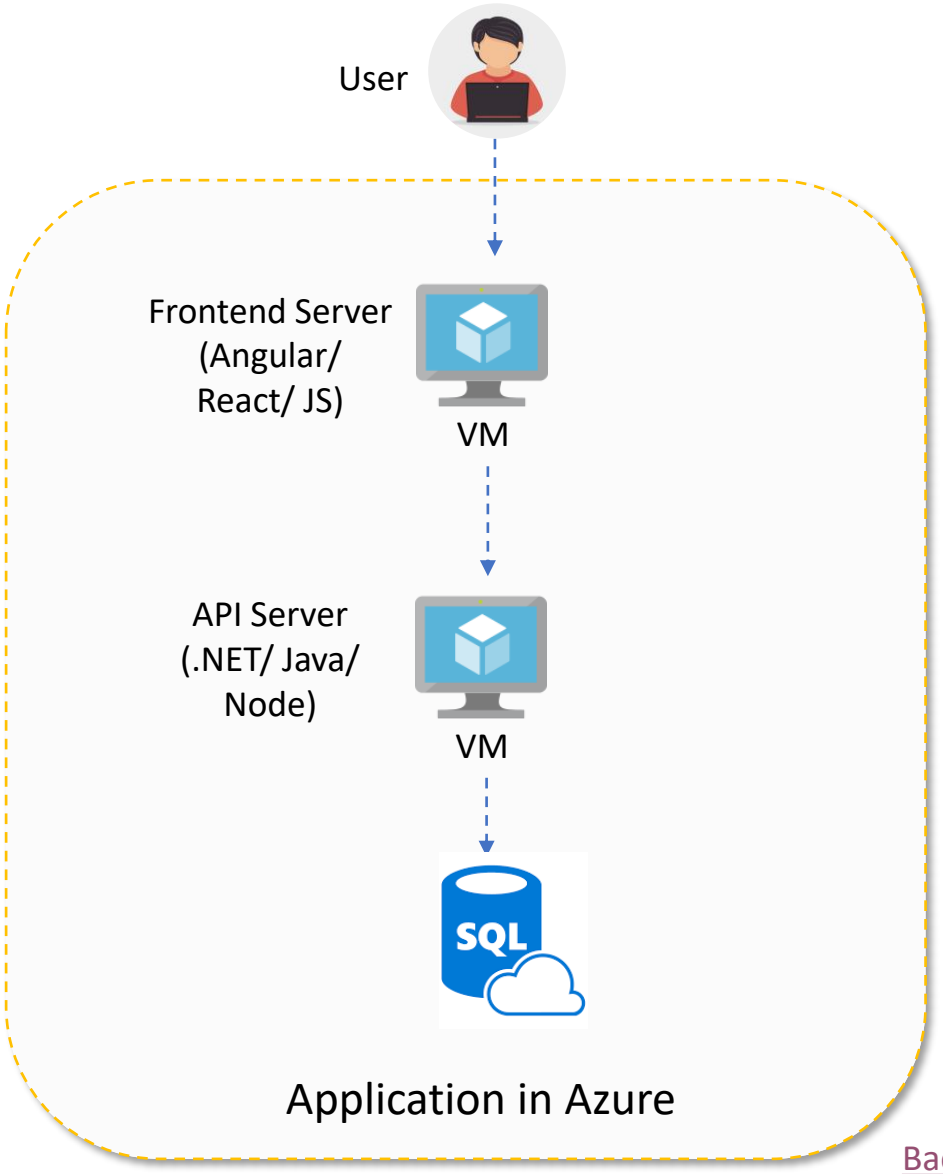
Q. What is Azure SQL Database? How is it different from regular SQL Server? V. IMP.

- Azure SQL Database is a fully managed, cloud-based relational database service provided by Microsoft on Azure.
- It is built on the SQL Server engine, but it is offered as a Platform as a Service (PaaS).



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Q. What is the difference between Azure SQL Managed Instance and Azure SQL Database?

Azure SQL Managed Instance	Azure SQL Database
1. Azure SQL Managed Instance is a fully managed, nearly 100% **SQL Server-compatible**.	Azure SQL Database is a **single database service**, ideal for cloud-native applications, but lacks instance-level control.
2. Use Azure SQL Managed Instance if **migrating an existing on-prem SQL Server** with complex dependencies (SQL Agent, cross-DB queries, etc.).	Use Azure SQL Database for **modern cloud applications** needing automatic scaling, high availability, and lower maintenance.

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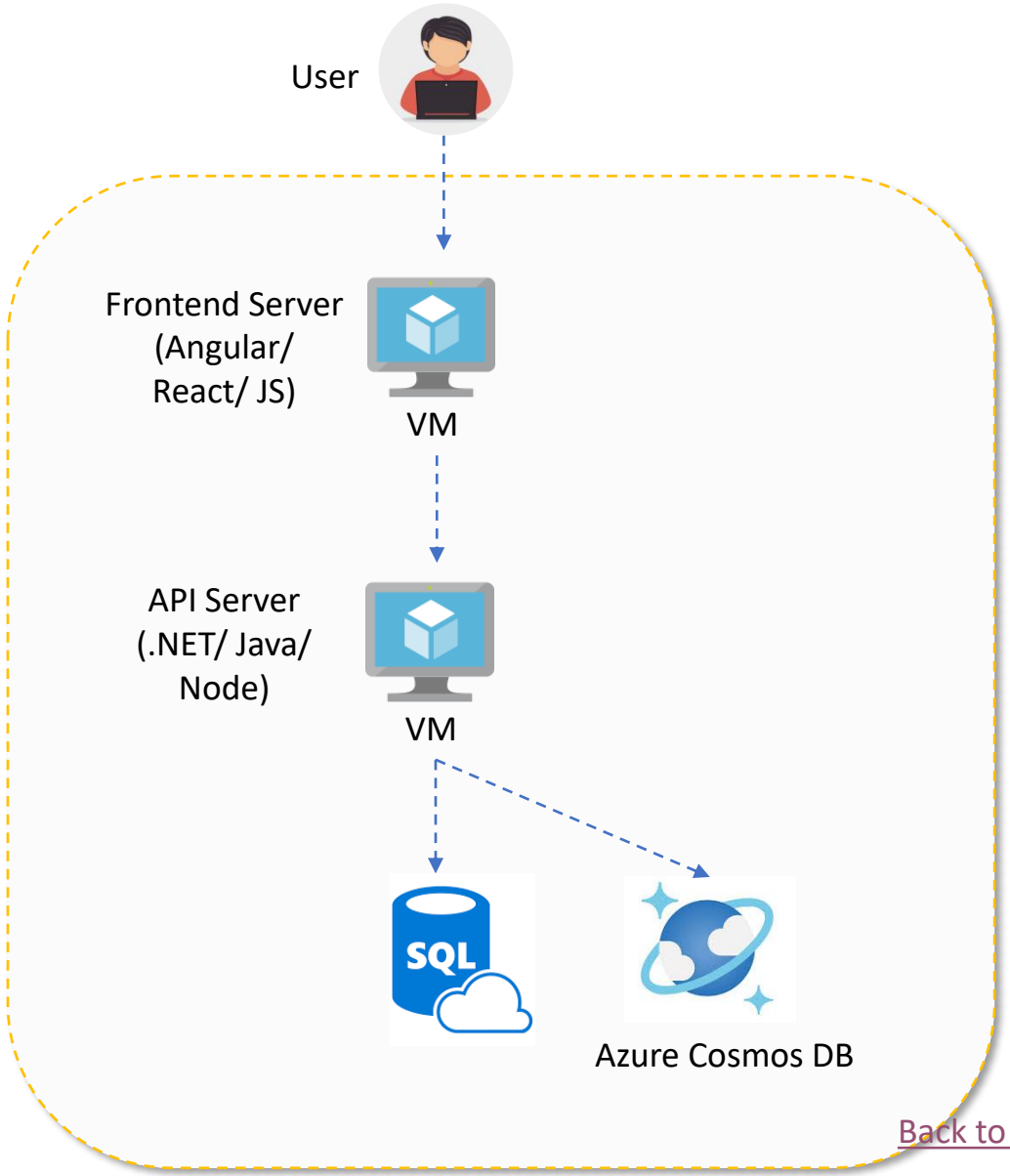
Q. What is Azure Cosmos DB? When should you use it in your project? V. IMP.

- Azure Cosmos DB is a fully managed, globally distributed **NoSQL database** service from Microsoft Azure which is designed for high availability and scalability.



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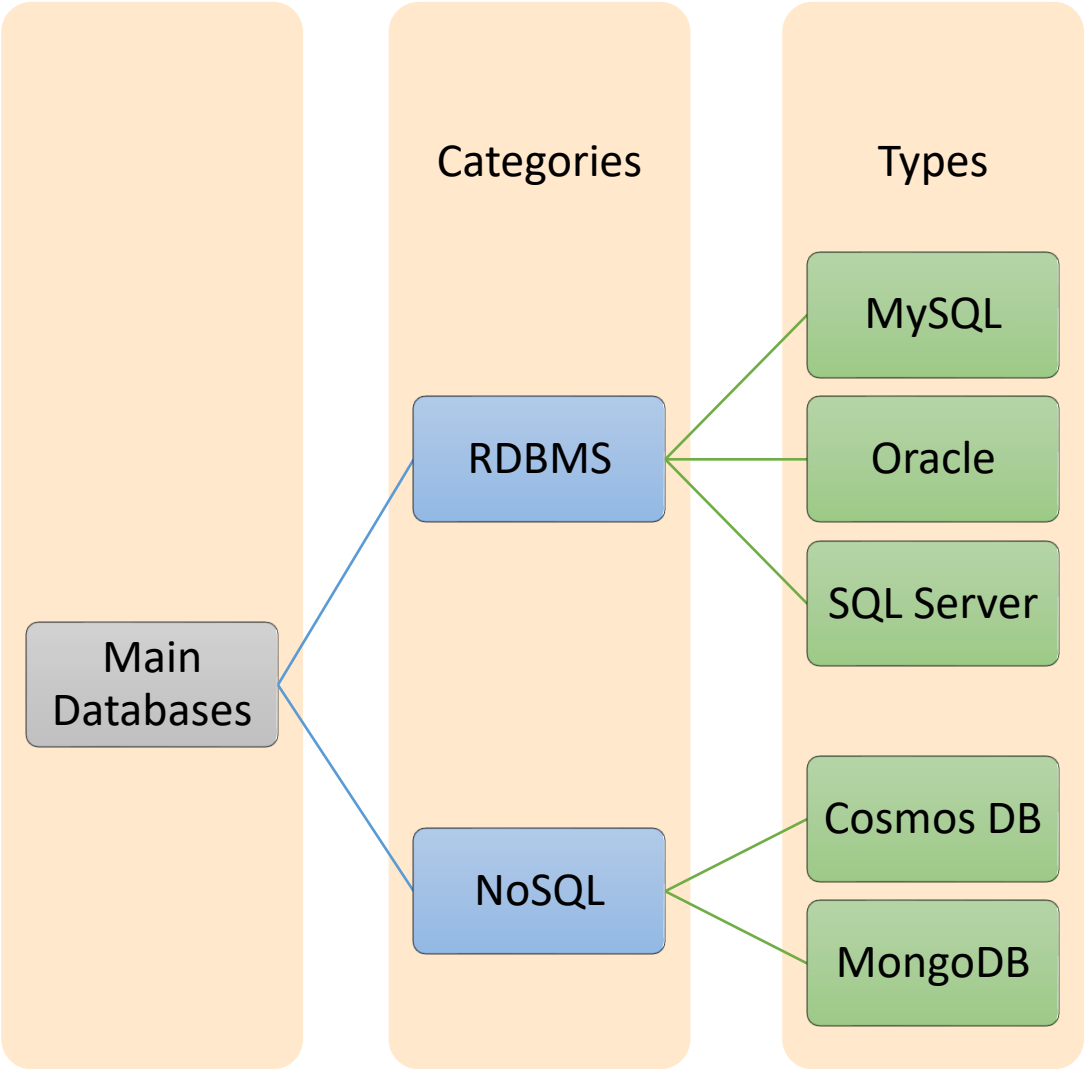
Q. What is NoSQL?

- NoSQL(Not Only SQL) databases are designed to handle structured, semi-structured, and unstructured data.



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Q. What is the difference between NoSQL & RDBMS?

❖ RDBMS(SQL)

	ID	Name	Passion
Row 1	1	Happy	Football
Row 2	2	Amit	null

Users (Table)

❖ NoSQL

```
{
  "ID": 1,
  "Name": "Happy" ,
  "Passion": "Football"
}
```

```
{
  "ID": 2,
  "Name": "Amit"
  "Age": 40
}
```

Document 1 Document 2

Users (Collection)

❖ RDBMS are designed to handle **structured data** (predefined and fixed schemas).

❖ NoSQL databases are designed to handle **structured, semi-structured, and unstructured data**.

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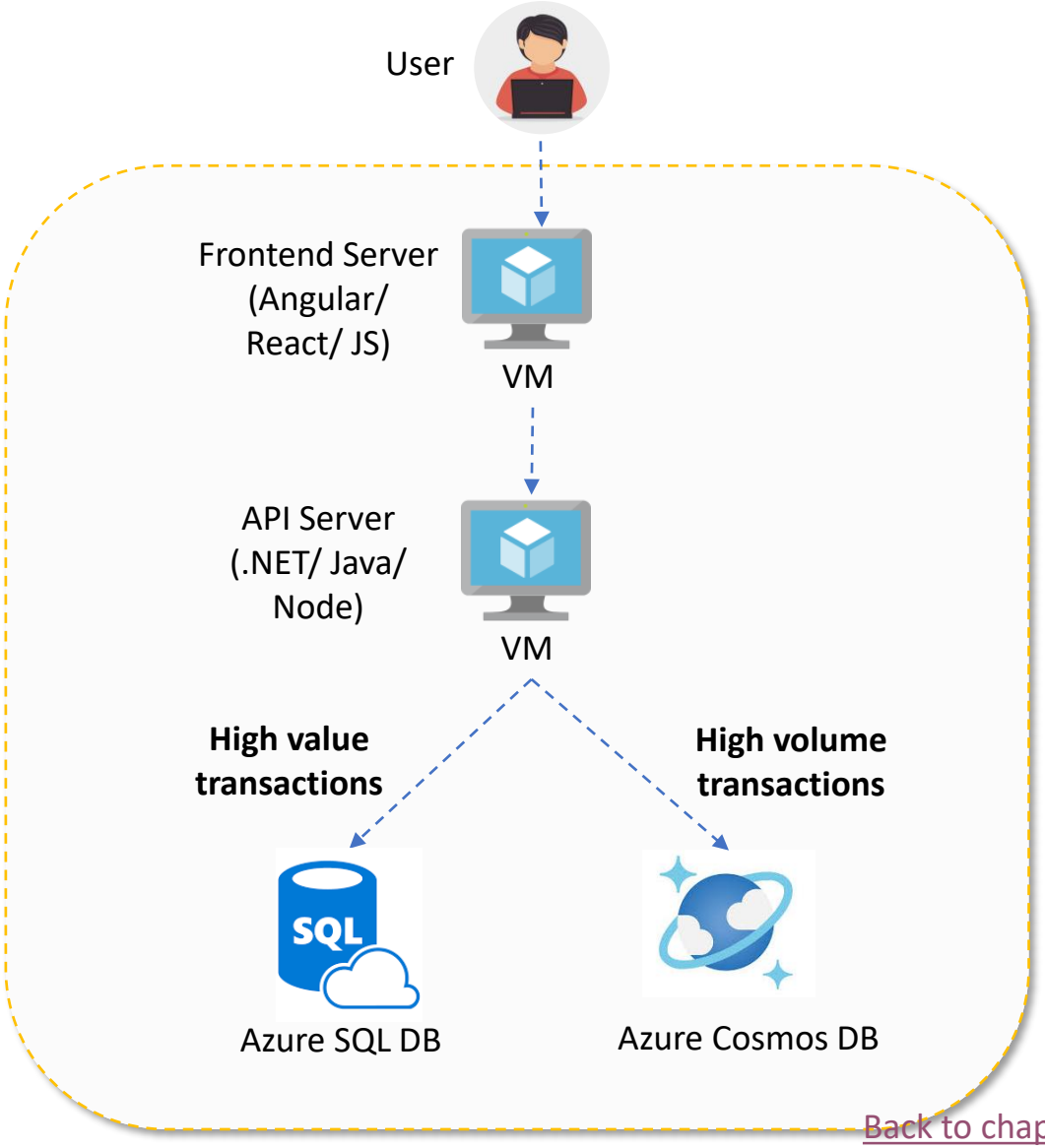
Q. When to use Azure SQL (RDBMS) and when Azure Cosmos (NoSQL) in your project?

- Use RDBMS () are suitable for applications that require **complex transactions and data integrity(ACID)**, such as banking, finance, and e-commerce.

- NoSQL databases are designed to handle **large volumes of data** with high-speed read and write operations, such as social media, IoT, and gaming.



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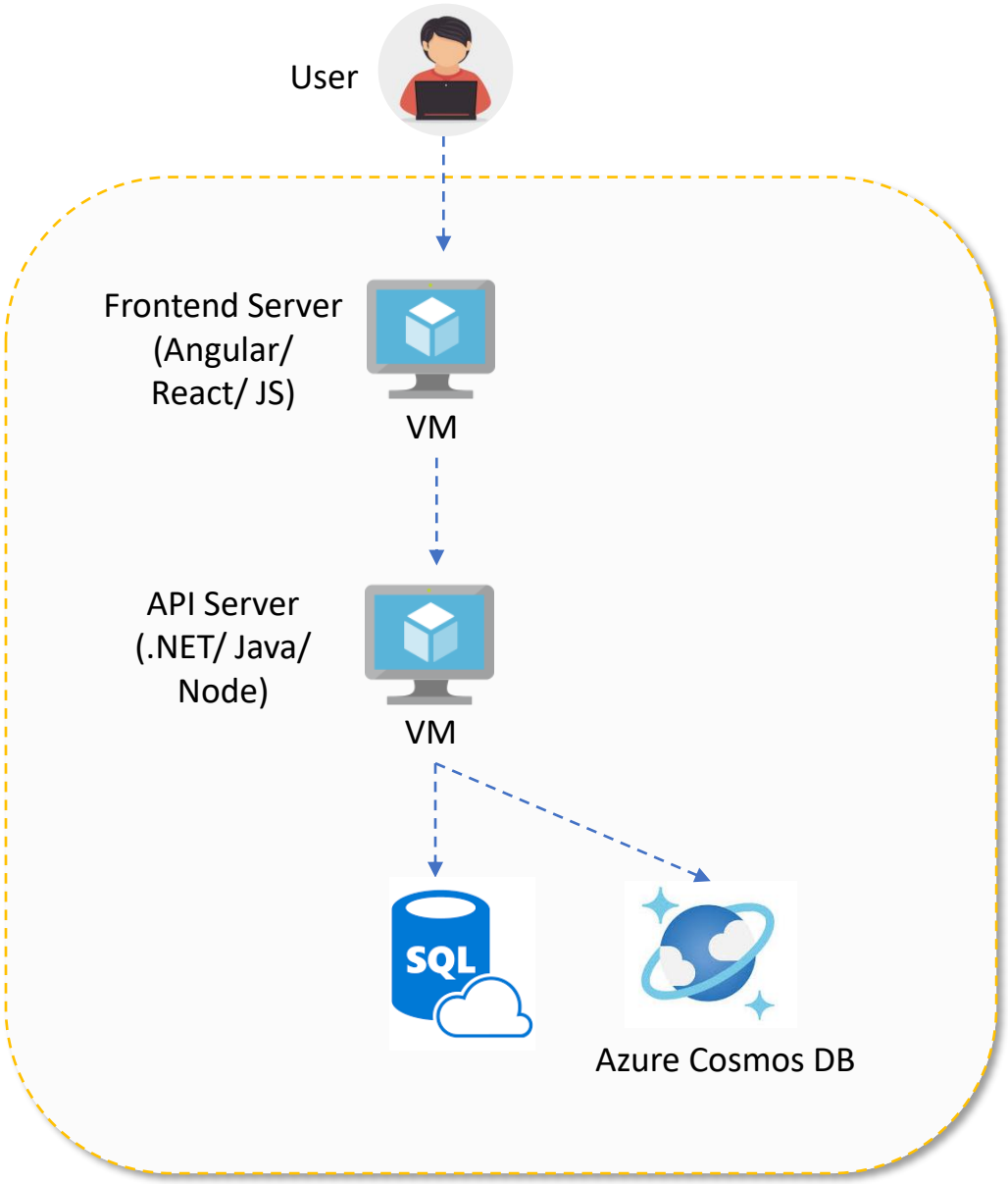
Azure Database Service
(Scenario Based Questions)



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Q. Your app requires auto-scaling, built-in backups, and high availability for a relational DB. Which Azure DB service would you use?

- **Use Azure SQL Database**.
- It's a PaaS offering with built-in backups, patching, high availability, and elastic scalability.

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Q. What is the key advantage of using Azure SQL Managed Instance over Azure SQL Database?

- Managed Instance offers broader SQL Server compatibility (agent jobs, cross-database queries), ideal for migrating on-prem SQL workloads.

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Q. Your global app needs a NoSQL database with great response times. What would you choose?

- **Use Azure Cosmos DB**.
- It's globally distributed, supports multiple APIs, and delivers low-latency reads/writes.

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Q. What is a real-world use case for NoSQL over RDBMS?

- Use NoSQL like Cosmos DB for chats, social media feeds, shopping carts, or IoT sensor data, where scalability and schema flexibility matter.

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Q. Can you use Cosmos DB for SQL-like queries?

- **Yes**.
- Cosmos DB's SQL API lets you run familiar SQL-style queries on JSON documents.

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Q. When should you prefer Azure SQL over Cosmos DB?

- Use Azure SQL when your app needs structured data, strong consistency, complex joins, and **relational features** like stored procedures.

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Q. How can you migrate on-prem SQL Server to Azure with minimal changes?

- **Use Azure SQL Managed Instance** with Data Migration Service (DMS). It supports near 100% compatibility for smooth lift-and-shift.

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7. Azure Networking - Part 1

Q. What is a Virtual Network (VNet) in Azure? Why do we need it? **V. IMP.**

Q. How are applications hosted on VMs inside a VNet accessible to outside users?

Q. What is the difference between Private IP and Public IP in Azure?

Q. What is the role of Azure Load Balancer? **V. IMP.**

Q. What is the role of Azure Application Gateway? **V. IMP.**

Q. When to use Azure Load Balancer and when to use Application Gateway?

Q. How to use Load Balancer and Application Gateway together in Azure?

- Scenario based Real Application Questions

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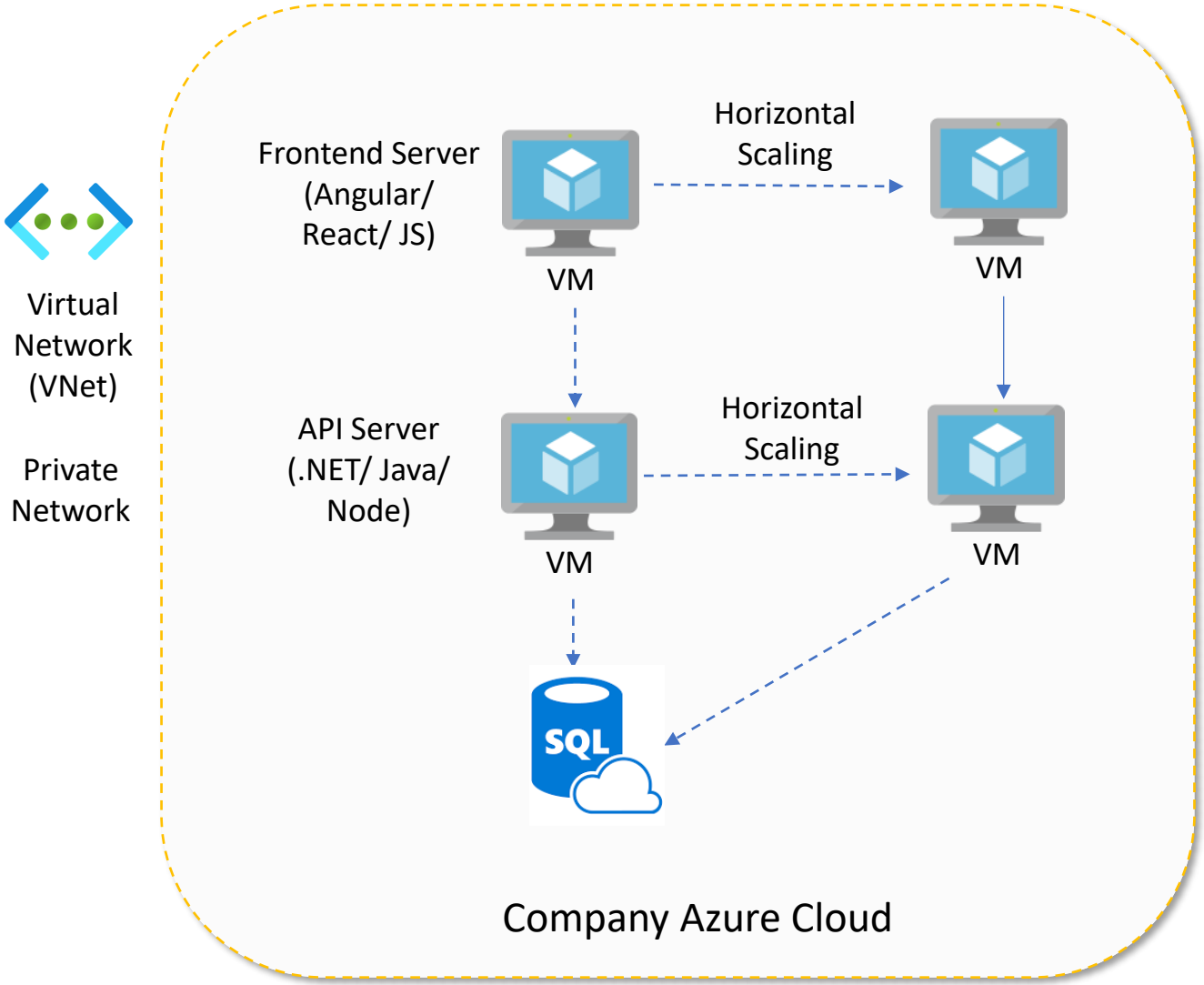
Q. What is a Virtual Network (VNet) in Azure? Why do we need it? V. IMP.

- Azure Virtual Network (VNet) is Azure's networking service that enables secure communication between Azure resources, on-premises networks, and the internet.
- It acts like a private network in the cloud.
- Azure automatically creates a default VNet if you don't specify one.



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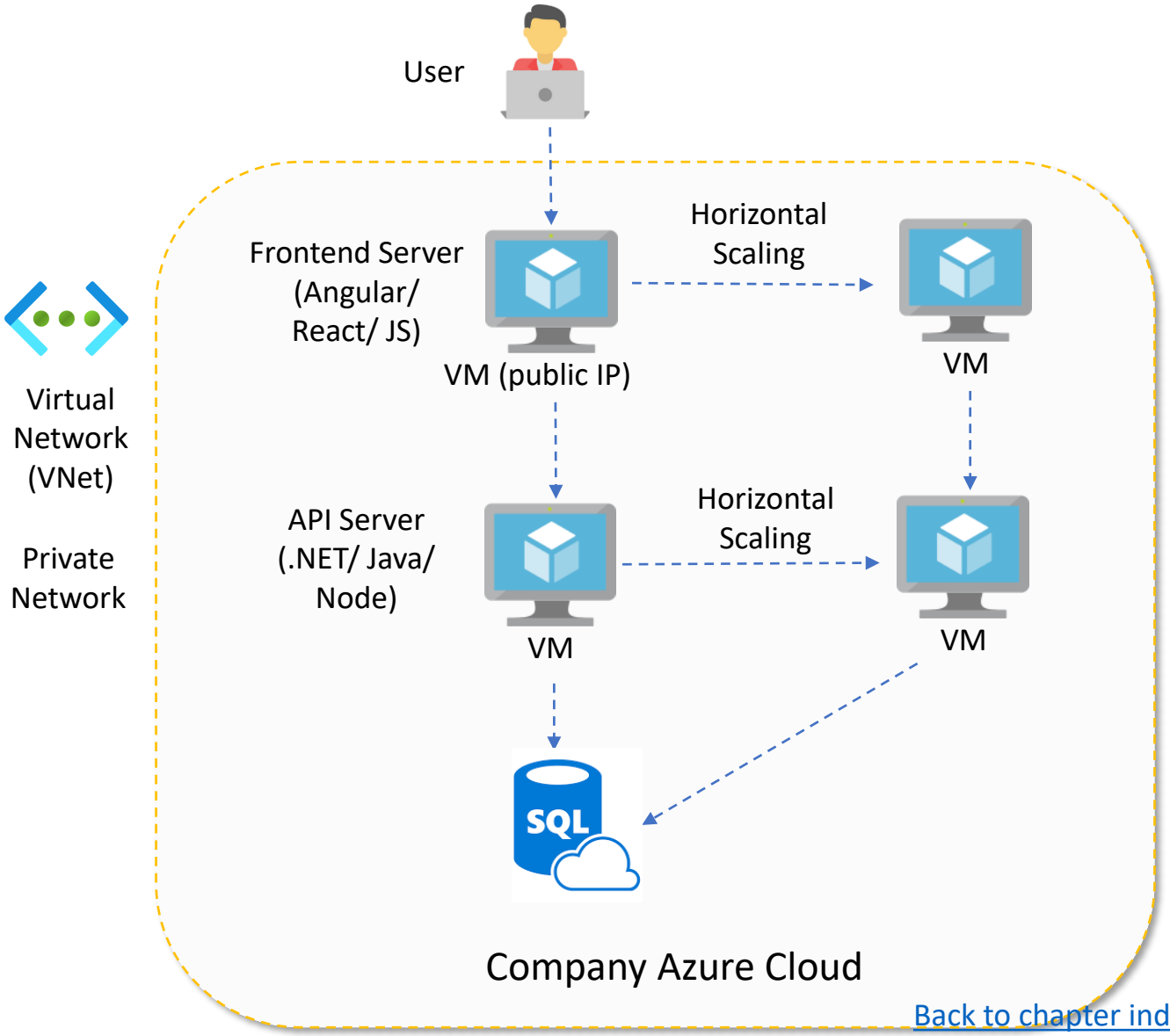
Q. How are applications hosted on VMs inside a VNet accessible to outside users?

- To allow access from outside, the VM must have a **Public IP address**.



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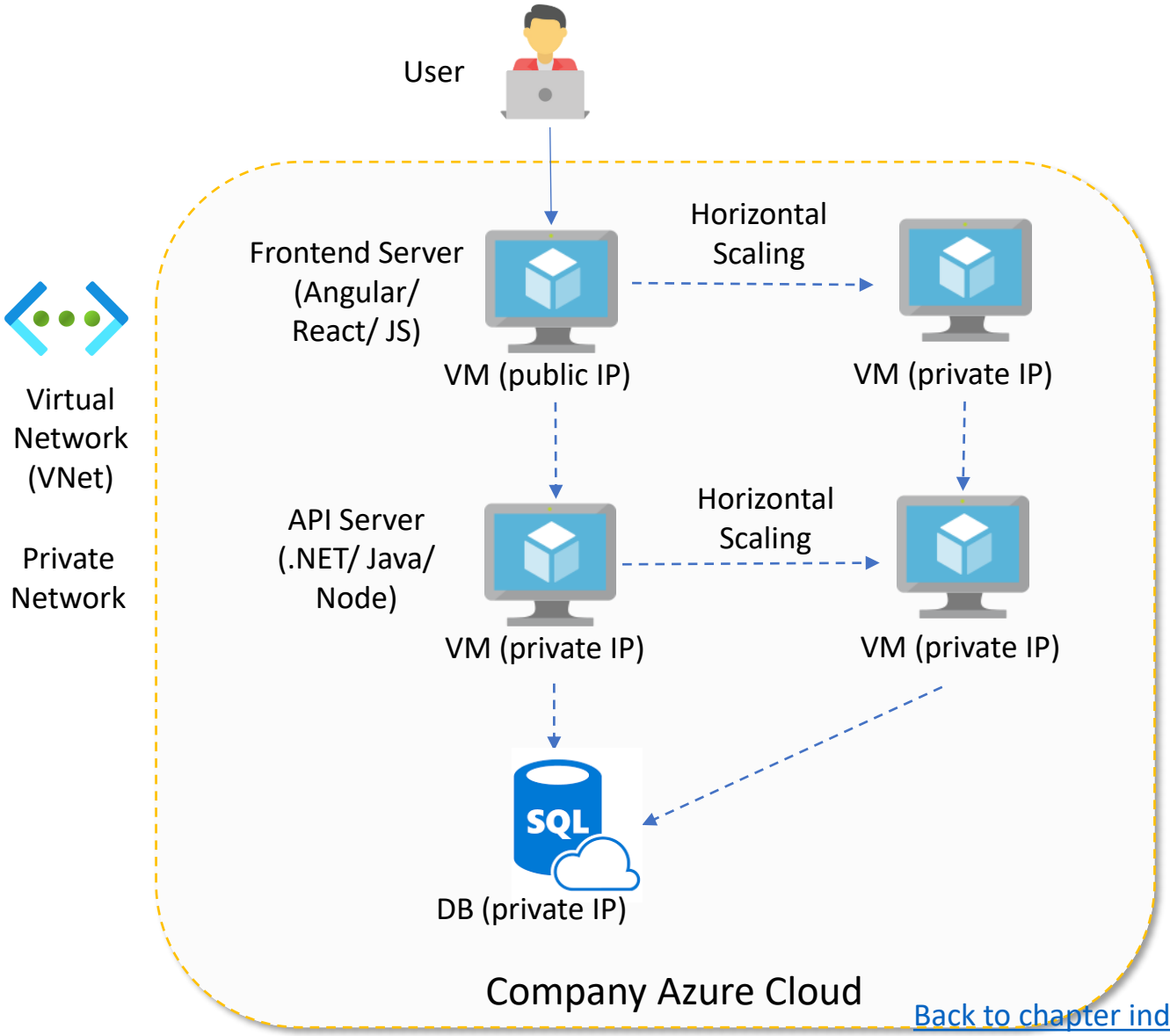
Q. What is the difference between Private IP and Public IP in Azure?

- A Private IP is an IP address assigned to resources inside a VNet by default, and it is not accessible from the internet.

- A Public IP is an IP address that you can assign to a resource to make it accessible from the internet.



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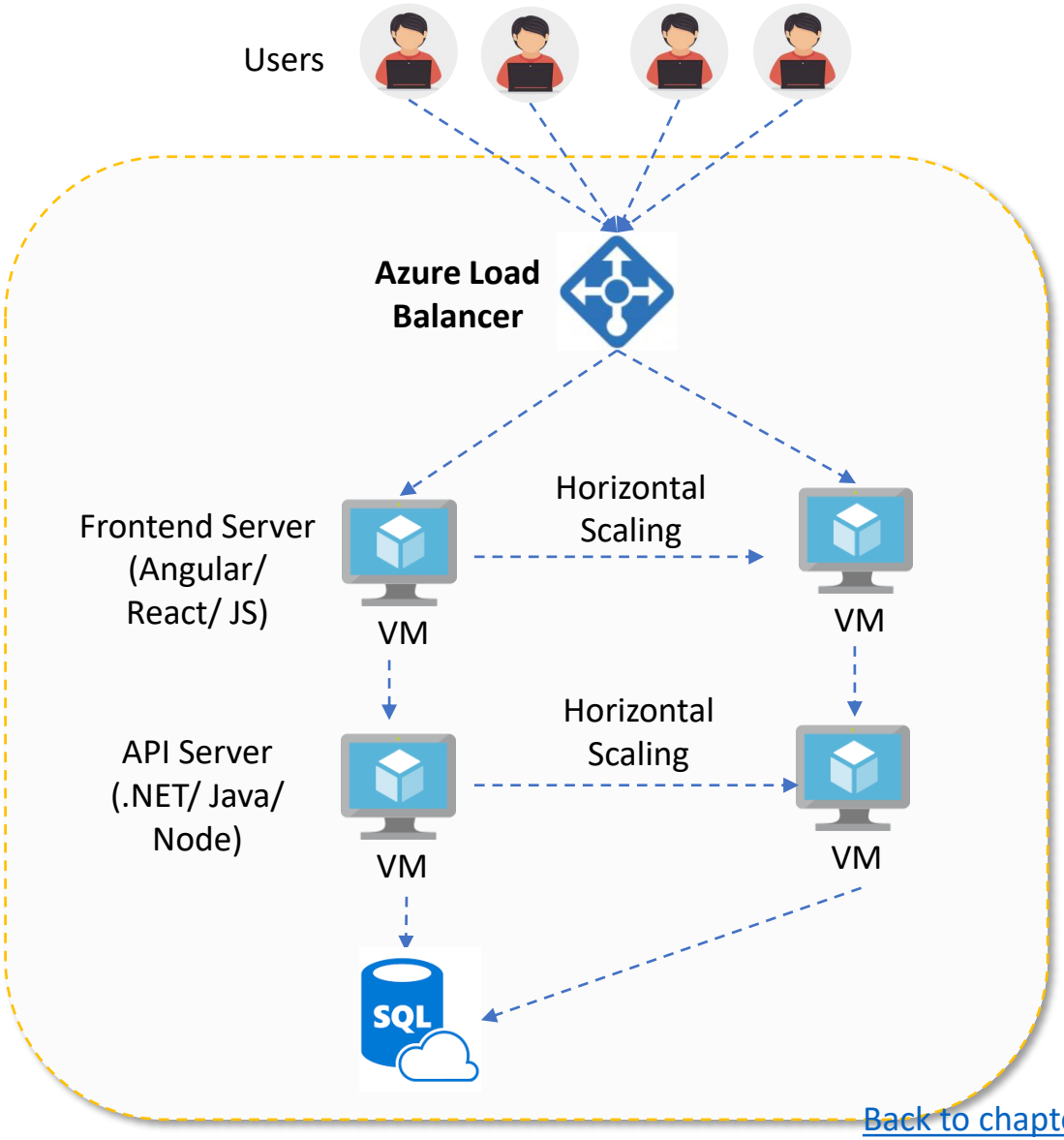
Q. What is the role of Azure Load Balancer? V. IMP.

- Azure Load Balancer **distributes incoming traffic** across multiple virtual machines (VMs) to ensure high availability, scalability, and fault tolerance.



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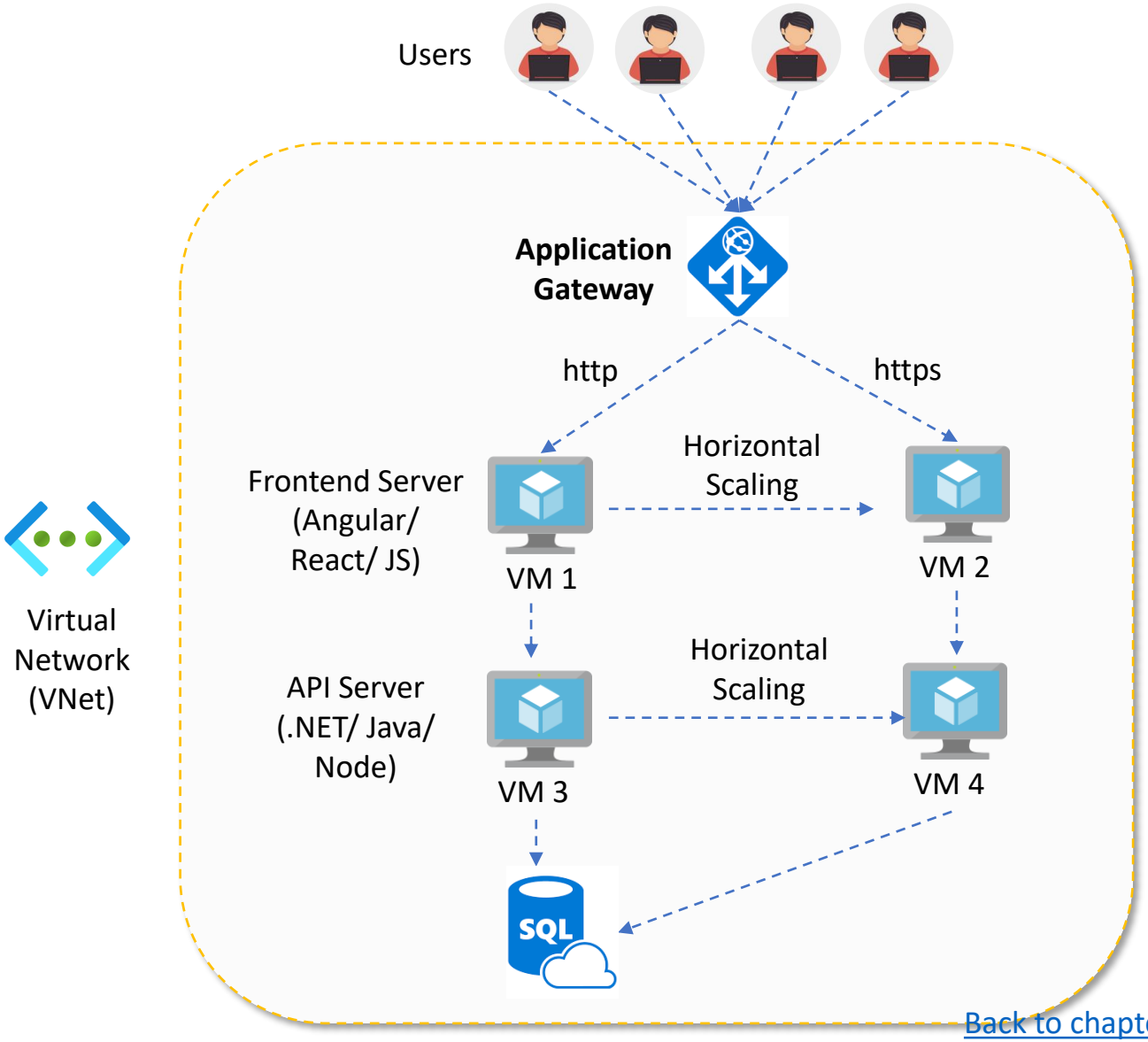
Q. What is the role of Azure Application Gateway? V. IMP.

- Azure Application Gateway distributes and route incoming traffic to your applications, **based on various rules** (example based on HTTP/HTTPS).



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Q. When to use Azure Load Balancer and when to use Application Gateway?

Scenario	Use Load Balancer	Use Application Gateway
Load balancing VMs	✅ Yes	❌ No
Handling Web Traffic (HTTP/HTTPS)	❌ No	✅ Yes
Path-based Routing (e.g., /api, /images)	❌ No	✅ Yes

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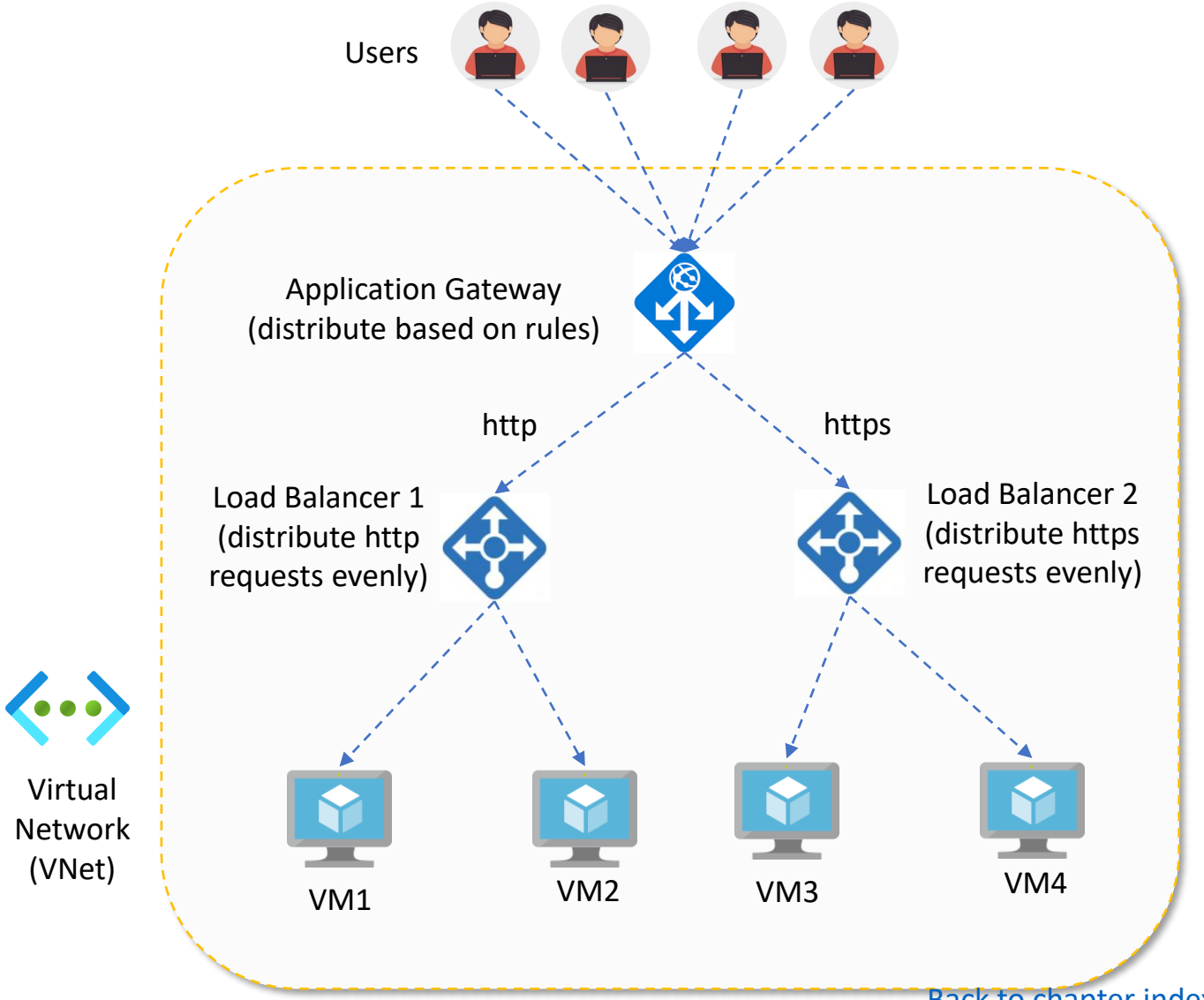
Q. How to use Load Balancer and Application Gateway together in Azure?

- Yes, in real-world applications, Azure Application Gateway and Load Balancer can be used together. Typically, the Application Gateway is placed in front of the Load Balancer.



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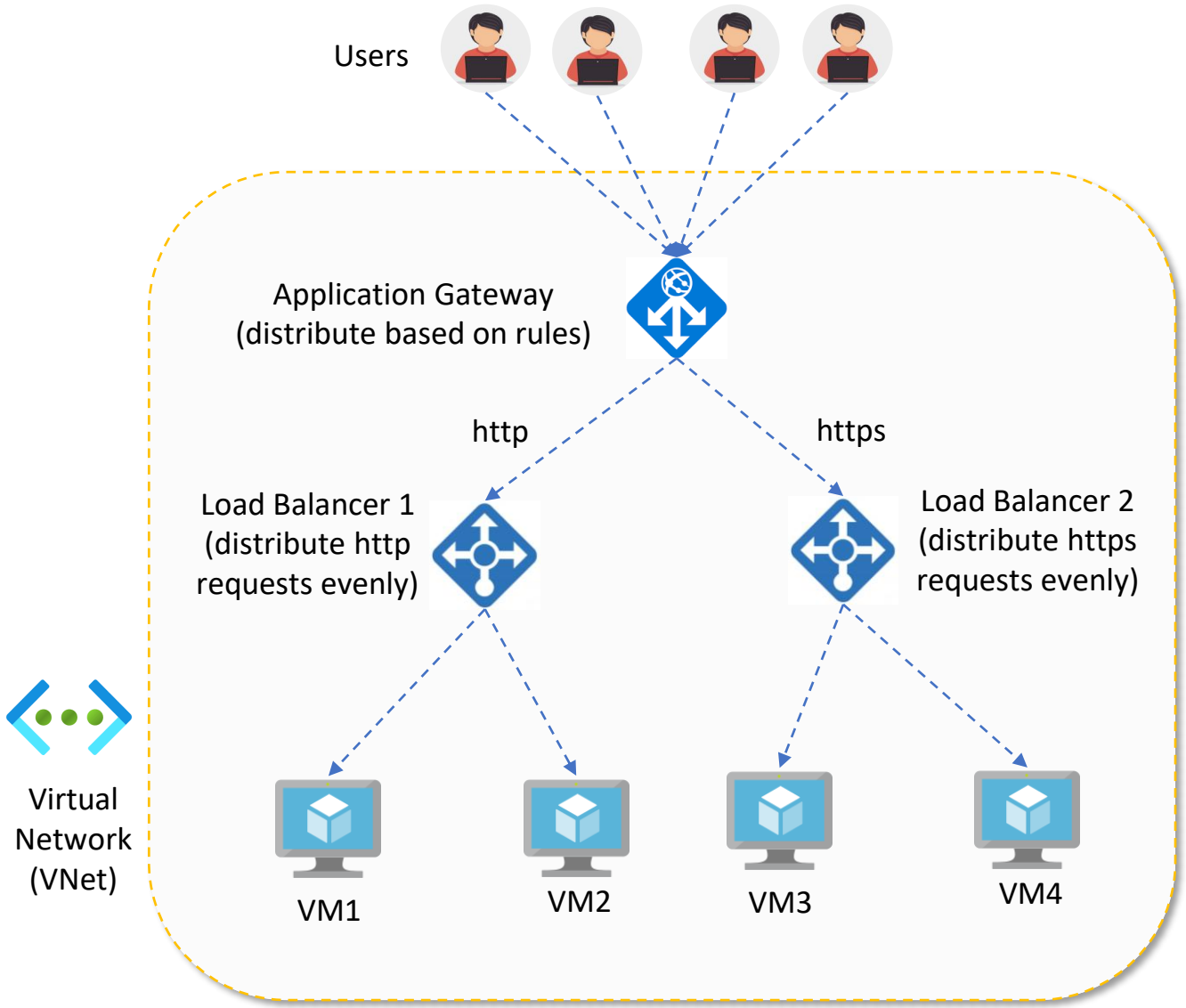
Azure Networking – I
(Scenario Based Questions)



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Q. Can two Azure VMs in different VNets communicate with each other?

- **Yes**, use **VNet Peering** to enable communication between two VNets, even across regions, with low latency and high bandwidth.

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Q. Your web app on a VM should be internet-facing. What do you assign?

- Assign a **Public IP address** to the VM

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Q. What is the default communication rule within a single VNet?

- All resources in a single VNet and subnet can communicate with each other by default, unless restricted by NSG rules.

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Q. Your app runs on 3 VMs and must evenly distribute traffic. What will you use?

- **Use an Azure Load Balancer** with a backend pool and health probes to distribute traffic among all healthy VMs.

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Q. You want to route requests based on URL path (e.g., /images, /videos). What Azure service fits?

- **Use Azure Application Gateway** with Path-based routing. It can direct requests to different backend pools based on URL segments.

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Q. What happens if your VNet has only private IPs but needs to access the internet?

- Use **Azure NAT Gateway** to allow secure internet access from private IP resources.

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Q. What's the role of Subnet in a VNet?

- A Subnet segments a VNet into smaller address spaces, allowing logical separation of resources like web, app, and DB layers.

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Q. Can a VM have both a private and public IP in Azure?

- **Yes**.
- A VM's can have a private IP for internal traffic and a public IP (static or dynamic) for internet access.

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8. Azure Networking - Part 2

Q. What is Azure Traffic Manager? How does it manage global traffic? **V. IMP.**

Q. What is a Subnet in Azure? What's the purpose of using it?

Q. What are Network Security Groups (NSGs)?

Q. What is the difference between Application Gateway and NSG?

Q. What is Azure CDN? When and why to use it? **V. IMP.**

Q. What is Azure Front Door? **V. IMP.**

Q. What is the difference between Azure Front Door & Azure Traffic Manager?

Q. Scenario based Real Application Questions

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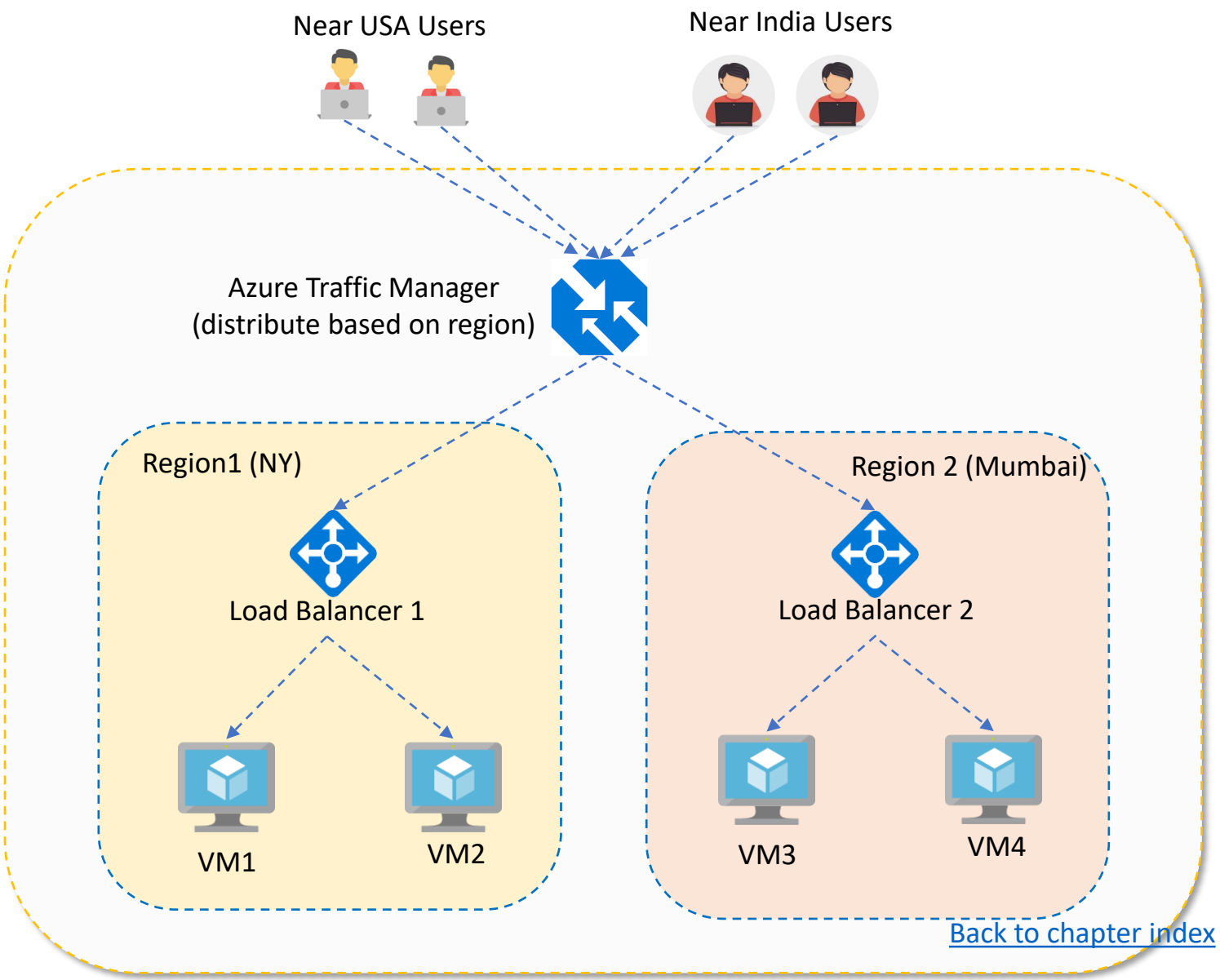
Q. What is Azure Traffic Manager? How does it manage global traffic?

- Azure Traffic Manager distributes user traffic **across multiple Azure regions** to improve availability, performance, and disaster recovery.



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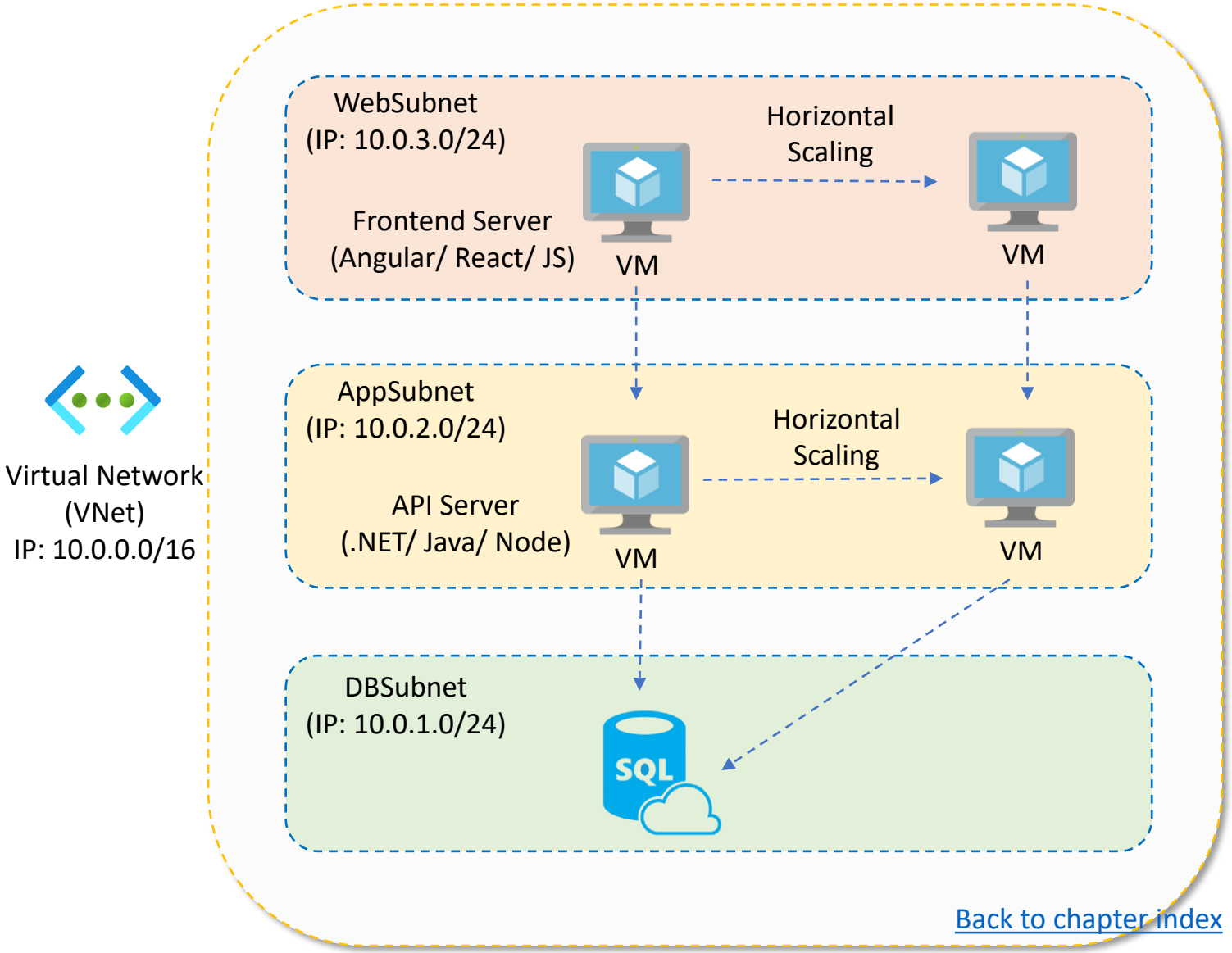
Q. What is a Subnet in Azure? What's the purpose of using it?

- A Subnet in Azure is a smaller logical segment of a Virtual Network (VNet).

- Why Use Subnets in Azure?

1. Segmentation: Separate tiers (web/app/db) for better control.
2. Security: Apply different NSGs to control traffic.



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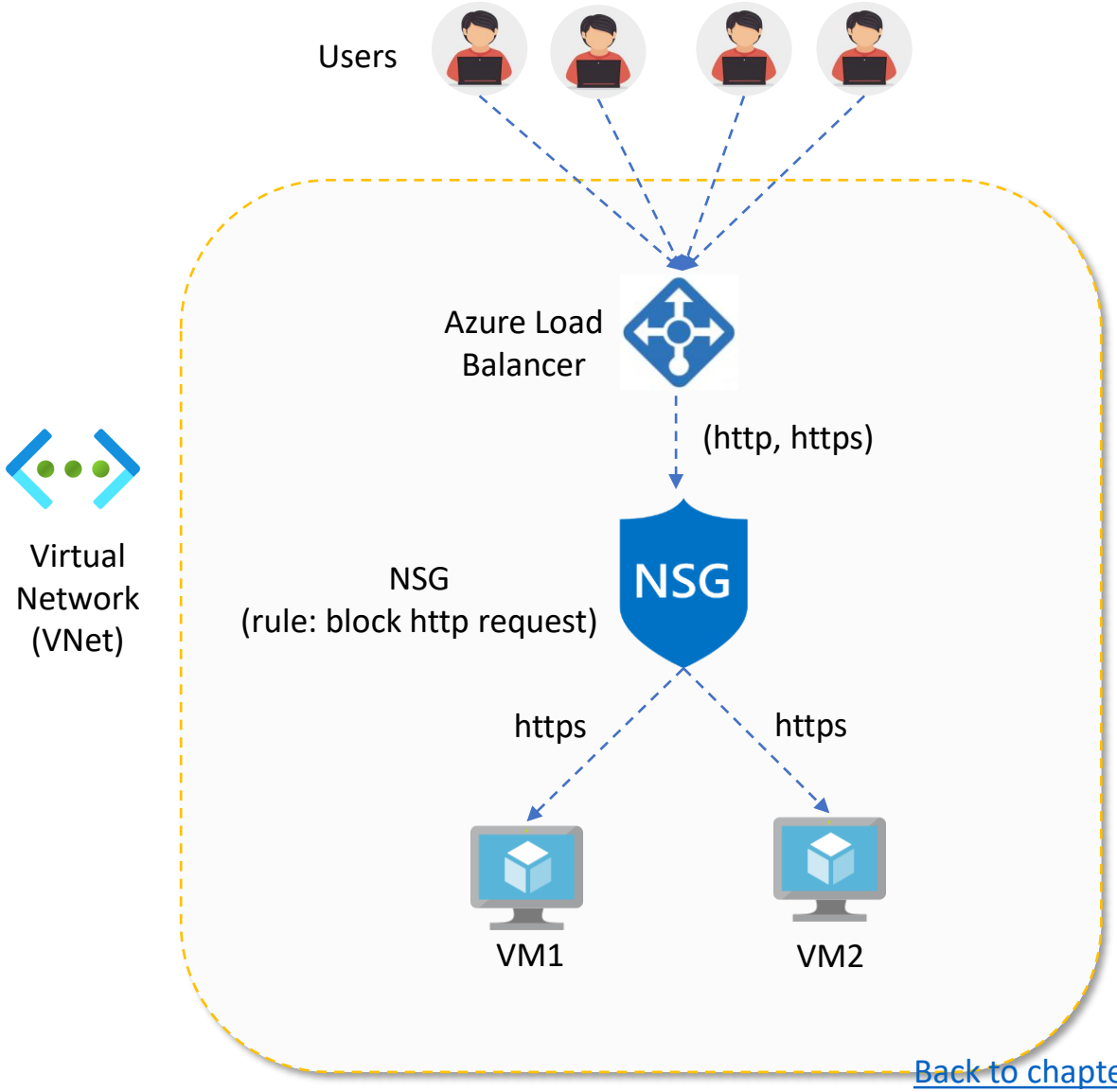
Q. What are Network Security Groups (NSGs)?

- Network Security Group (NSG) in Azure is a **virtual firewall** that controls inbound and outbound network traffic to and from Azure resources based on security rules.
- For example, NSGs can allow or block HTTP requests by configuring rules.



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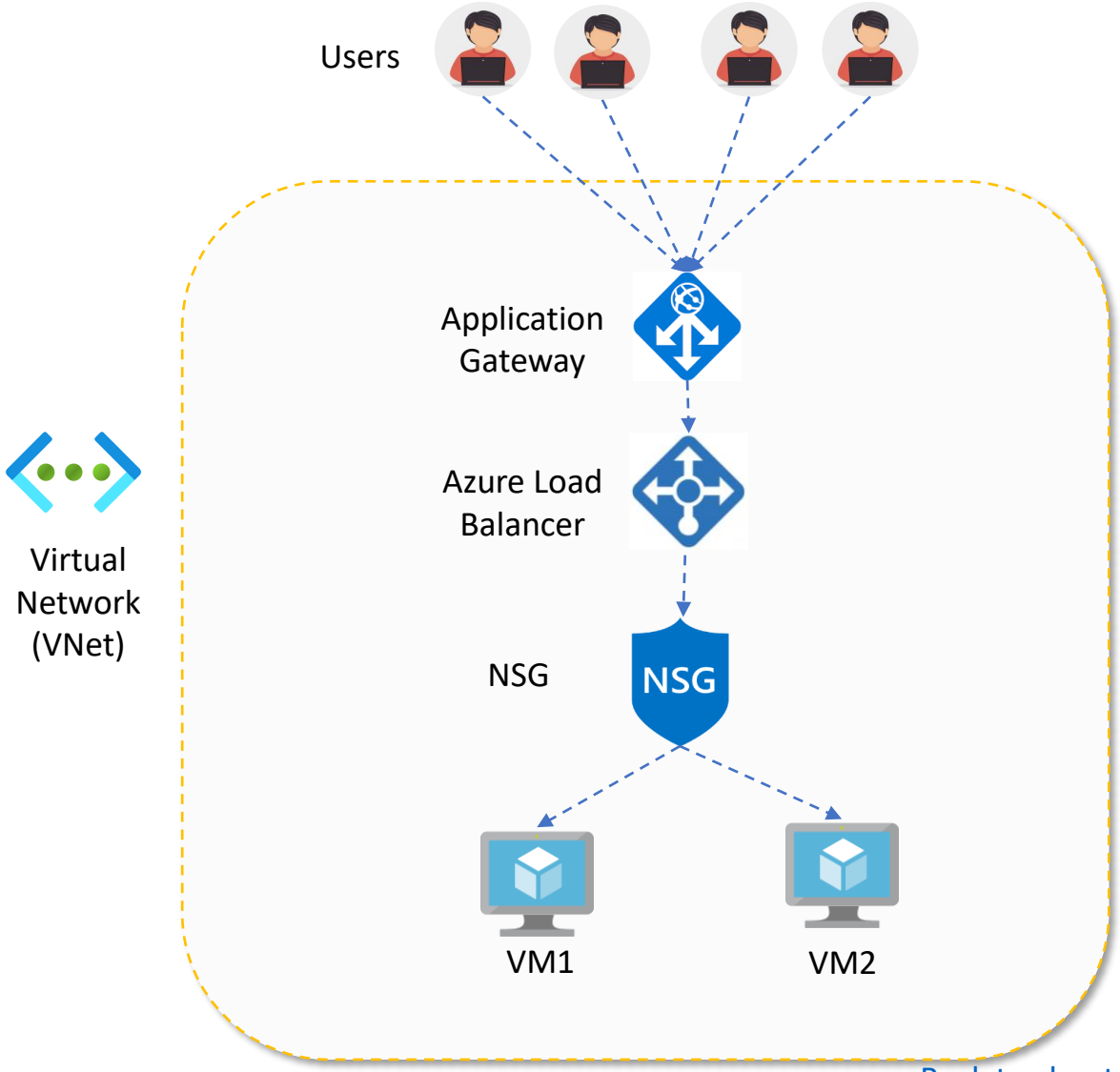
Q. What is the difference between Application Gateway and NSG?

- NSG (Network Security Group) works like a firewall at the network layer (Layer 3 & 4). It allows or blocks traffic based on IP addresses, ports, and protocols. It's used to secure subnets and VMs inside a VNet.
- Application Gateway works at the application layer (Layer 7). It can route traffic based on URL paths, host headers, HTTP/HTTPS rules.



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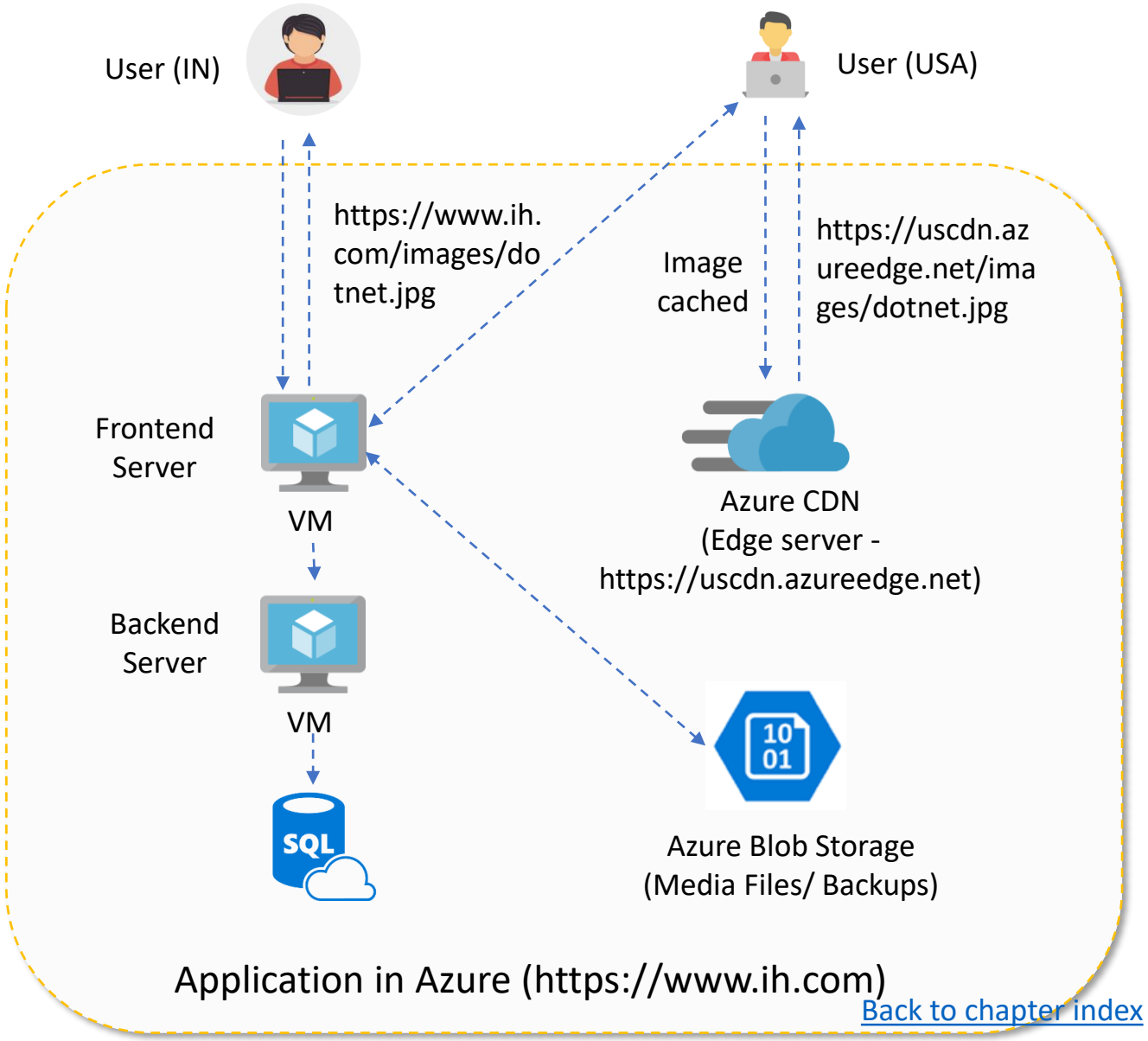
Q. What is Azure CDN? When and why to use it? V. IMP.

- Azure CDN (**Content Delivery Network**) is a service that helps deliver your website content - like images, videos, or files - faster to users by using Microsoft's global network of servers (called edge servers) located in different geographic locations.



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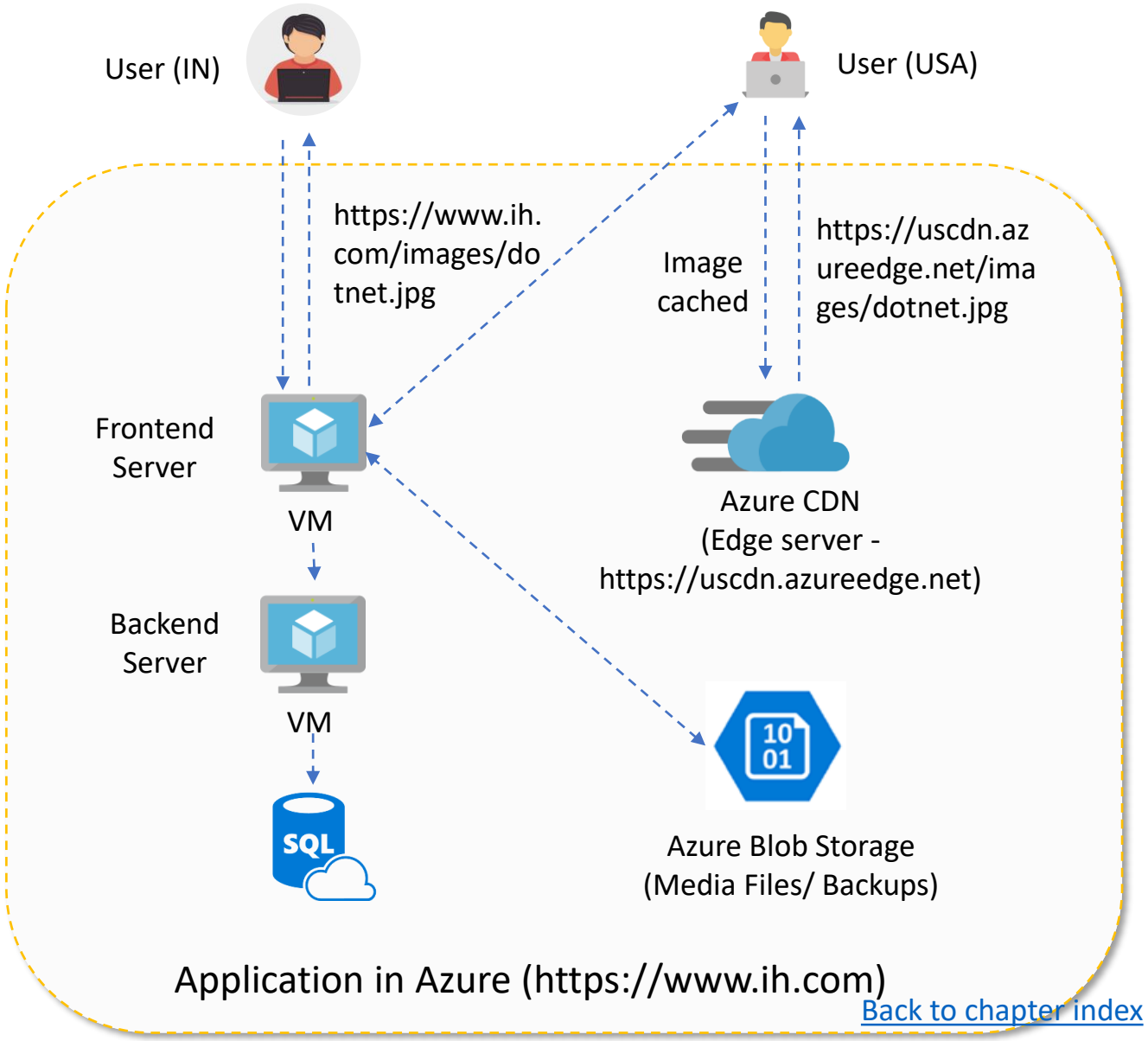
Q. What is Azure CDN? When and why to use it? V. IMP.

❖ Why use Azure CDN?

1. To reduce latency (delay in loading content)
2. To offload traffic from the main server
3. To improve performance and user experience
4. To scale globally without changing infrastructure



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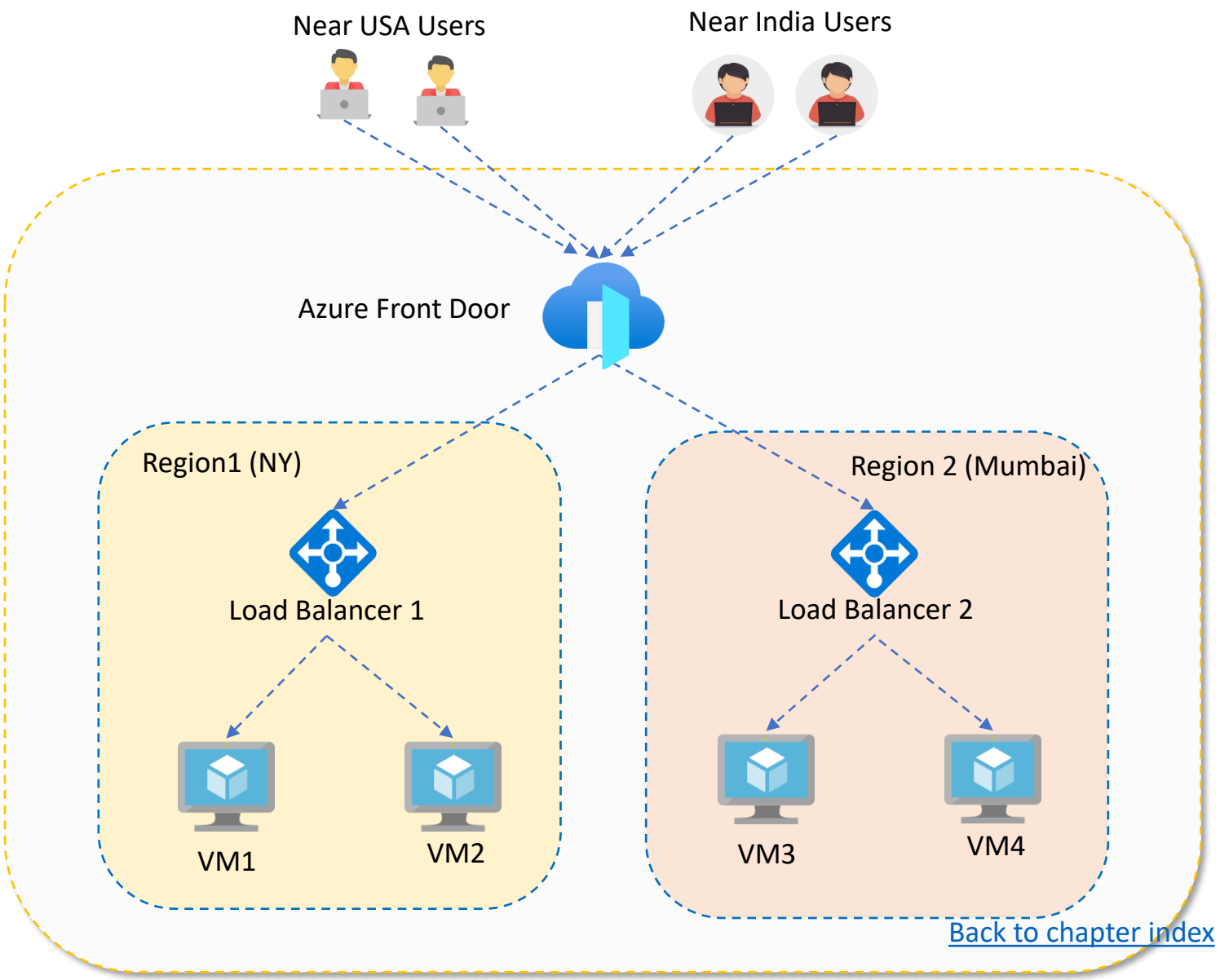
Q. What is Azure Front Door? V. IMP.

- Azure Front Door is an **entry point** for your web application. It works like a smart traffic controller that routes user requests to the fastest and healthiest backend.



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Q. What is the difference between Azure Front Door & Azure Traffic Manager? V. IMP.

Feature	Azure Traffic Manager	Azure Front Door
📍 Closest Server Selection	Based on DNS location (DNS-level - Layer 4)	Based on edge node and backend health (Application-level - Layer 7)
⚠️ Failover Speed	Slower (due to DNS caching)	Instant failover (real-time)
⚡ Caching at Edge	❌ Not available	✅ Yes (Built-in CDN)
🔒 Security Features	❌ None	✅ WAF, SSL Offloading
🛠️ URL-based Routing	❌ Not supported	✅ Yes (e.g., /images → one region, /api → another)
💰 Cost	Generally cheaper	Costlier (more features)

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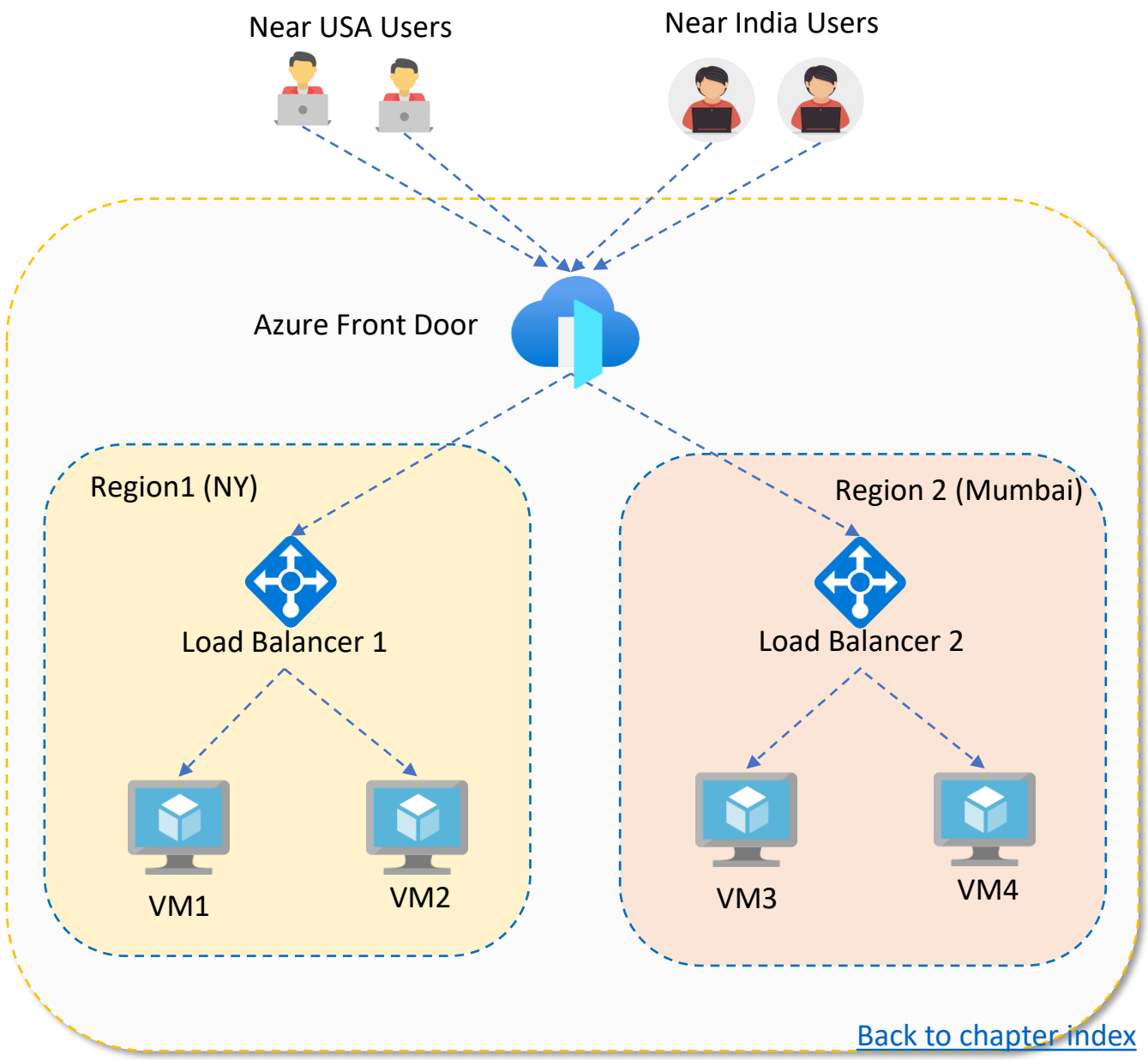
Azure Networking – II
(Scenario Based Questions)



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Q. Your app is used globally. How do you route users to the nearest region?

- **Use Azure Traffic Manager**.
- It routes users based on DNS and chooses the closest healthy endpoint using methods like performance or geographic routing.

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Q. What is the main use of a Subnet in Azure VNet?

- A Subnet logically **separates workloads** in a VNet (e.g., web, app, DB). You can apply different security rules and route settings per subnet.

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Q. You want to block port 22 (SSH) access to VMs. What would you use?

- **Use a Network Security Group (NSG)** to create an inbound rule that blocks traffic on port 22.

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Q. Your app needs to cache static images closer to users worldwide. What Azure service helps?

- **Use Azure CDN.**
- It caches static content like images and videos at edge locations, improving performance and reducing load on origin server.

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Q. You want to host a web app globally with fast failover. What would you choose?

- **Use Azure Front Door.**
- It offers global routing, health checks, edge caching, and WAF for security.

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Q. Can Azure CDN work with custom domains and SSL?

- **Yes**.
- Azure CDN supports custom domains with HTTPS using built-in or your own SSL certificates.

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Q. Your app is hosted in multiple regions. You want to ensure HTTPS traffic is routed smartly. Which service you will use?

- **Use Azure Front Door** for smart Layer 7 routing and global HTTPS load balancing with built-in SSL offloading.

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Q. When should you use Azure Front Door over Traffic Manager?

- Use Front Door for real-time health checks, edge caching, WAF, and faster failover. Traffic Manager is better for DNS-level routing scenarios.

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9. Identity, Security & Access

Q. What is Azure Active Directory? How is it different from on-prem AD? **V. IMP.**

Q. What are Authentication and Authorization?

Q. What is Multi-Factor Authentication (MFA) in Azure?

Q. What is Azure Key Vault? Share an example of when you used it your project? **V. IMP.**

Q. What is Service Principal in Azure? **V. IMP.**

Q. What is Managed Identity in Azure? **V. IMP.**

Q. How do you securely connect Azure services without credentials?

Q. What is Azure API Management (APIM)? Why and when should you use it? **V. IMP.**

Q. Scenario based Real Application Questions

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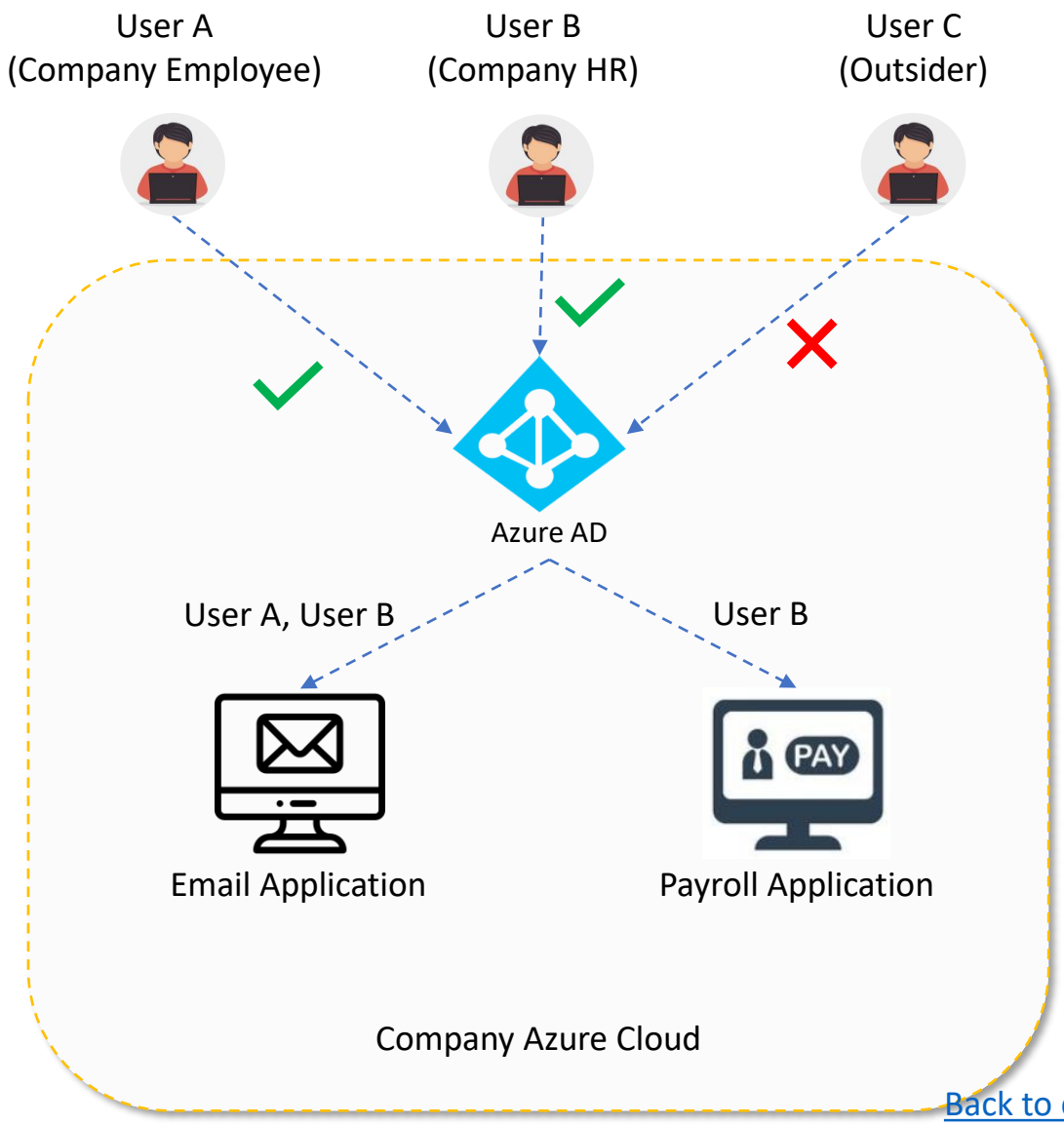
Q. What is Azure Active Directory? How is it different from on-prem AD? V. IMP.

- Azure Active Directory (Azure AD) is Microsoft's cloud-based **identity and access management** service that helps organizations manage user authentication and secure access to cloud applications.



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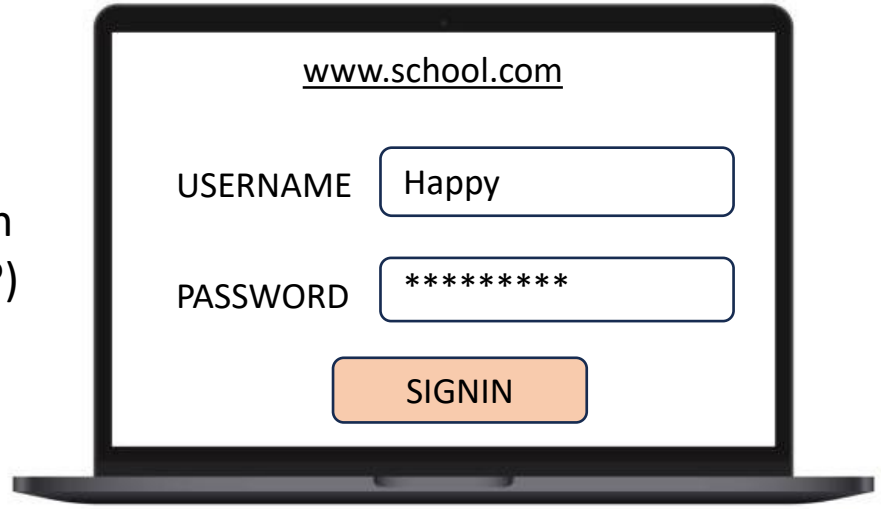
Q. What are Authentication and Authorization?

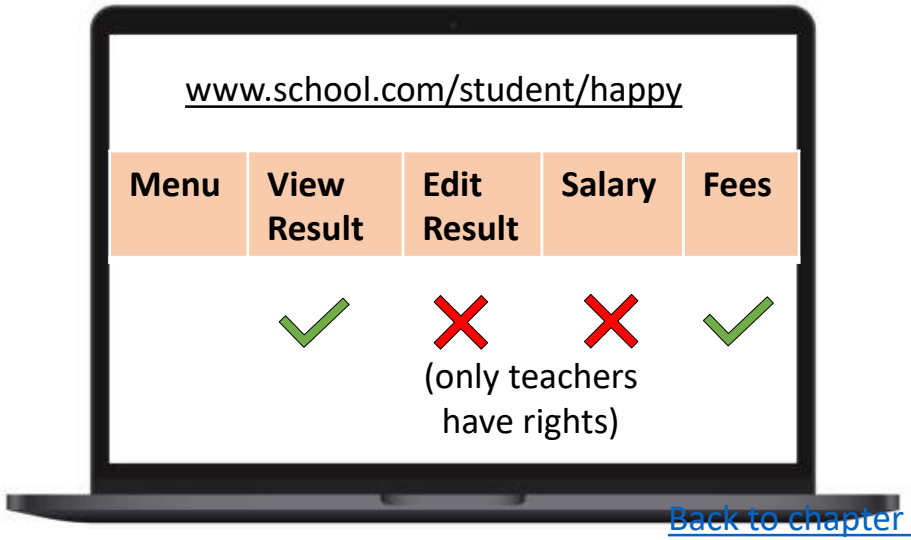
- Authentication is the process of **verifying the identity** of a user by validating their credentials such as username and password.

- Authorization is the process of allowing authenticated user **access to resources**.
- Authentication always precedes to Authorization.

Authentication (Who are you?)



Authorization (Your rights/ access?)



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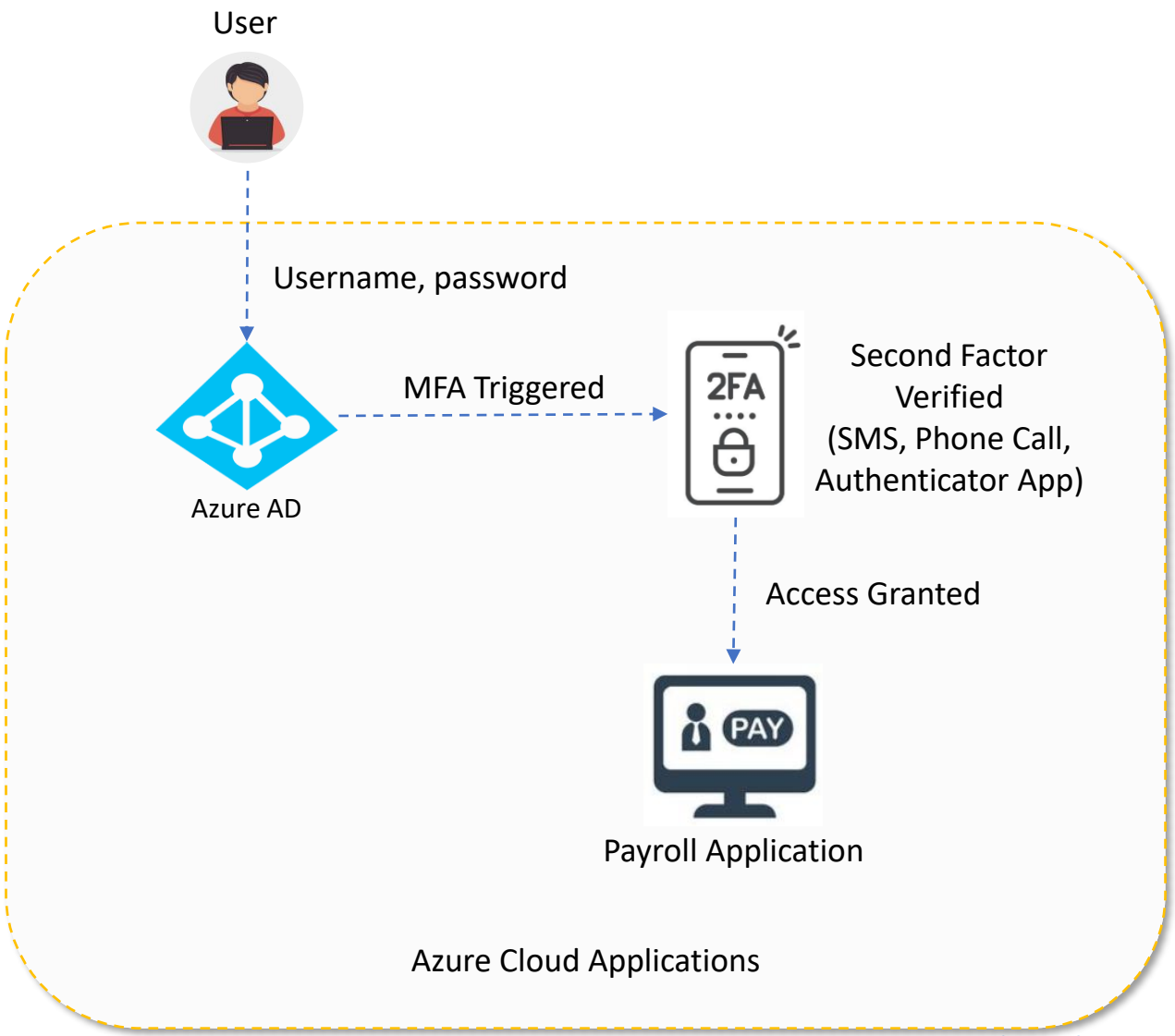
Q. What is Multi-Factor Authentication (MFA) in Azure?

- Azure Multi-Factor Authentication (MFA) is a security feature that enhances user authentication by requiring **two or more verification factors** before granting access to Azure resources, applications, and services.



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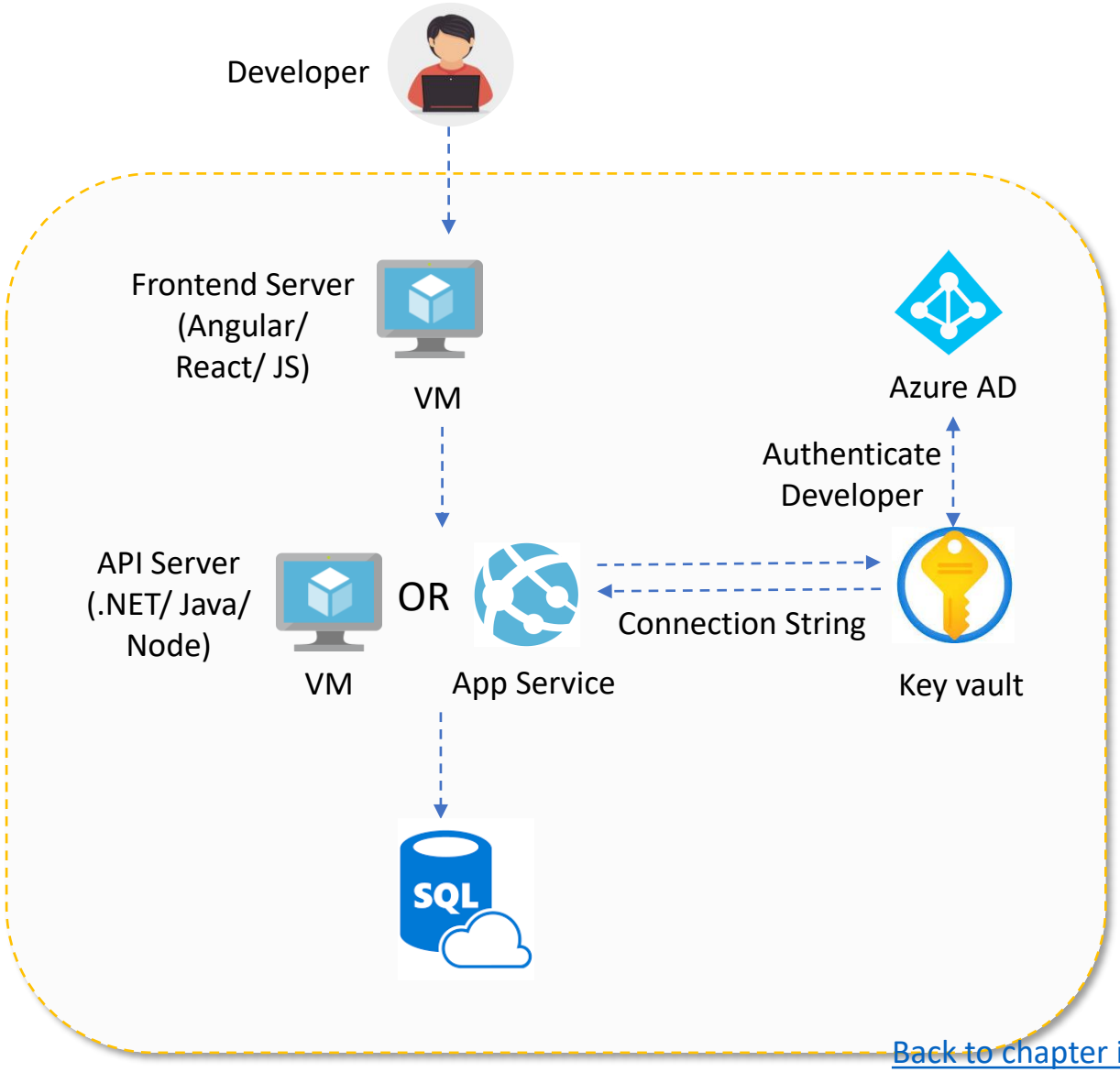
Q. What is Azure Key Vault? Share an example of when you used it your project? V. IMP.

- Azure Key Vault is a cloud-based security service that helps to **securely store** and manage secrets like connection strings, certificates, and passwords in Azure.

- Storing connection strings in Azure Key Vault keeps them secure, centrally managed, and protected from exposure in code or configuration files.



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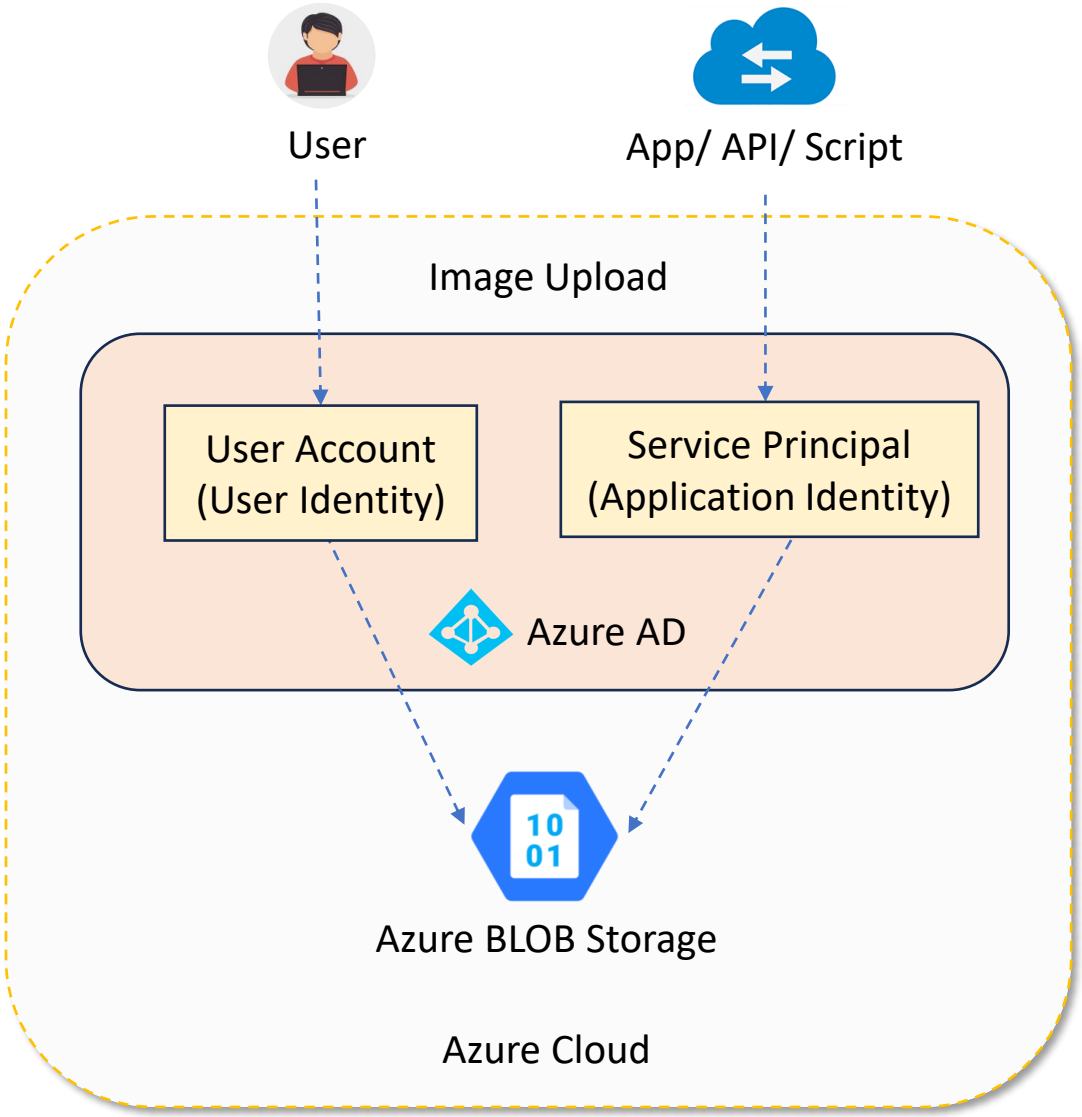
Q. What is Service Principal in Azure? V. IMP.

User Identity = Azure AD User Account

Service Principal = Application Identity

- A Service Principal is an **identity created for an application** in Azure Active Directory (Azure AD), like how a user identity is created for a person.
- It allows the application to securely authenticate and access Azure resources with specific permissions.



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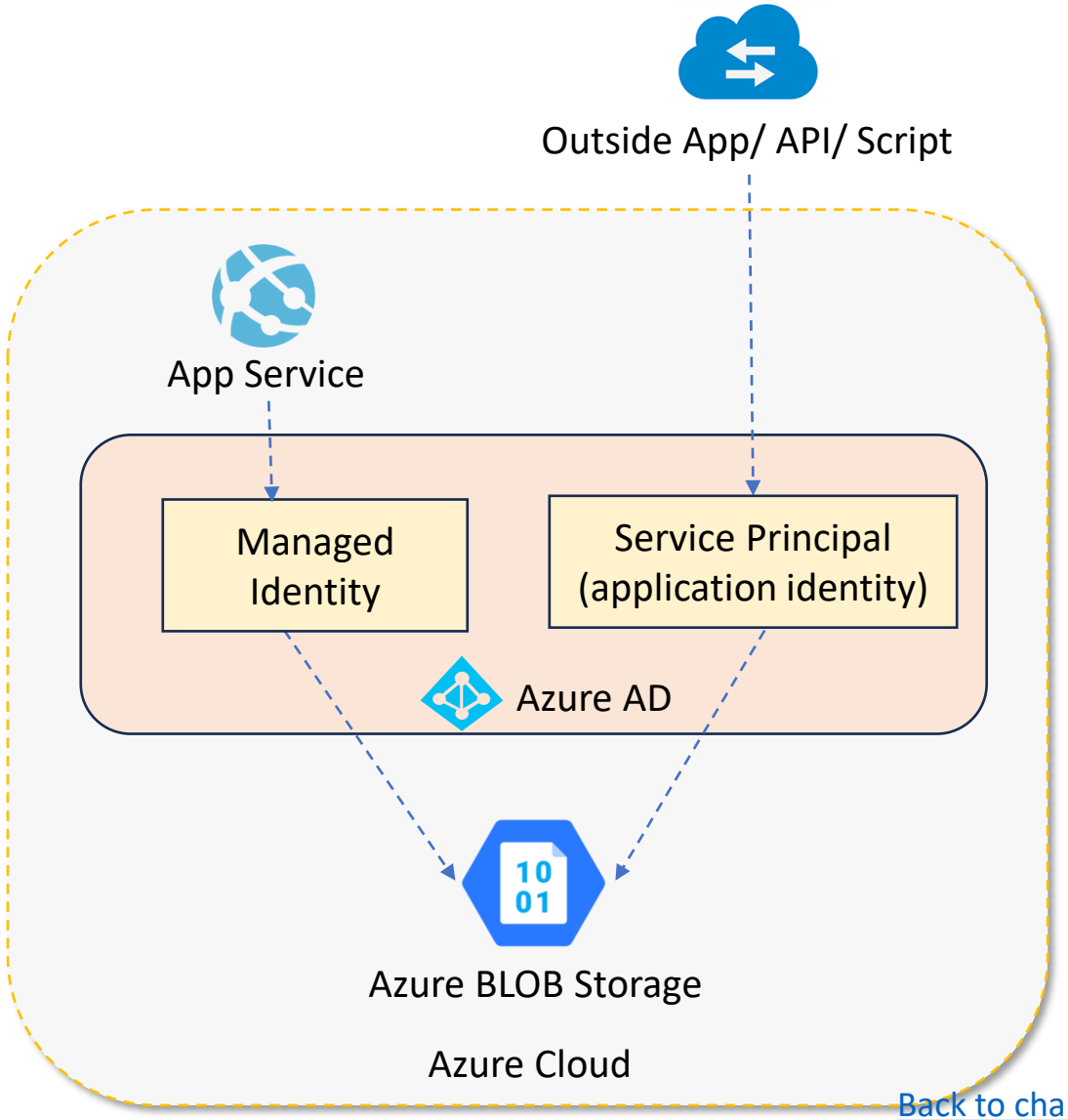
Q. What is Managed Identity in Azure? V. IMP.

- Managed Identity is a feature in Azure that provides an **automatically managed identity to Azure resources** (like Virtual Machines, App Services, Logic Apps, Azure Functions, etc.) so they can securely access other Azure services without needing secrets or credentials.



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Q. How do you securely connect Azure services without credentials?

- You can securely connect Azure services without using credentials like usernames, passwords, or secrets by using **Managed Identities**.

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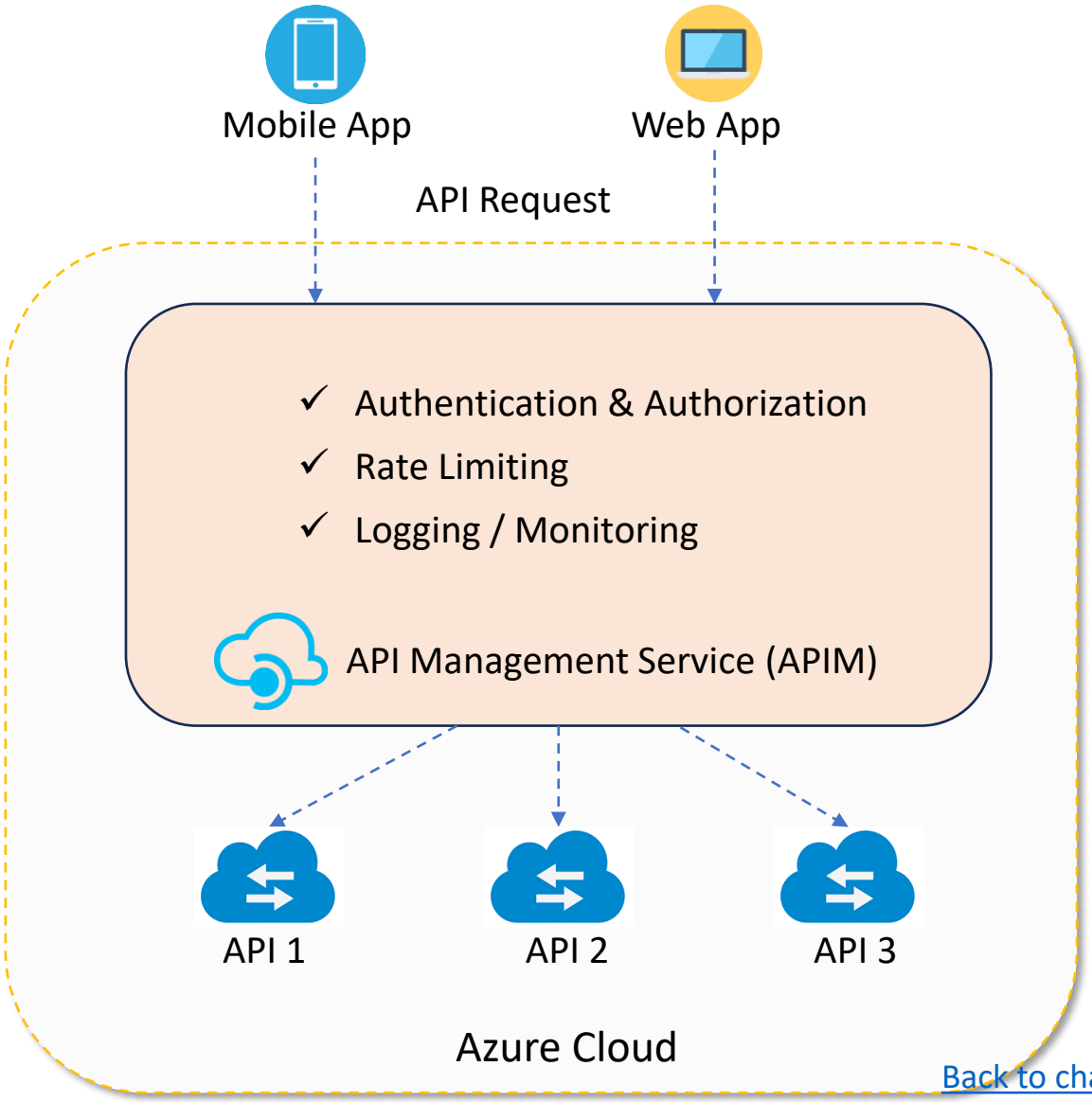
Q. What is Azure API Management (APIM)? Why and when should you use it? V. IMP.

- Azure API Management service acts like a **gatekeeper for your APIs**. APIM helps you publish, secure, monitor, and manage all your APIs in one place.



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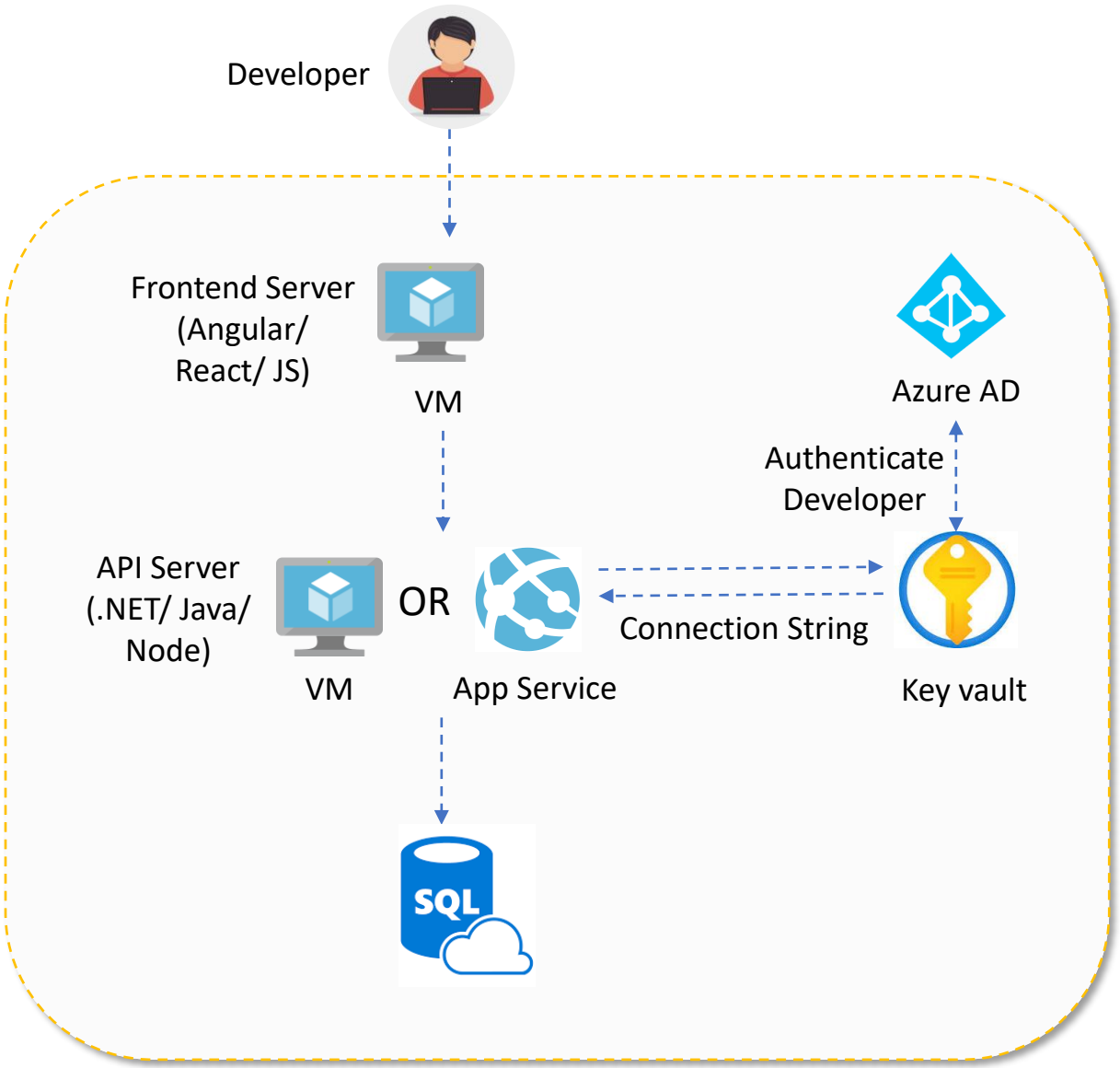
Identity, Security & Access
(Scenario Based Questions)



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Q. What is the key difference between Azure AD and on-prem Active Directory?

- On-prem AD manages devices and users **inside a network**.
- Azure AD is **cloud-based**, designed for managing users, apps, and access to cloud services like Microsoft 365, Azure, and SaaS apps.

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Q. A user enters a password and OTP to log in. What security concept is applied?

- This is Multi-Factor Authentication (**MFA**) — combining something you know (password) and something you have (OTP or mobile device).

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Q. What's the difference between Authentication and Authorization?

- Authentication confirms **who you are** (e.g., login).
- Authorization controls **what you can do** (e.g., read vs. write access).

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Q. You want an app to access a secret in Key Vault without storing credentials. What do you use?

- **Use Managed Identity**.
- It enables your app to authenticate with Azure services securely without storing secrets or credentials.

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Q. What is the difference between Managed Identity and Service Principal?

- Service Principal requires **manual secret** or certificate handling.
- Managed Identity is **auto-managed** by Azure and doesn't need manual credential storage.

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Q. Your team needs to expose APIs securely to external partners. What Azure service will you use?

- Use Azure API Management (**APIM**).
- It allows secure exposure, throttling, transformation, and monitoring of APIs with developer portal access.

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Q. Can Azure Key Vault trigger alerts if someone accesses a secret?

- **Yes**.
- Enable diagnostic logs and send them to Azure Monitor or Log Analytics to trigger alerts on secret access events.

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Q. How can you give limited access to only one secret in Key Vault?

- **Use Access Policies** or RBAC at the secret level. Assign the user or service only the "Get" permission for that specific secret.

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10. Real-Time Analytics

Q. What is Click Stream Analysis? How is it useful? **V. IMP.**

Q. What are Azure Event Hubs? When do we use them in our projects?

Q. What is Azure Stream Analytics? Can you explain a basic use case?

Q. What is Azure Data Lake Storage? When to use it?

Q. What is Big Data Analytics? What is the role of Azure Synapse Analytics? **V. IMP.**

Q. Summarize the click stream analysis steps and tools?

Q. Scenario based Real Application Questions

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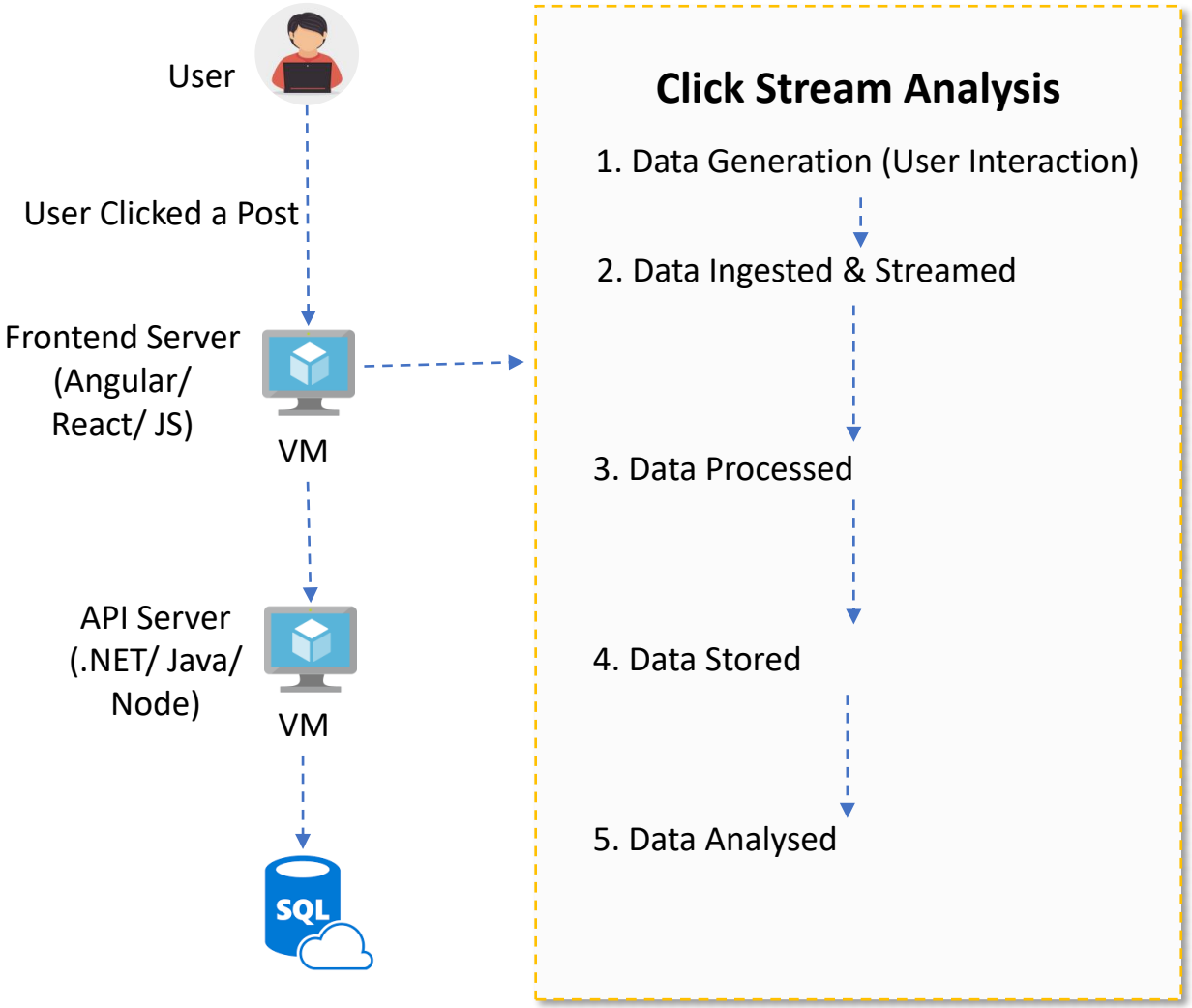
Q. What is Click Stream Analysis? How is it useful? V. IMP.

- Click Stream Analysis is the process of **capturing, processing, and analyzing user interactions** (clicks, page views, searches, etc.) on a website or application to gain insights for personalization, targeted ads, and user behavior trends.



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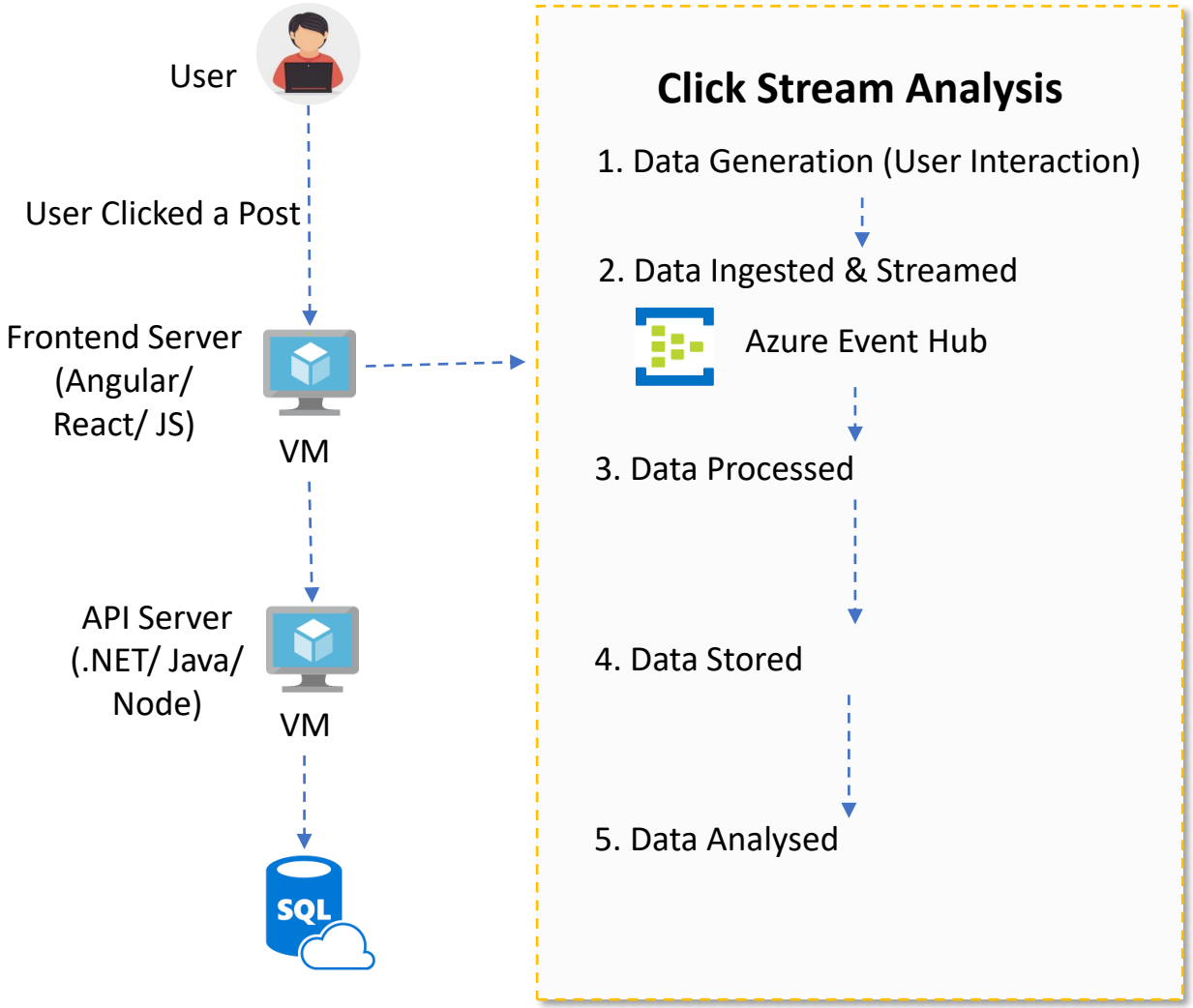
Q. What are Azure Event Hubs? When do we use them in our projects?

- Azure Event Hubs is a big data streaming platform and **event ingestion service** from Microsoft Azure.



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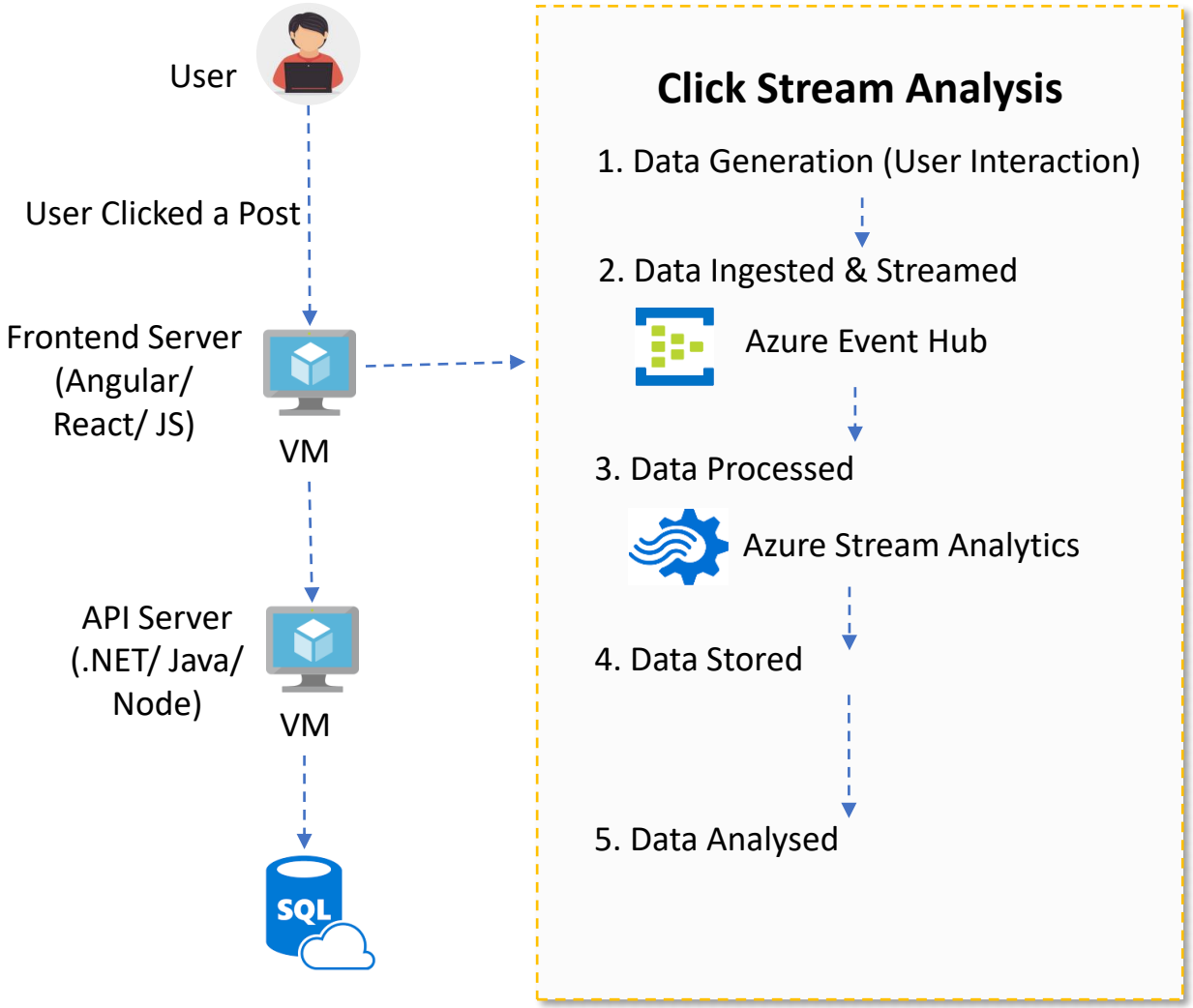
Q. What is Azure Stream Analytics? Can you explain a basic use case?

- Azure Stream Analytics is a **real-time data processing** service from Microsoft Azure that lets you process streaming data as it arrives.



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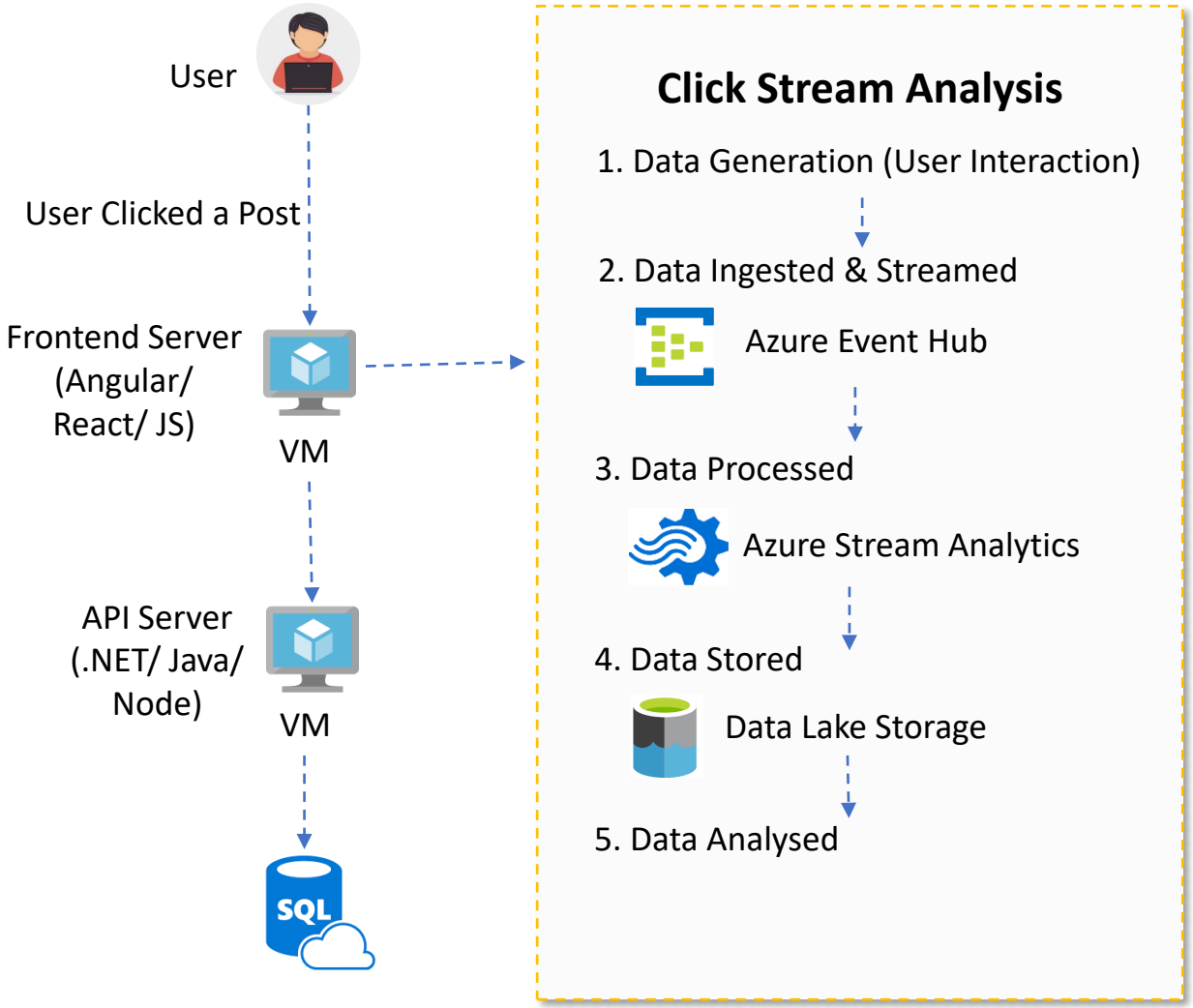
Q. What is Azure Data Lake Storage? When to use it?

- Azure Data Lake Storage (ADLS) is a scalable, secure, and high-performance **cloud storage service** designed for big data analytics.



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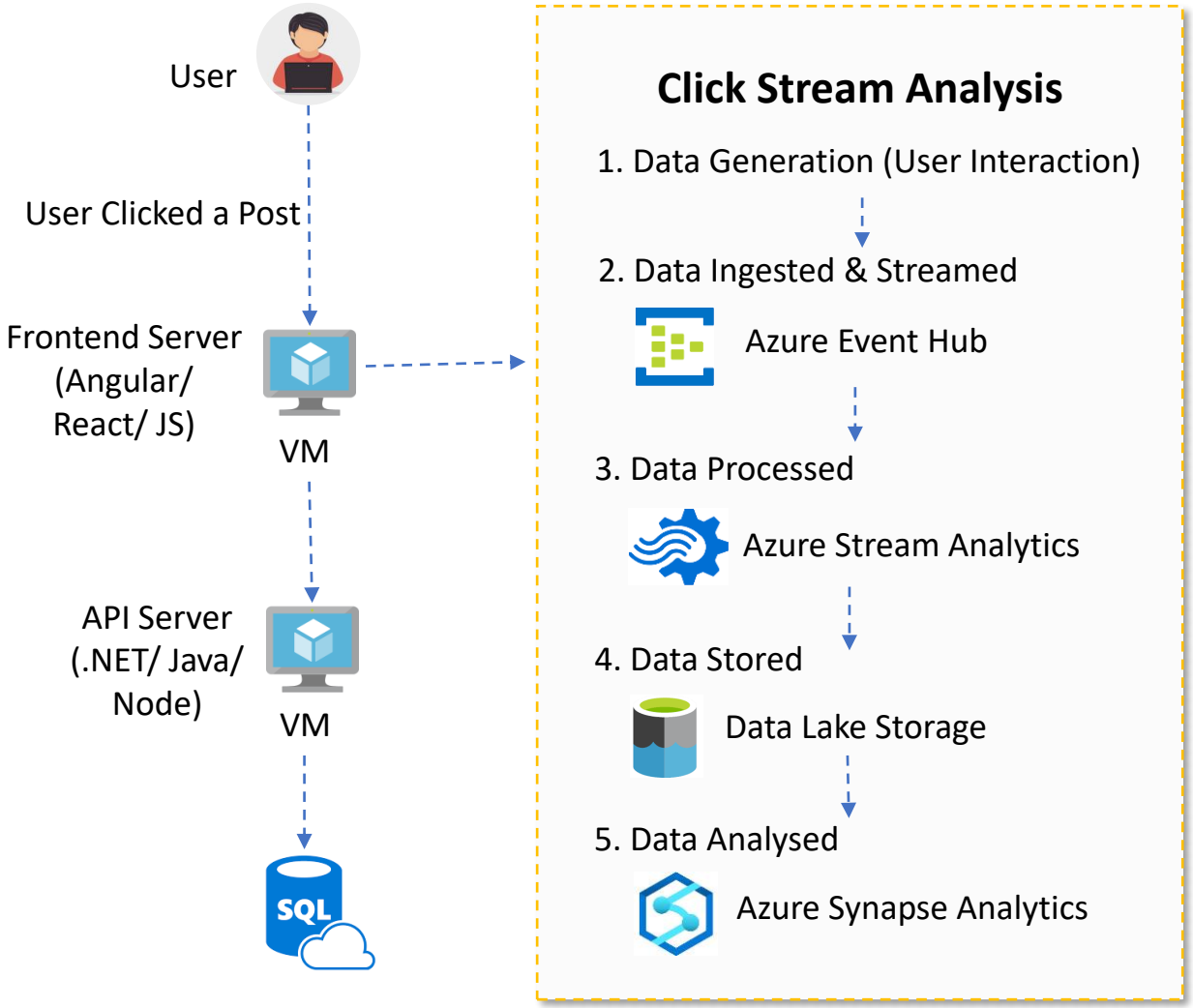
Q. What is Big Data Analytics? What is the role of Azure Synapse Analytics?

- Big Data Analytics is the process of **examining large and complex data sets** to find patterns, trends, and insights that help in decision-making and predictions.

- Azure Synapse Analytics is a unified cloud service for big data processing, data warehousing, and **real-time analytics**.



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Q. Summarize the click stream analysis steps and tools?

Step	What Happens	Azure Services Involved
1. User Interaction	User clicks, scrolls, browses	-
2. Data Ingestion & Streaming	Events are captured and streaming into Azure	Event Hubs, Kafka
4. Data Processing	Clean, filter, transform data	Data Factory, Stream Analytics
5. Data Storage	Save data for querying and analysis	Data Lake Storage, SQL DB
6. Data Analysis	Analyse user behaviour and patterns	Power BI, Synapse Analytics

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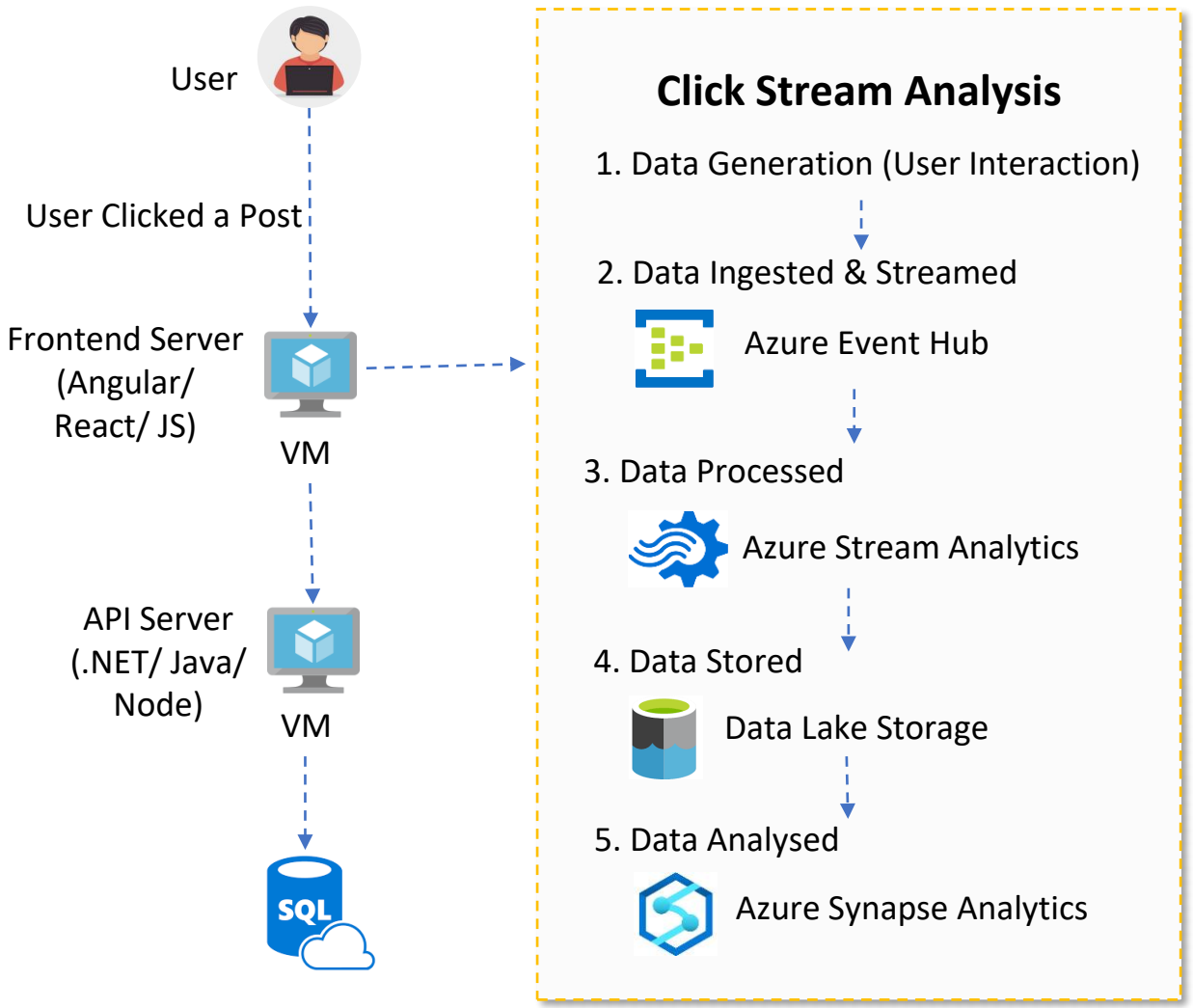
Real-Time Analytics
(Scenario Based Questions)



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Q. Where Click Stream Analysis is used?

- It’s the process of **tracking user clicks and behavior on a website/app**. Used in e-commerce, media, and SaaS apps to understand user interests, personalize content, and improve UX.

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Q. Your app generates high-speed event data from multiple sources. How would you capture it?

- **Use Azure Event Hubs**.
- It's built to handle millions of events per second and ingest telemetry or log data from distributed apps/devices.

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Q. How do you analyze real-time data from Event Hub?

- **Use Azure Stream Analytics**.
- It connects to Event Hubs, processes streaming data in real time using SQL-like queries, and pushes results to dashboards, databases, or alerts.

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Q. You need a solution that combines big data analytics with SQL. What do you use?

- **Use Azure Synapse Analytics**.
- It combines data warehousing and big data processing in one service with support for SQL, Spark, and pipelines.

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Q. What is a real-world use case for Azure Stream Analytics?

- Monitor IoT devices in factories. If temperature crosses a threshold, trigger an alert instantly using real-time rules in Stream Analytics.

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Q. What tool stores the raw event data for batch processing after real-time analysis?

- **Use Azure Data Lake Storage** or Blob Storage as long-term storage after processing the stream for future batch analytics.

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Q. Summarize the steps in clickstream analysis.

- Ingest events using Event Hub
- Process using Stream Analytics
- Store in Data Lake
- Visualize in Power BI Or analyze using Synapse Analytics

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11. DevOps, CI/CD & IaC

Q. What is DevOps and Azure DevOps? What is the difference? **V. IMP.**

Q. What is a CI/CD pipeline? How have you used it in your project? **V. IMP.**

Q. What are the main components of Azure DevOps?

Q. How Azure Boards and Azure Repos are used in Azure DevOps?

Q. What are Azure Pipelines? How do they help with automation in projects?

Q. What are Azure Test Plans and Azure Artifacts?

Q. What is Infrastructure as Code (IaC)? Explain how it works? **V. IMP.**

Q. What are the benefits of using IaC? Is there any time when you would not use it?

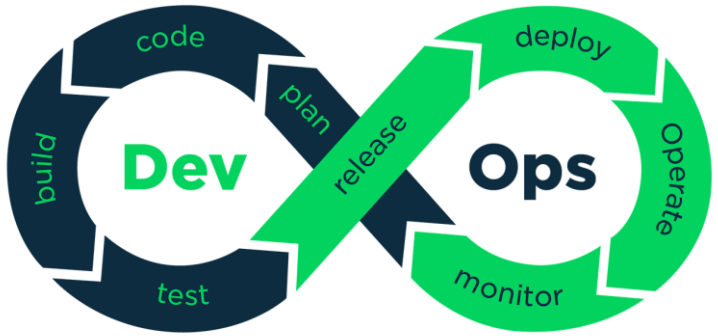
Q. Scenario based Real Application Questions

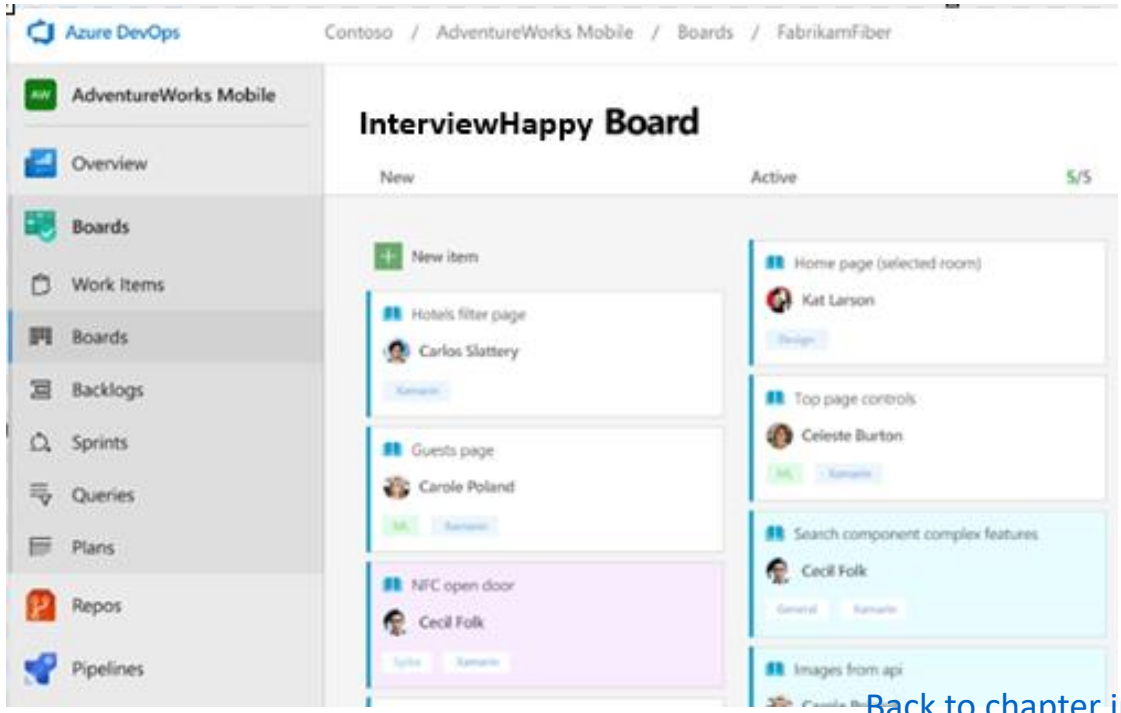
Back to main index

Q. What is DevOps and Azure DevOps? What is the difference? V. IMP.

- DevOps = Development + Operations

- Azure DevOps is a **cloud-based DevOps platform** from Microsoft that provides end-to-end tools for software development, CI/CD, testing, and project management.
- It enables teams to plan, develop, build, test, and deploy applications efficiently.





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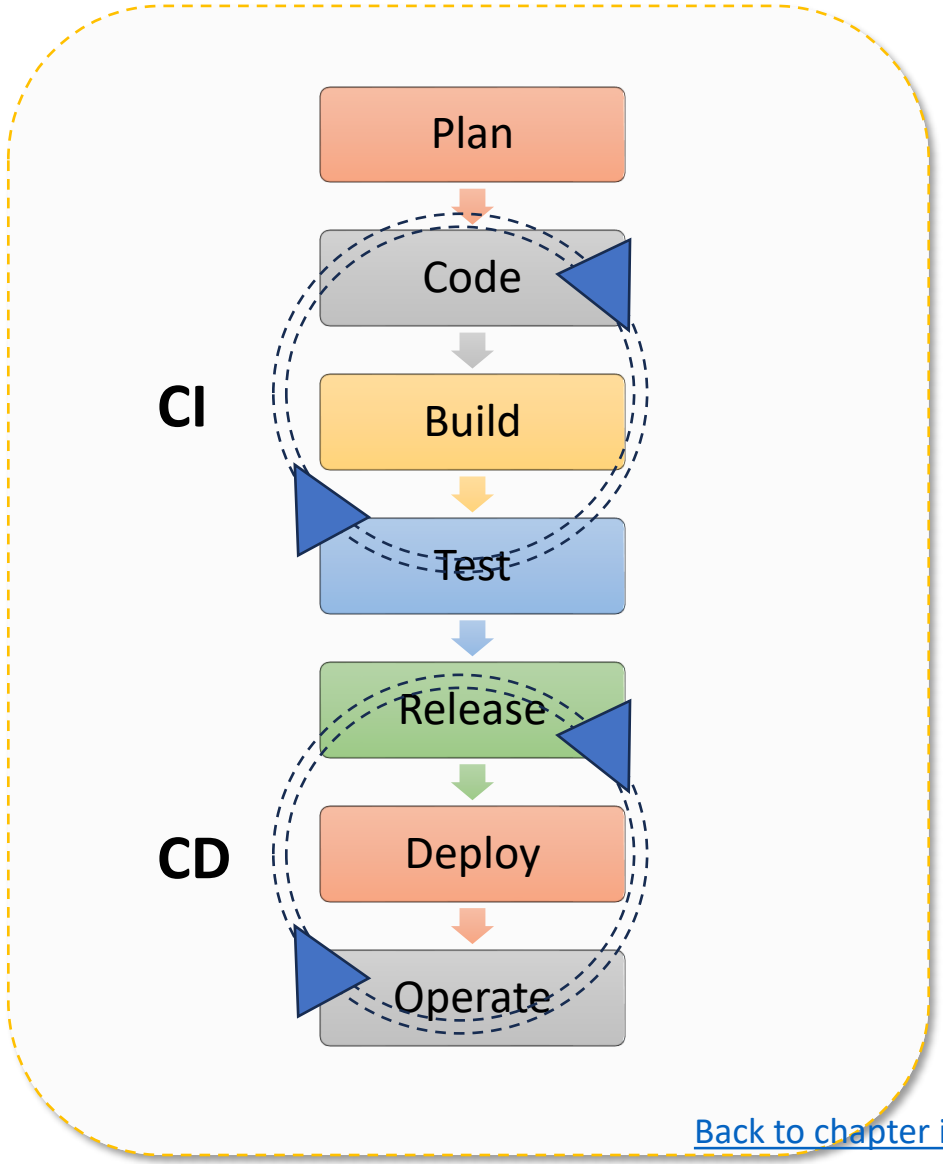
Q. What is a CI/CD pipeline? How have you used it in your project? V. IMP.

- A CI/CD pipeline is an automated process used by developers and DevOps teams to build, test, and deploy code quickly and reliably.



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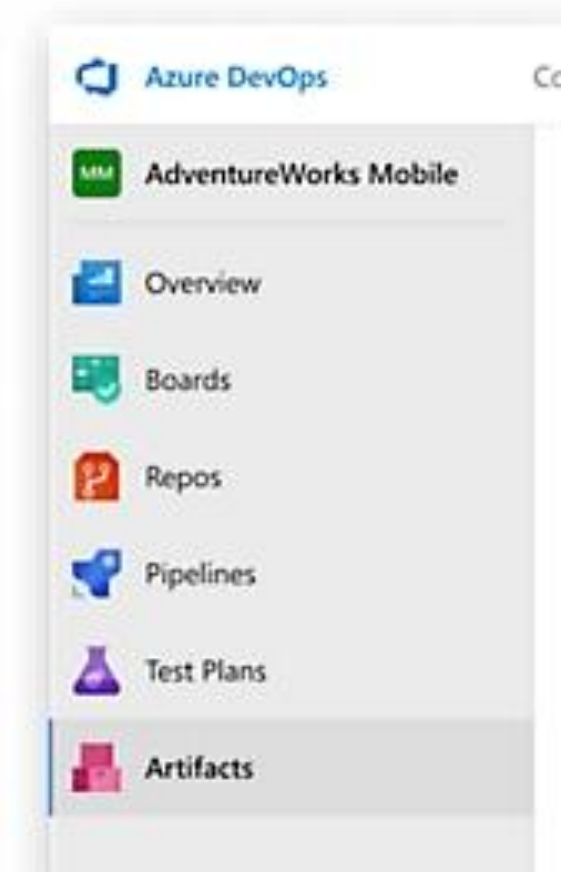
Q. What are the main components of Azure DevOps?

Components of Azure DevOps

1. Azure Boards
2. Azure Repos
3. Azure Pipelines
4. Azure Test Plans
5. Azure Artifacts



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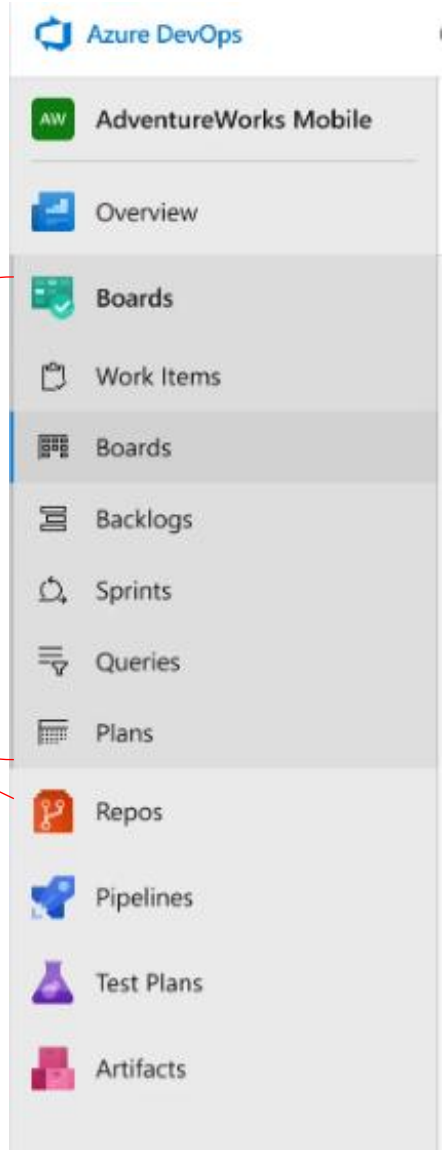
Q. How Azure Boards and Azure Repos are used in Azure DevOps?

- Azure Boards are used for **project management**. Teams use it to plan work, create user stories, track tasks, manage sprints, and follow up on bugs. It's like Jira - but built into Azure DevOps.

- Azure Repos is where you **store your source code** using Git. It supports version control, branching, and pull requests so multiple developers can work on the code efficiently.



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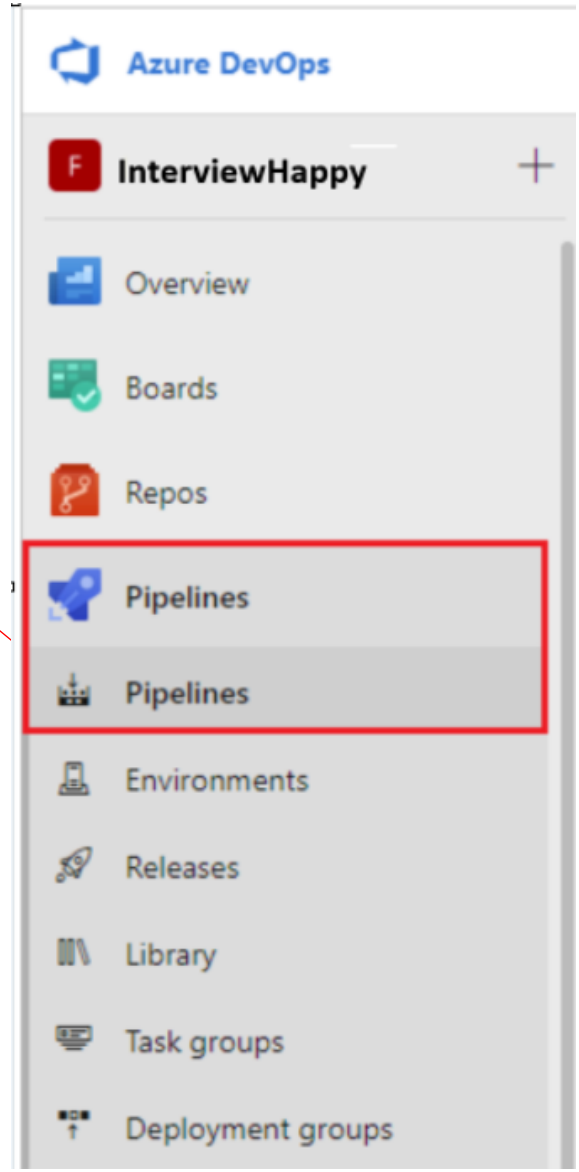
Q. What are Azure Pipelines? How do they help with automation in projects?

- Azure Pipelines are **used for CI/CD**. It helps you automate the build, test, and deployment process. You can use it to continuously integrate code and deploy applications to various environments like Dev, QA, and Production.



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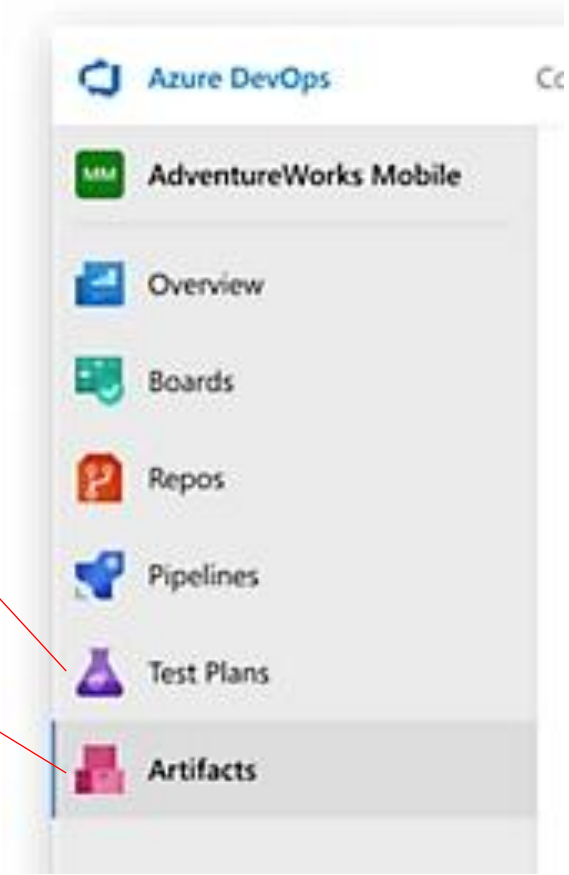
Q. What are Azure Test Plans and Azure Artifacts?

- Azure Test Plans is used for **manual and automation testing**. It allows your QA team to create test cases, execute them, report bugs, and track test coverage.

- Azure Artifacts is a **package management tool** (NuGet, npm, Maven, Python) for dependency management and versioning.



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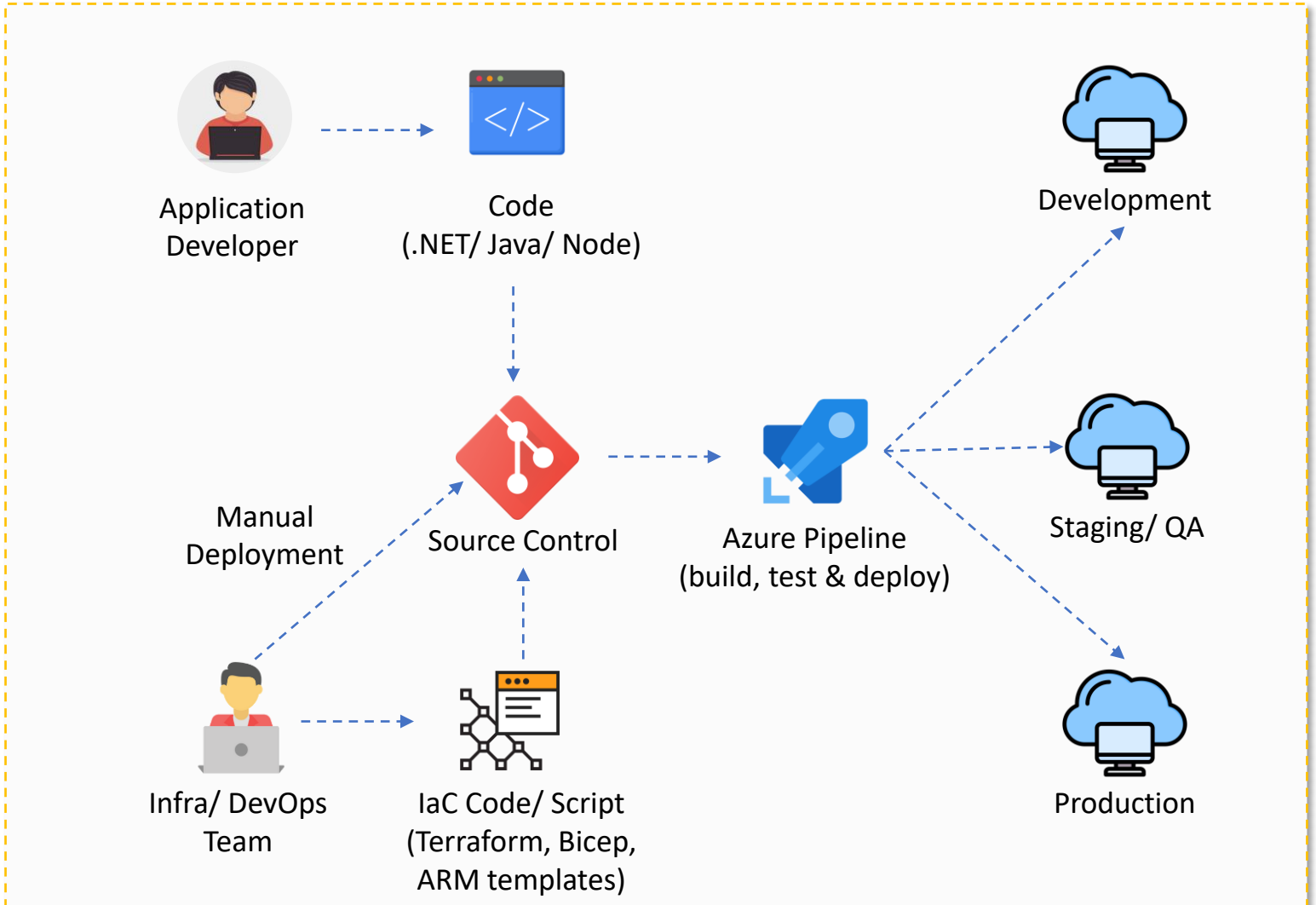
Q. What is Infrastructure as Code (IaC)? Explain how it works? V. IMP.

- Infrastructure as Code (IaC) is a DevOps practice that allows you to define, provision, and manage **infrastructure using code** instead of manual processes.



Infrastructure as Code (IaC)

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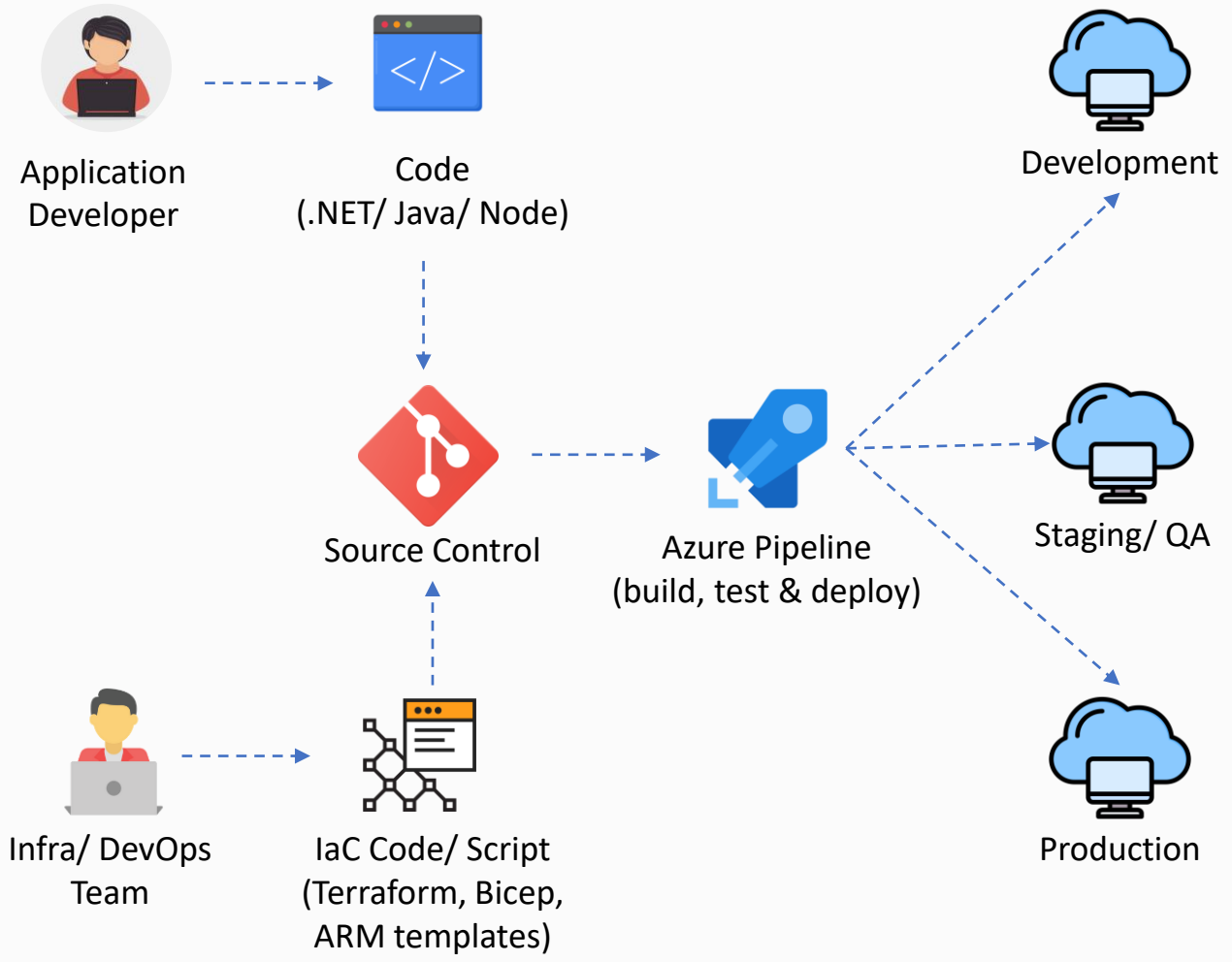
Q. What are the benefits of using IaC? Is there any time when you would not use it?

- IaC enables automation and consistency in cloud environments.

- For small applications, simple manual process can be used, but for large applications, IaC is used to deploy in Dev, QA and Production to **maintain consistency** in all environments.



Infrastructure as Code (IaC)

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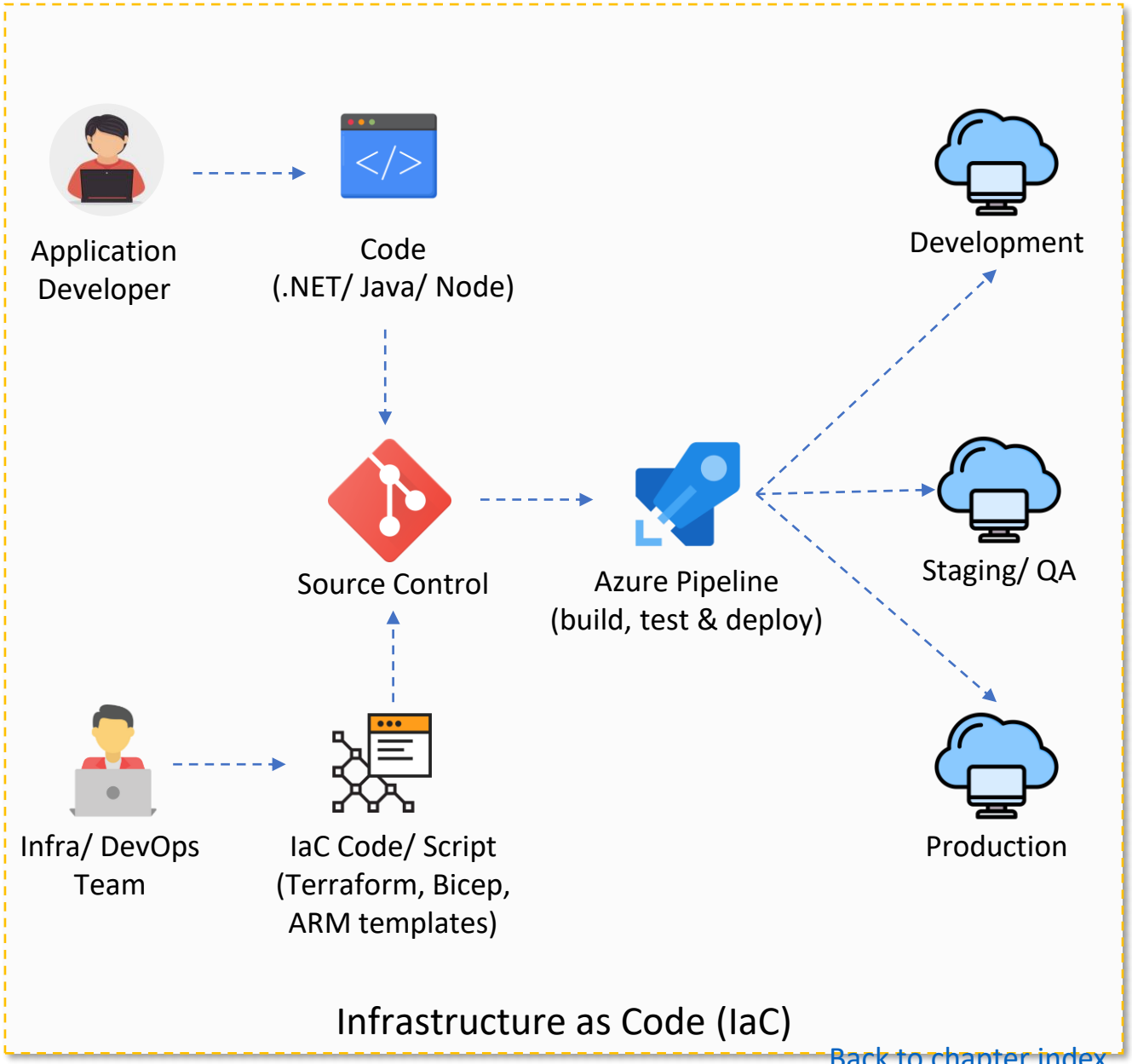
DevOps, CI/CD & IaC
(Scenario Based Questions)



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Q. What is the key difference between DevOps and Azure DevOps?

- DevOps is a practice **combining development and operations** for faster delivery. Azure DevOps is a toolset that helps implement DevOps using services like Boards, Repos, Pipelines, and more.

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Q. How would you automate deployment after every code commit?

- **Use a CI/CD pipeline** in Azure Pipelines.
- It builds the code, runs tests, and deploys to staging or production automatically after each commit.

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Q. What does a basic Azure DevOps pipeline consist of?

- CI = Build stage (compilation, tests)
- CD = Release stage (deployment).

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Q. What is the role of Azure Boards in a team project?

- Azure Boards helps manage work items, tasks, bugs, and sprint planning using Kanban boards or Scrum.

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Q. How are Azure Repos used in a DevOps setup?

- Azure Repos is a Git repository to **store source code**. Developers push code to Repos, triggering builds in the pipeline.

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Q. Your app works fine in dev but fails in prod due to infra mismatch. How do you solve it?

- Use Infrastructure as Code (IaC) like ARM templates or Bicep to create consistent environments across dev, test, and prod.

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Q. What are the benefits of using IaC in projects?

- IaC ensures repeatability, version control, automation, and faster provisioning of infrastructure. It reduces human error and config drift.

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Q. Can you list tools used for Infrastructure as Code in Azure?

- Tools include **ARM Templates, Bicep, Terraform, and Pulumi**. All define infrastructure as code for automation and consistency.

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Q. When should you avoid using IaC?

- For **very small** or temporary environments, using manual setup may be quicker. But even then, IaC is preferred for future-proofing.

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12. Monitoring & Logs

Q. What is Azure Monitor? How does it help developers? **V. IMP.**

Q. What is Application Insights in Azure? **V. IMP.**

Q. What types of application issues can you troubleshoot using Azure Application Insights?

Q. What is KQL? How does it help developers?

Q. Scenario based Real Application Questions

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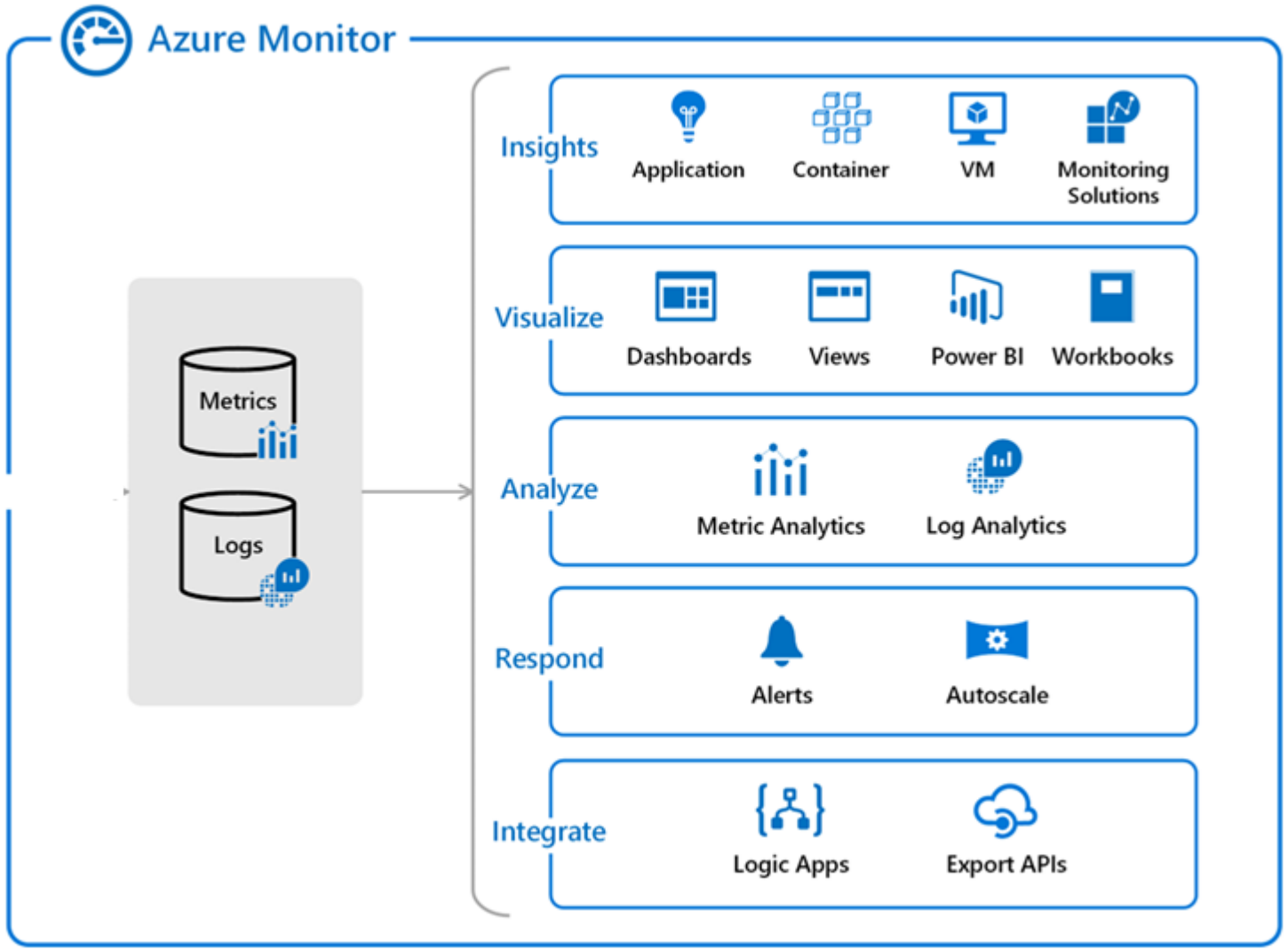
Q. What is Azure Monitor? How does it help developers? V. IMP.

- Azure Monitor is a **cloud-based monitoring** and observability service that collects, analyzes, and visualizes performance data, logs, and metrics from Azure applications and environments.



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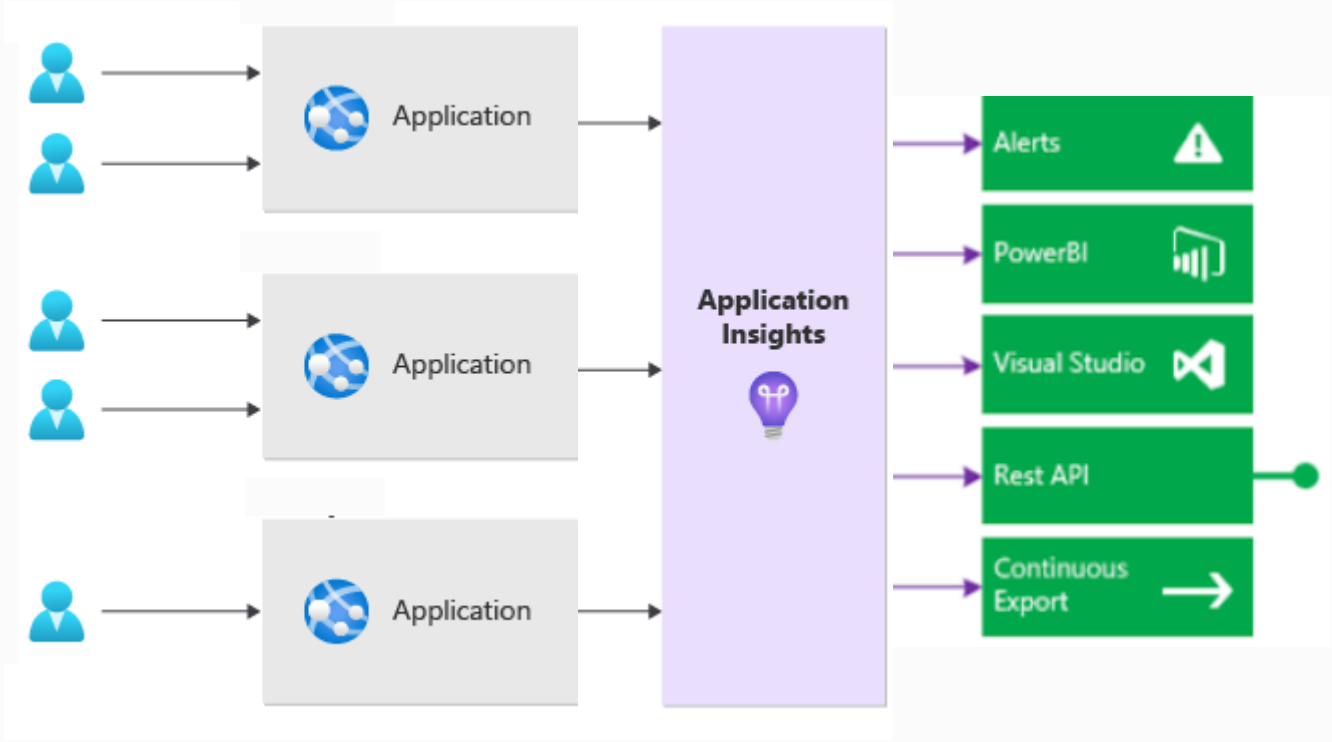
Q. What is Application Insights in Azure? V. IMP.

- Azure Application Insights is an **Application Performance Monitoring (APM) service** that helps developers detect, diagnose, and optimize application performance in real-time. It is a part of Azure Monitor.



Applications in Azure

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Q. What types of application issues can you troubleshoot using Azure Application Insights?

1. Performance problems	Like slow page loads, delayed API responses, or high server response times.
2. Failures and exceptions	Including unhandled exceptions, HTTP 500 errors, or failed dependency calls like database or third-party APIs.
3. Usage anomalies	Such as unexpected drops in user activity or traffic spikes. Application crashes – you can track when and where the app crashed and under what conditions.
4. Incorrect or inconsistent outputs	By checking telemetry data for specific user sessions.
5. Regional or environment-specific issues	Like one deployment working fine, but another showing frequent timeouts.

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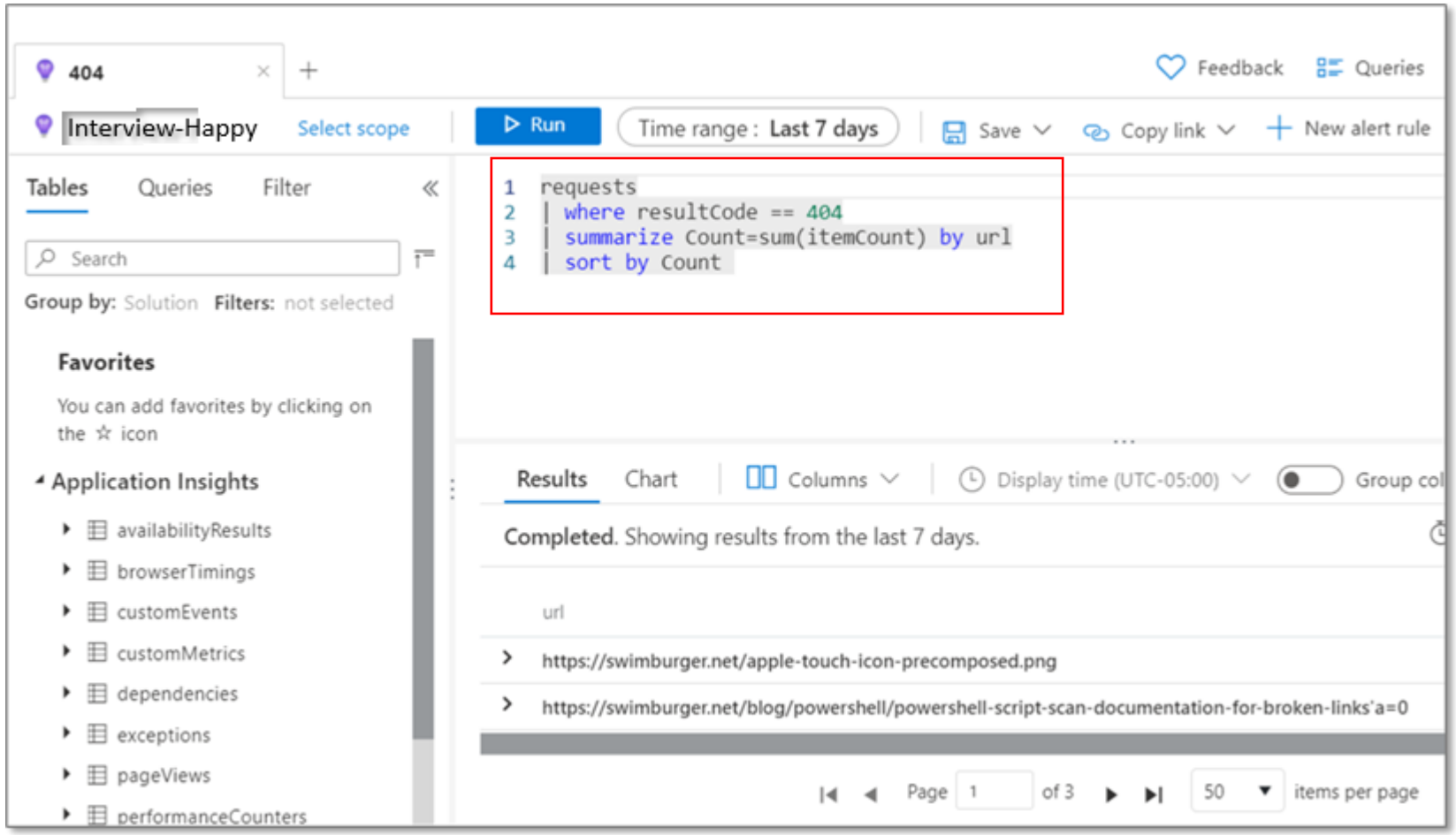
Q. What is KQL? How does it help developers?

- KQL (Kusto Query Language) is a read-only query language used to explore and analyze large datasets in Azure services like Application Insights, Log Analytics, and Azure Monitor.



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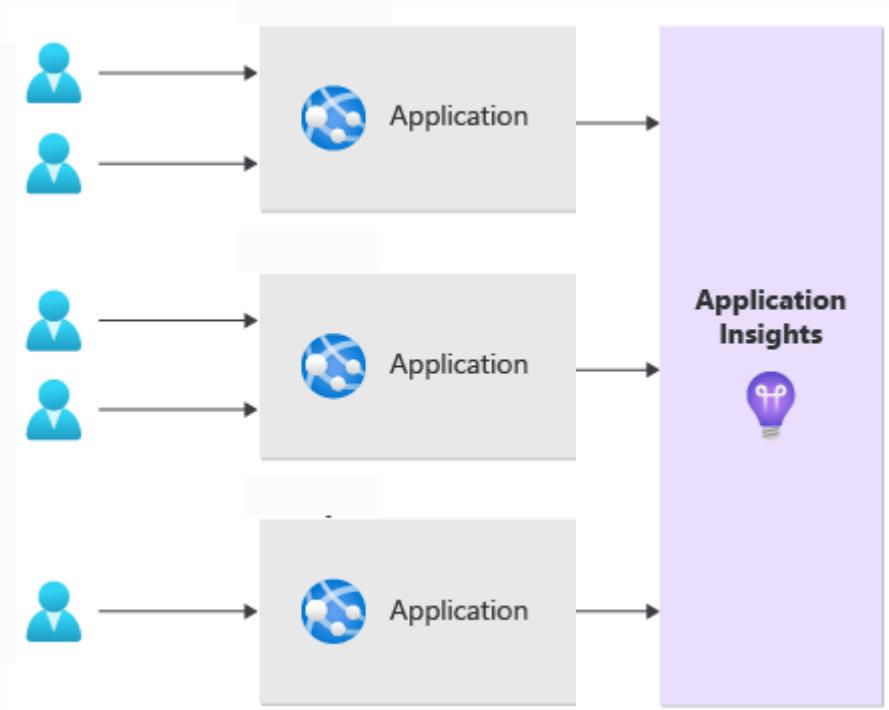
Azure

Monitoring & Logs
(Scenario Based Questions)



Applications in Azure

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Q. How does Application Insights help in a production web app?

- It **automatically tracks requests**, response times, exceptions, dependencies, and usage. It helps diagnose issues and monitor app performance in real-time.

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Q. Your app is facing random slowdowns. Which Azure service helps diagnose it?

- **Use Application Insights** to identify slow requests, failed dependencies, and performance bottlenecks using telemetry and smart diagnostics.

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Q. What is a real-world use case of KQL?

- A developer can write a KQL query to find all **failed requests** in the past 24 hours for a specific API route.

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Q. Can Azure Monitor trigger alerts automatically?

- **Yes**.
- You can create alert rules in Azure Monitor that trigger emails, SMS, or run automation scripts based on metric thresholds or log queries.

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Q. How do Application Insights and Azure Monitor differ?

- App Insights is focused on **application-level** telemetry.
- Azure Monitor is broader — includes **infrastructure metrics, logs, and alerts**.

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13. Event-Driven Messaging

Q. What is Azure Service Bus? Share a real-world example of how it is used? **V. IMP.**

Q. What is the format in which a message is stored in Azure Service Bus?

Q. What is the Pub-Sub model in Azure Service Bus?

Q. What is Topic in Azure Service Bus? What is the difference btw Topic and Queue?

Q. What is the role Azure Event Grid? When to use it in projects? **V. IMP.**

Q. Explain the difference between Azure Functions, Event Grid, Event Hubs, and Service Bus? When should each be used? **V. IMP.**

- Scenario based Real Application Questions

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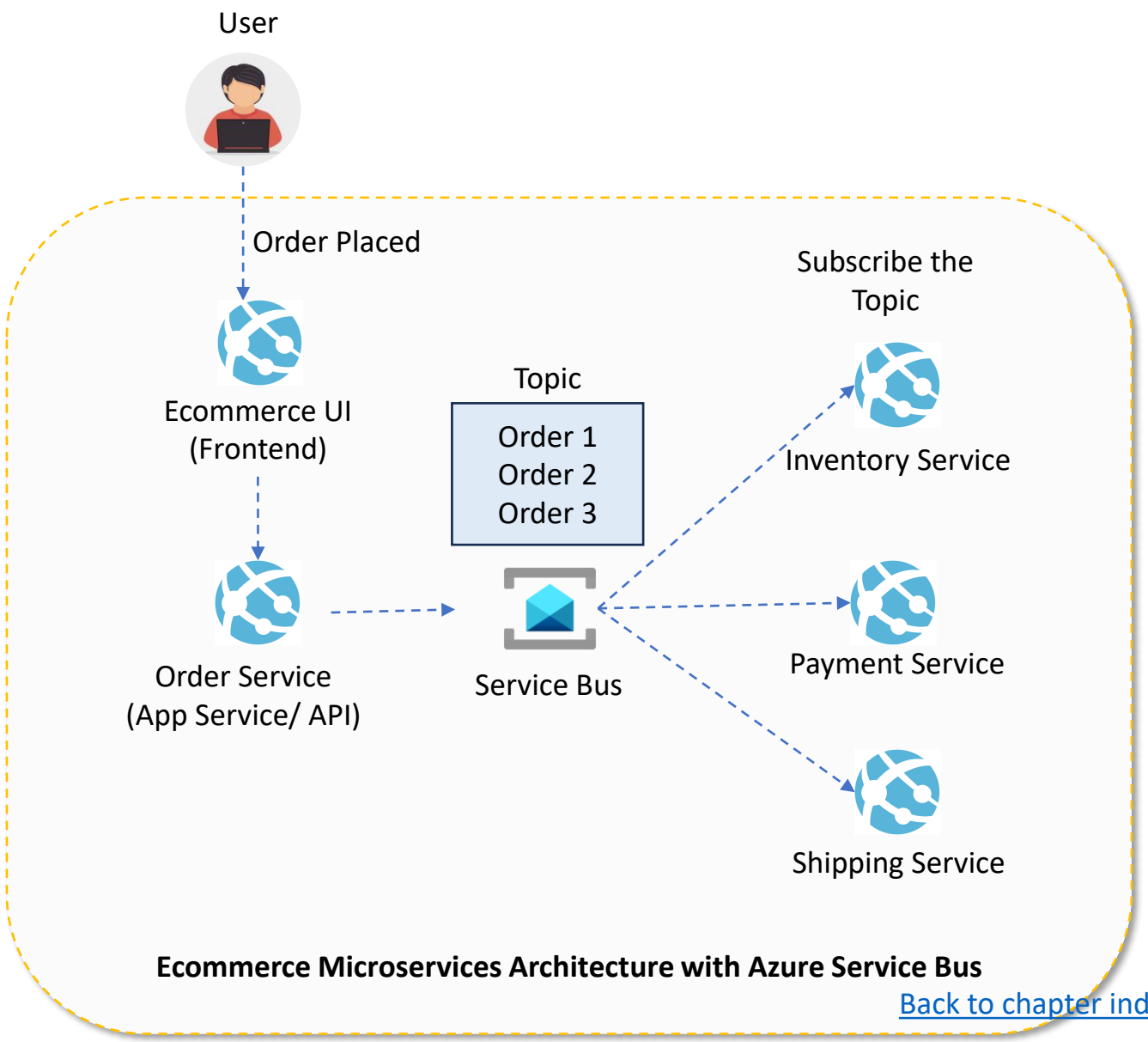
Q. What is Azure Service Bus? Share a real-world example of how it is used? V. IMP.

- Azure Service Bus is a fully managed **message broker** (Message Queue and Pub/Sub).

- Why to use it?

It enables asynchronous messaging between applications and services.



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Q. What is the format in which a message is stored in Azure Service Bus?

- Azure Service Bus receive the message in a readable format like JSON, XML, or plain text. This content is then serialized into **bytes** before being stored in the Service Bus.

```
{
    "orderId": "ORD123456",
    "customerId": "CUST7890",
    "orderDate": "2025-04-04T10:30:00Z",
    "items": [
        {
            "productId": "PROD001",
            "productName": "Wireless Mouse",
            "quantity": 2,
            "price": 499
        }
    ],
    "totalAmount": 1997,
    "paymentStatus": "Pending"
}
```

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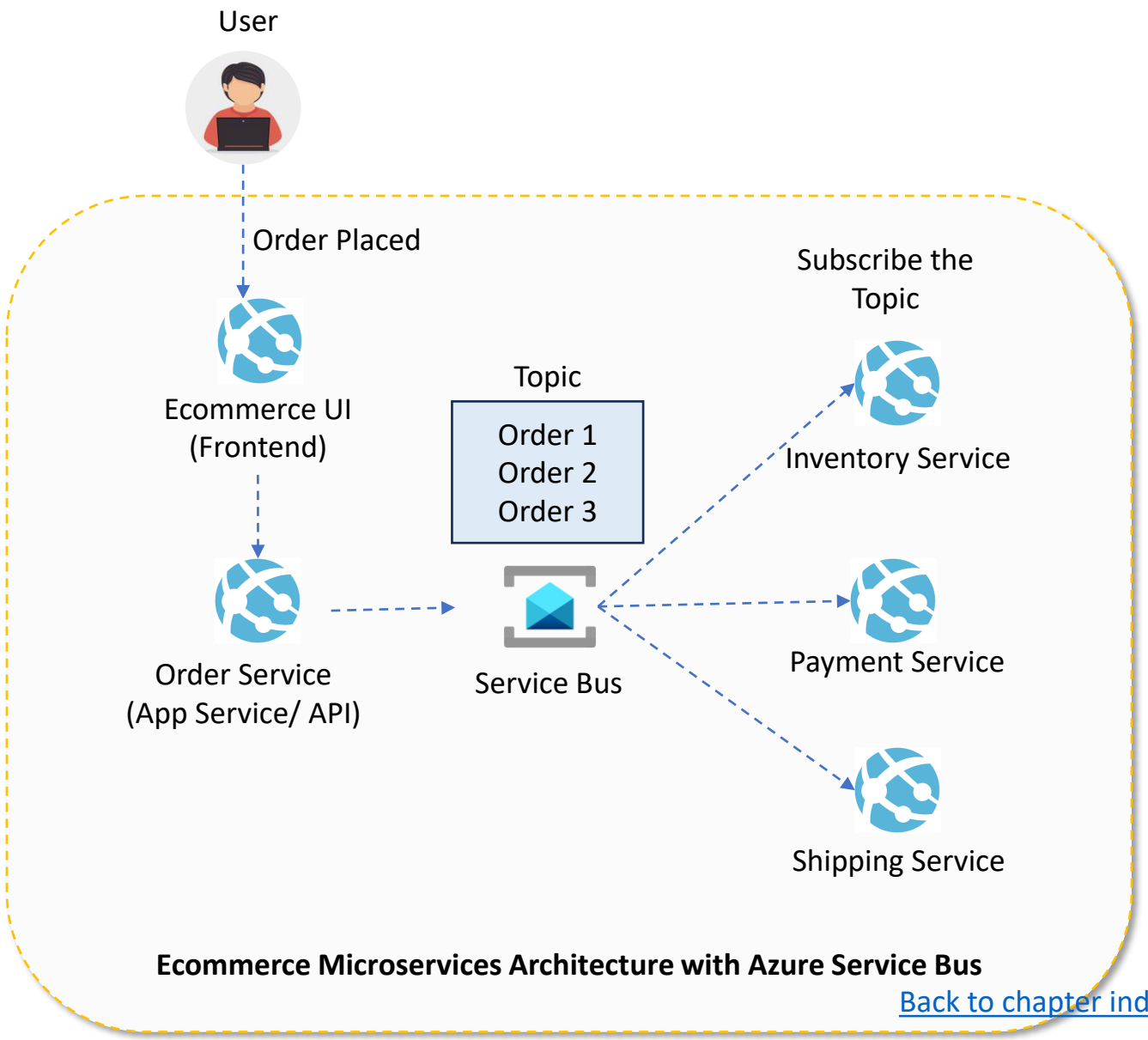
Q. What is the Pub-Sub model in Azure Service Bus?

- The Publish-Subscribe (Pub-Sub) model in Azure Service Bus means:

1. One service (the publisher) sends messages to a Topic.
2. Multiple services (the subscribers) can listen to that topic by creating their own Subscriptions.
3. Each subscriber gets its own copy of every message published to the topic.



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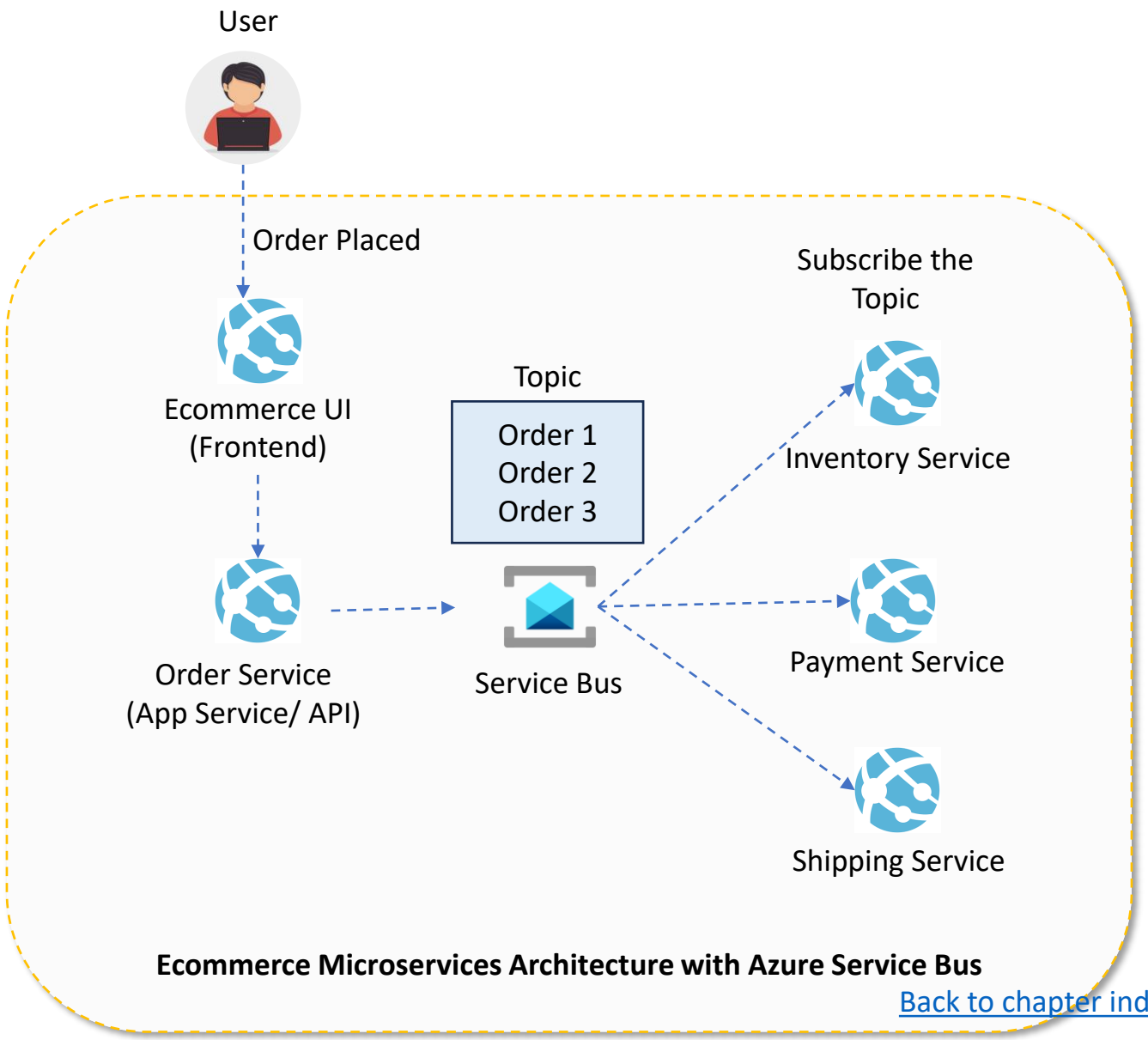
Q. What is Topic in Azure Service Bus? What is the difference btw Topic and Queue?

- A Topic is like a **message broadcasting station** - one message goes in, and many subscribers can receive it independently.

- Topic = One sender (publisher) -> Multiple receivers (subscribers)
- Queue = One sender -> one receiver



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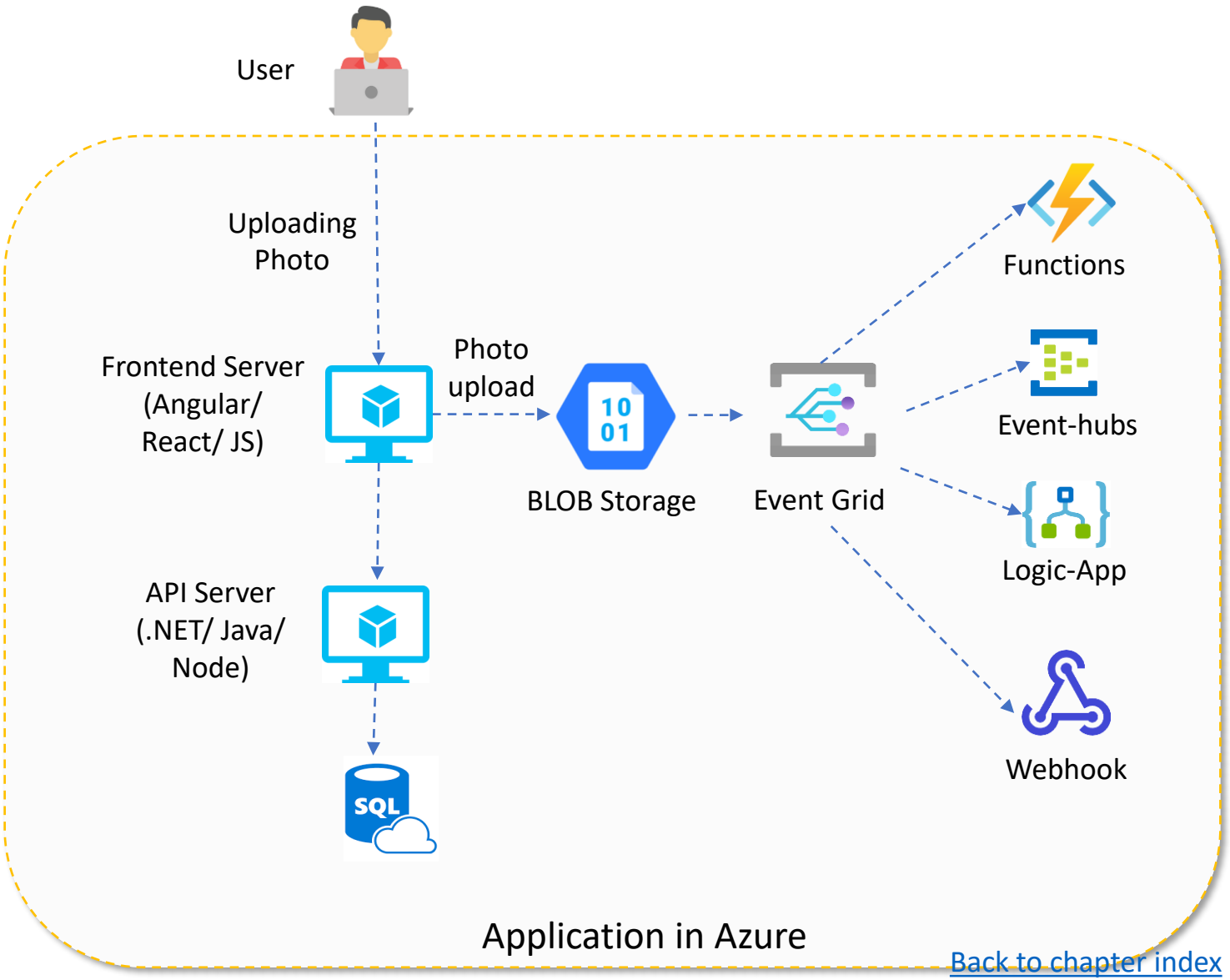
Q. What is the role Azure Event Grid? When to use it in projects? V. IMP.

- Azure Event Grid is an **event routing service** that enables real-time, event-based communication between applications and services using a publish-subscribe model.



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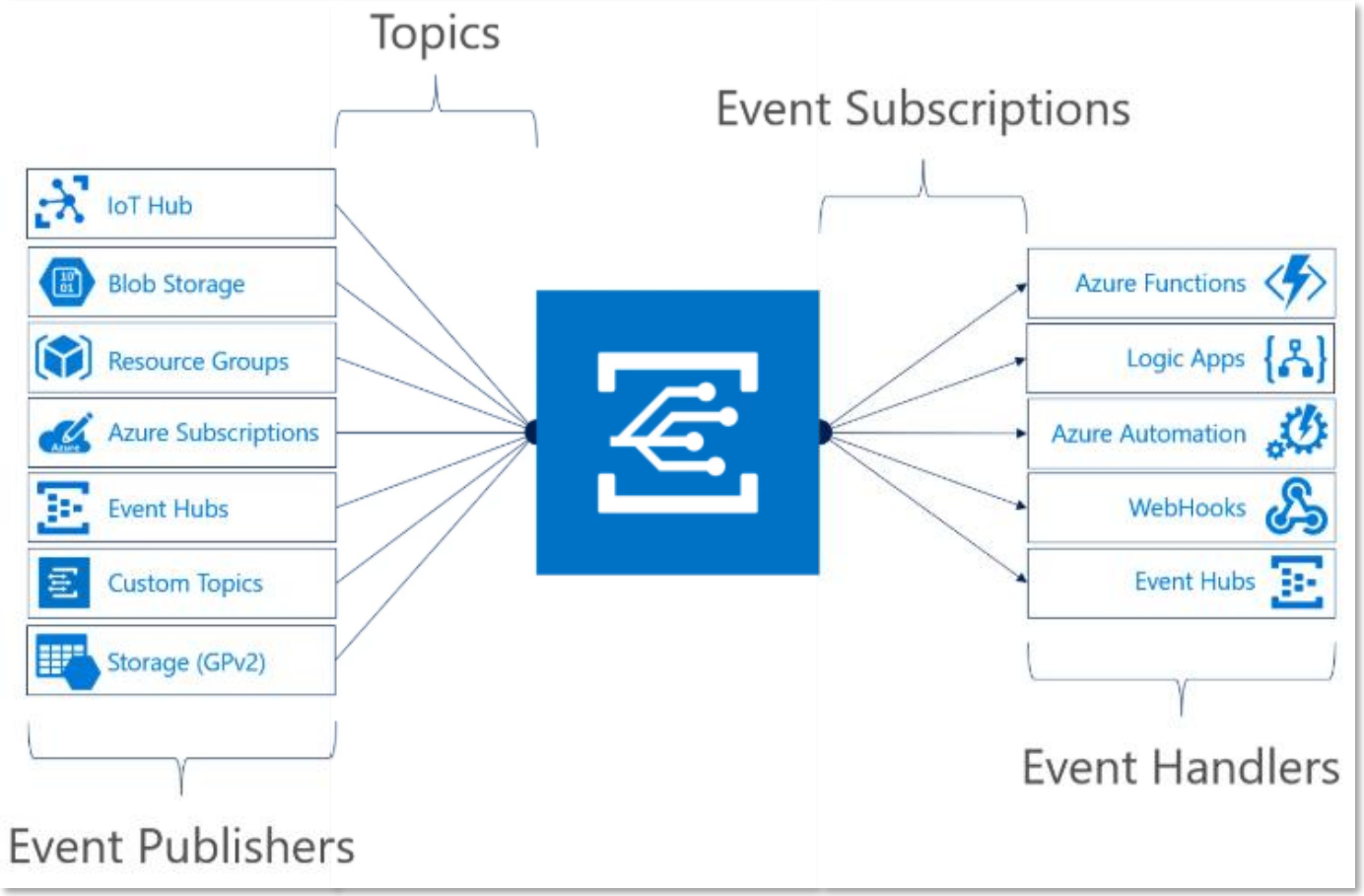
Q. What is the role Azure Event Grid? When to use it in projects? V. IMP.

- When should you use Azure Event Grid?

When you want to trigger multiple services from a single event.



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Q. Explain the difference between Azure Functions, Event Grid, Event Hubs, and Service Bus? When should each be used? V. IMP.

Service	What It Is?	Purpose / When to Use?
Azure Functions	Serverless compute service	When you want to run code in response to events without managing servers. For example, running business logic after triggers like HTTP, timers, and Blob.
Event Grid	Event routing service using pub/sub model	When you want to react to events from many sources and route to multiple handlers. For example, from a single event you can trigger functions, Logic-App, web-hook etc.
Event Hubs	Big data streaming service	When you need to collect and stream massive real-time data from many sources. For example, IoT data ingestion or live analytics
Service Bus	Enterprise message broker	When you need reliable messaging, decoupling, FIFO, or message durability. For example, inter-service communication.

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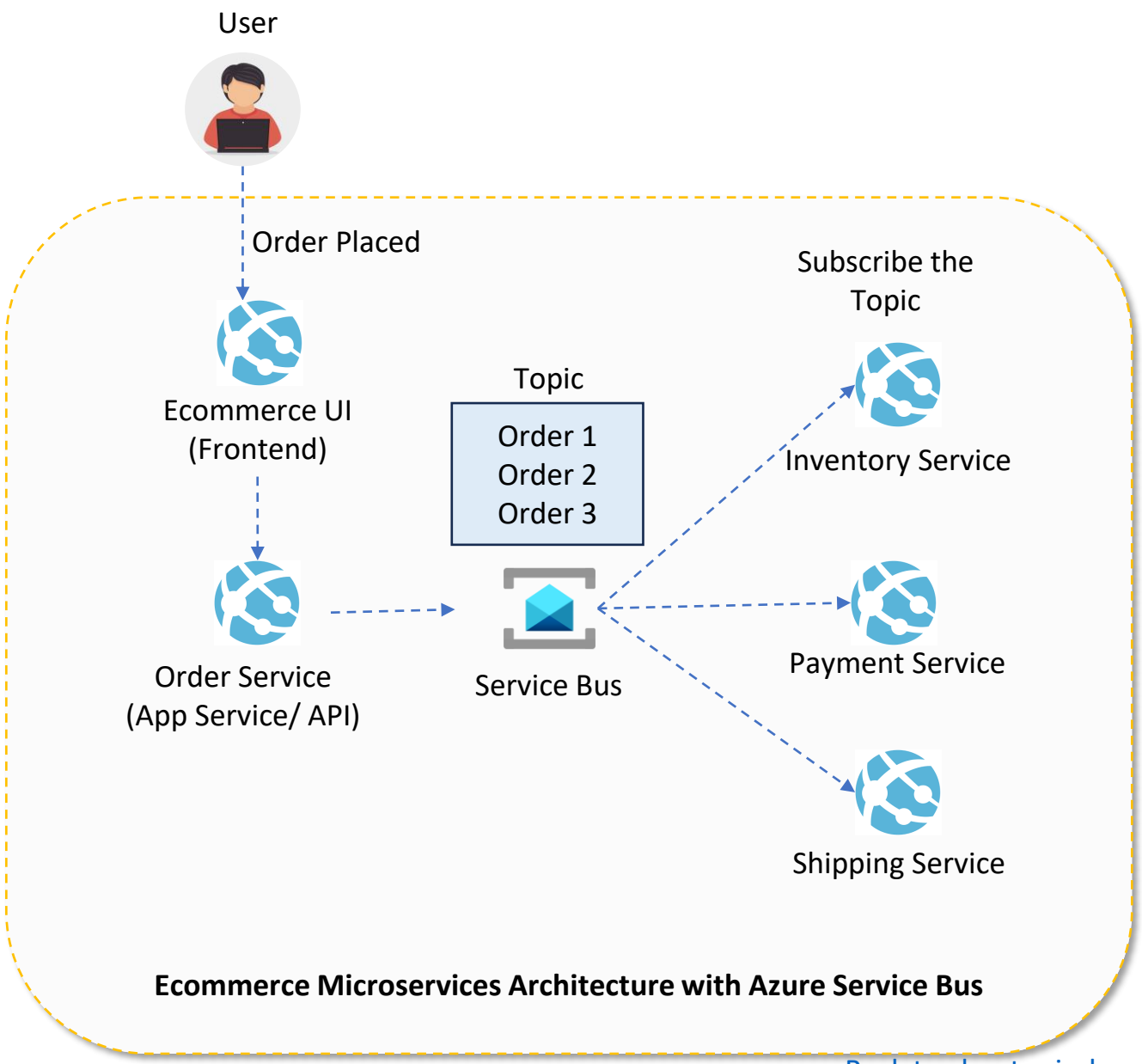
Event-Driven Messaging
(Scenario Based Questions)



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Q. How is Azure Service Bus used in real projects?

- Azure Service Bus is an enterprise-grade **messaging service**.
- Example: In an e-commerce app, order confirmation and shipping updates are sent via Service Bus to different backend services. In microservices based projects.

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Q. What is the Pub-Sub model in Service Bus?

- In Publish-Subscribe, one message is sent to a Topic, and multiple Subscribers can independently receive and process the message.

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Q. You want to notify 3 microservices when a new invoice is created. What would you use?

- **Use Service Bus** Topic with 3 subscriptions, one for each microservice. All get the same message independently.

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Q. What is the difference between Queue and Topic?

- Queue delivers the message to **one receiver**.
- Topic delivers the message to **multiple subscribers**, each with its own rule/filter.

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Q. When should you use Azure Event Grid?

- Event Grid **routes lightweight event notifications** (e.g., Blob created) to Azure services like Functions. Use it for serverless event-driven apps.

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Q. How is Azure Event Grid different from Service Bus?

- Event Grid is for high-volume, lightweight notifications, usually **one-way**.
- Service Bus is for reliable, ordered delivery with advanced messaging patterns.

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Q. You need to process a high-speed stream of telemetry from IoT devices. Which service?

- **Use Azure Event Hubs**.
- It's designed for streaming millions of events per second from distributed devices or applications.

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Q. Summarize when to use Functions, Event Grid, Event Hubs, and Service Bus.

- Azure Functions – Event handler
- Event Grid – Lightweight events (e.g., blob created)
- Event Hubs – Telemetry/log streaming
- Service Bus – Reliable business messages (order placed)

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14. Azure Cognitive Services & ML

Q. What are Azure Cognitive Services?

Q. Share the name of some Azure Cognitive Services?

Q. What is Azure Machine Learning Service? How does it work?

Q. Scenario based Real Application Questions

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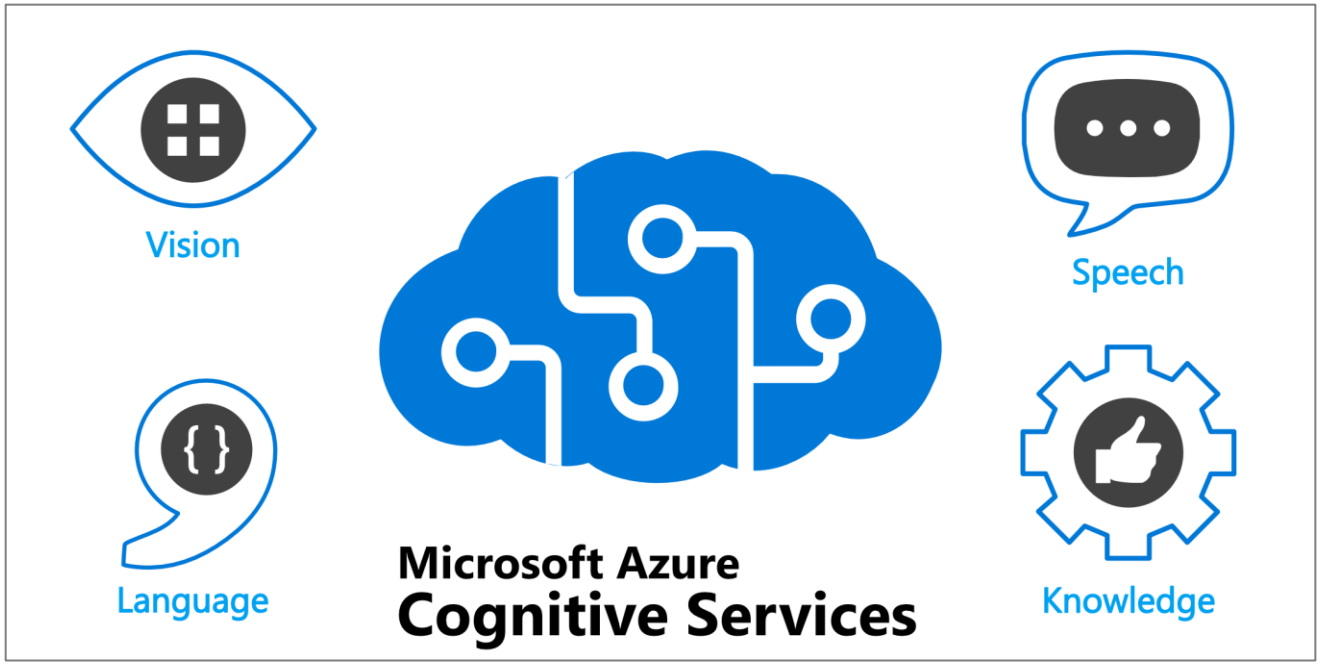
Q. What are Azure Cognitive Services?

- Azure Cognitive Services is a **collection of AI-powered APIs** that enable applications to perform vision, speech, language, and decision-making tasks without requiring deep machine learning expertise.



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Q. Share the name of some Azure Cognitive Services?

Category	Service	Description
Vision	Computer Vision API	Analyse images, detect objects, extract text (OCR), generate descriptions.
	Face API	Detect, identify, and analyze human faces (age, emotion, etc.).
	Form Recognizer	Extract text, key-value pairs, and tables from forms, receipts, etc.
	Custom Vision	Train custom image classifiers with your own data.
	Video Indexer	Extract speakers, objects, and sentiments from video files.
Speech	Speech-to-Text	Convert spoken audio into text.
	Text-to-Speech	Convert text into natural-sounding speech.
	Speech Translation	Translate spoken language in real time.
	Speaker Recognition	Identify or verify speakers by voice.
Language	Text Analytics	Sentiment analysis, key phrase extraction, entity recognition.
	Language Understanding (LUIS)	Understand user intent and extract entities from natural language.
	Translator	Translate text across 90+ languages.
	QnA Maker	Create FAQ-based question answering bots.
	Azure Language Service	Unified service for QnA, summarization, classification, etc.
Decision (Knowledge)	Personalizer	Deliver personalized content using reinforcement learning.
	Content Moderator	Detect offensive or unwanted content in text, images, and videos.
	Anomaly Detector	Detect unusual patterns or anomalies in time-series data.

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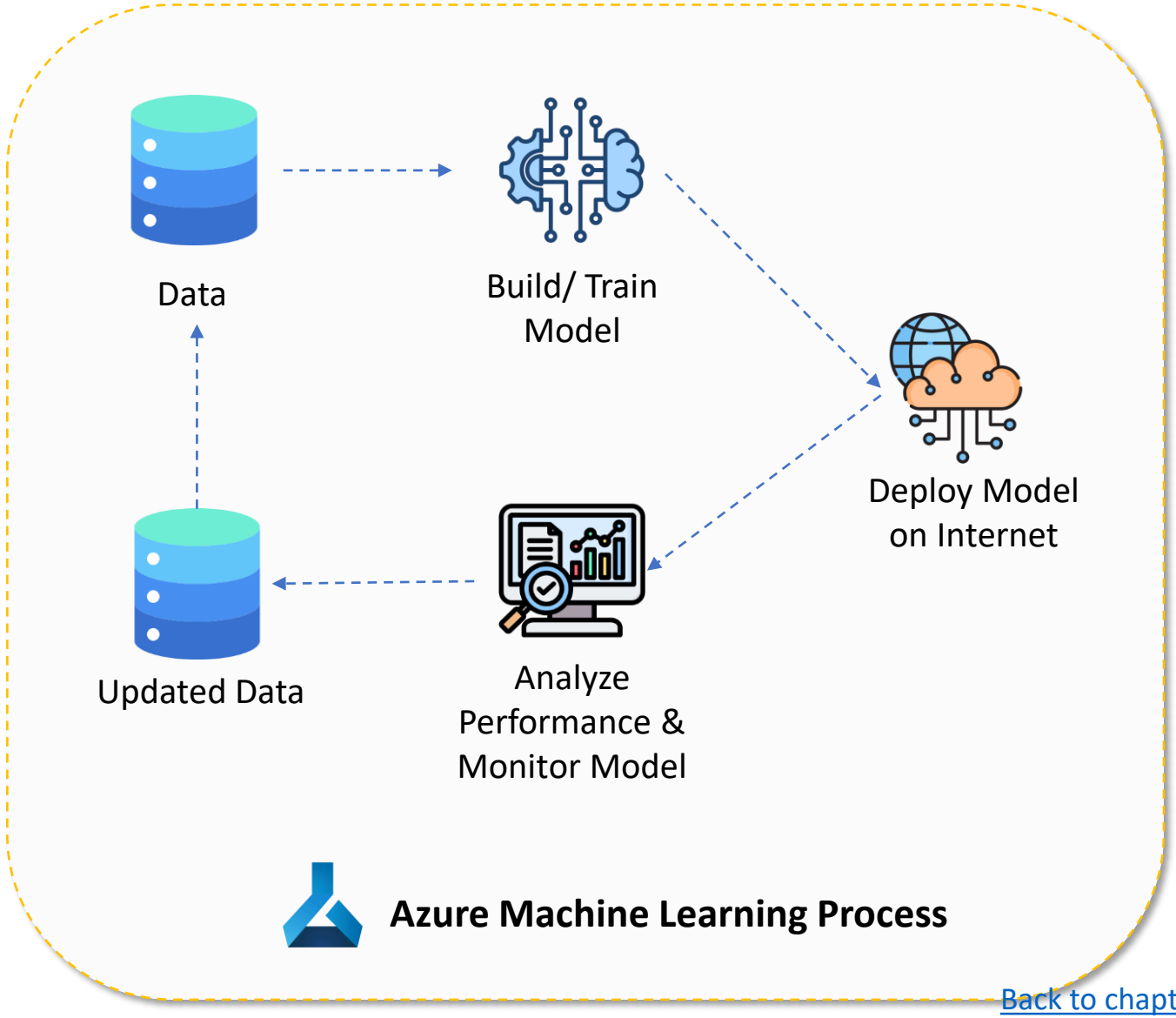
Q. What is Azure Machine Learning Service? How does it work?

- Azure Machine Learning (Azure ML) is a cloud-based platform provided by Microsoft for building, training, and deploying machine learning models at scale. It's designed for data scientists, ML engineers, and developers to create intelligent applications easily.



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Q. What is Azure Machine Learning Service? How does it work?

❖ How Does Azure ML Work?

1. Data Preparation – Import data from Azure Storage, SQL, or external sources and preprocess it.
2. Model Training – Use AutoML, custom Python/R scripts, or deep learning frameworks (TensorFlow, PyTorch, Scikit-learn).
3. Model Deployment – Deploy trained models as REST APIs in Azure Kubernetes Service (AKS), Azure Functions, or Edge devices.
4. Model Monitoring & MLOps – Track model performance, automate retraining, and manage versions using Azure ML Pipelines.
5. Integration – Works with Power BI, Databricks, and Azure Synapse Analytics for advanced analytics.

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15. Azure Performance Optimization

Q. How do you design highly available applications in Azure?

Q. What are the 7 best practices for application developers working on Azure? **V. IMP.**

Q. Share 5 ways to improve performance or reduce the latency of applications? **V. IMP.**

Q. What are the key factors to consider when designing a scalable application in Azure?

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Q. How do you design highly available applications in Azure? V. IMP.

1. Use Availability Zones and Availability Sets	Distribute VMs or services across Availability Zones (physically separate data centers) or Availability Sets (within the same region) to avoid single points of failure.
2. Use Load Balancers	Place an Azure Load Balancer or Application Gateway in front of your services to automatically route traffic to healthy instances.
3. Use Azure App Services	For web apps or APIs, use Azure App Service with Auto-Scaling and deploy it in multiple regions if needed.
4. Use Traffic Manager/ Front Door	Use Traffic Manager or Front Door to route traffic across multiple regions for regional failover.
5. Monitor & Auto-Recovery	Enable Azure Monitor and Application Insights for health checks and automatic alerts.

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Q. What are the 7 best practices for application developers working on Azure? V. IMP.

1. Design for High Availability	Use Availability Zones, Load Balancers, and Traffic Manager to ensure uptime.
2. Use Managed Services	Prefer services like App Service, Azure SQL, or Cosmos DB to reduce maintenance and increase scalability.
3. Implement Secure Identity with Azure AD	Use Azure Active Directory, RBAC, and Managed Identities for secure authentication and authorization.
4. Use Application Insights and Azure Monitor	Continuously monitor performance, track failures, and diagnose issues with telemetry data.
5. Implement DevOps and CI/CD Pipelines	Automate build, test, and deployment using Azure DevOps or GitHub Actions for faster, reliable releases.
6. Secure Application Data	Use **Key Vault** for secrets, encryption at rest and in transit, and network security groups (NSGs).
7. Follow Infrastructure-as-Code (IaC)	Use ARM templates, Bicep, or Terraform to automate and version your infrastructure.

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Q. Share 5 ways to improve performance or reduce the latency of applications? V. IMP.

1. Enable Azure CDN (Content Delivery Network)	Cache static content closer to users to reduce latency.
2. Use Azure Redis Cache	Store frequently accessed data in-memory to speed up response times.
3. Use Application Insights & Azure Monitor	Track performance metrics and quickly identify bottlenecks or slow endpoints.
4. Optimize Database Queries	Use indexing, stored procedures, and performance tuning in Azure SQL or Cosmos DB.
5. Use Azure Front Door or Traffic Manager	Distribute traffic efficiently across regions for global applications.
6. Run background tasks with Azure Functions or Azure Logic Apps	Offload long-running or non-critical tasks to background services for faster user-facing responses.

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Q. What are the key factors to consider when designing a scalable application in Azure?

1. Use **Azure App Services** with auto-scaling
2. Design for **statelessness** so instances can scale easily
3. Store shared data in **Azure Storage or SQL Database**
4. Use **Azure Load Balancer** or **Azure Front Door** for traffic distribution
5. Implement message queues like **Azure Service Bus** to decouple services (microservices based architecture)

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All the best for your interviews

Never ever give up